

This section discusses the primary design considerations common in an MPLS L3VPN environment, along with the possible design options of each.

One of the common connectivity models of enterprise customers to the MPLS L3VPN-based WAN is multihoming. Enterprise customers with this connectivity model often need to load balance traffic across both WAN links (this model includes both, one CE with two links or one site with two CEs and links), as shown in [Figure 6-7](#).



As shown in [Figure 6-7](#), there is an MP-BGP route reflector (RR) part of the MP-BGP control plane architecture. Normally, the RR will advertise only the best route to its clients (other PEs) from the RR point of view to the topology, which will usually break the requirement of load balancing or sharing for those multihomed enterprise customers. One simple solution is to remove the RR and use a full mesh of M-iBGP sessions. However, this might not be an ideal solution for many carrier networks because it may introduce MP-BGP scalability limitations on the provider network. The other common and simple solution to this requirement is to configure the multihomed VPNs/VRFs of the multihomed sites with different RDs, where each route will appear as a unique VPN route to the RR. Consequently, the RR will send these VPN routes to the other remote PEs (PE-1 in [Figure 6-7](#), with PE-2 and PE-3 as the MP-BGP next hops).

Note

The BGP multipathing feature must be enabled within the relevant BGP VRF address family at the remote PE routers (for example, PE-1 in the preceding example). Similarly, enabling BGP multipathing is required in a single CE dual-attached use case to enable the load balancing/sharing from the CE end as well.

MPLS L3VPN Topologies

As covered earlier in this section, RT values enable you to control the import and export of VRF routes, which can control VPN membership per customer. This facilitates the creation of different L3VPN overlaid topologies based on customer requirements. The following are the most common L3VPN WAN topologies, which are controlled by RTs.

Full Mesh

The full-mesh topology shown in [Figure 6-8](#) is the simplest and most common topology that represents the typical MPLS L3VPN layout. Simply, the any-to-any communications model between different customer sites that normally belong to the same customer (under a single VPN or multiple VPNs) must carry the same RT values of the import and export among them (among the relevant PEs).

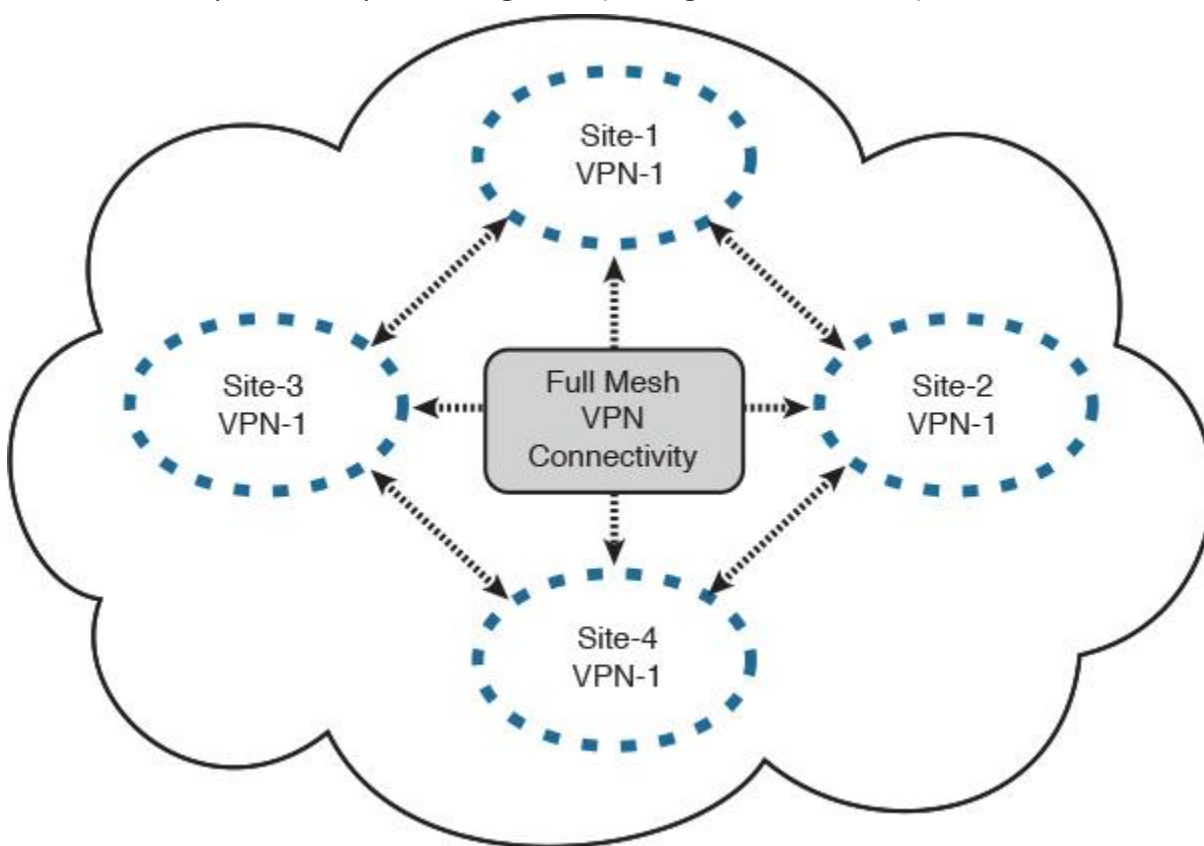


Figure 6-8 MPLS L3VPN Full-Mesh Topology

This design model logically can be shown as one large router with all other locations connected directly to it, as shown in [Figure 6-9](#), where the big router in the middle is the MPLS L3VPN cloud and all other sites are directly attached to it in a star topology.

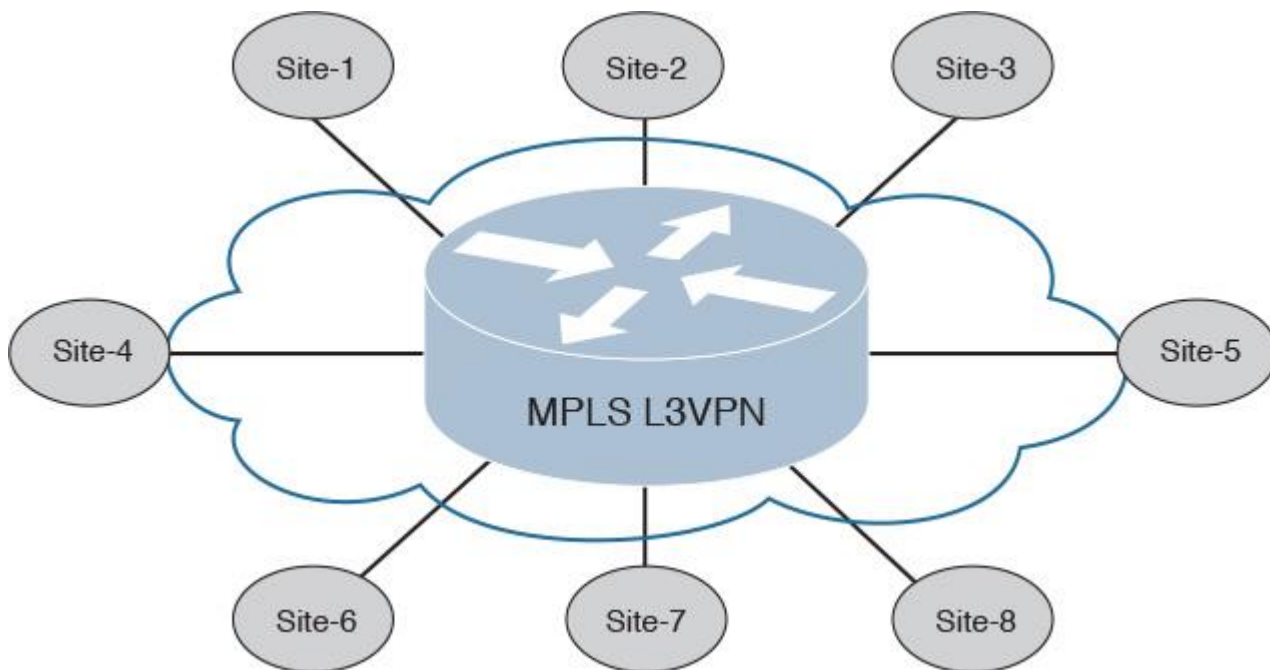


Figure 6-9 *MPLS L3VPN Conceptual View*

Hub-and-Spoke L3VPN Service

In some situations, MPLS L3VPN customers require, for different reasons, that the communication between remote sites has to go through the main or hub site (for example, to align with the enterprise security policy requirements). From the L3VPN service provider point of view, a hub-and-spoke topology can be provisioned for this type of requirement by controlling MP-BGP VPN route propagation (using RT import and export), as shown in [Figure 6-10](#).

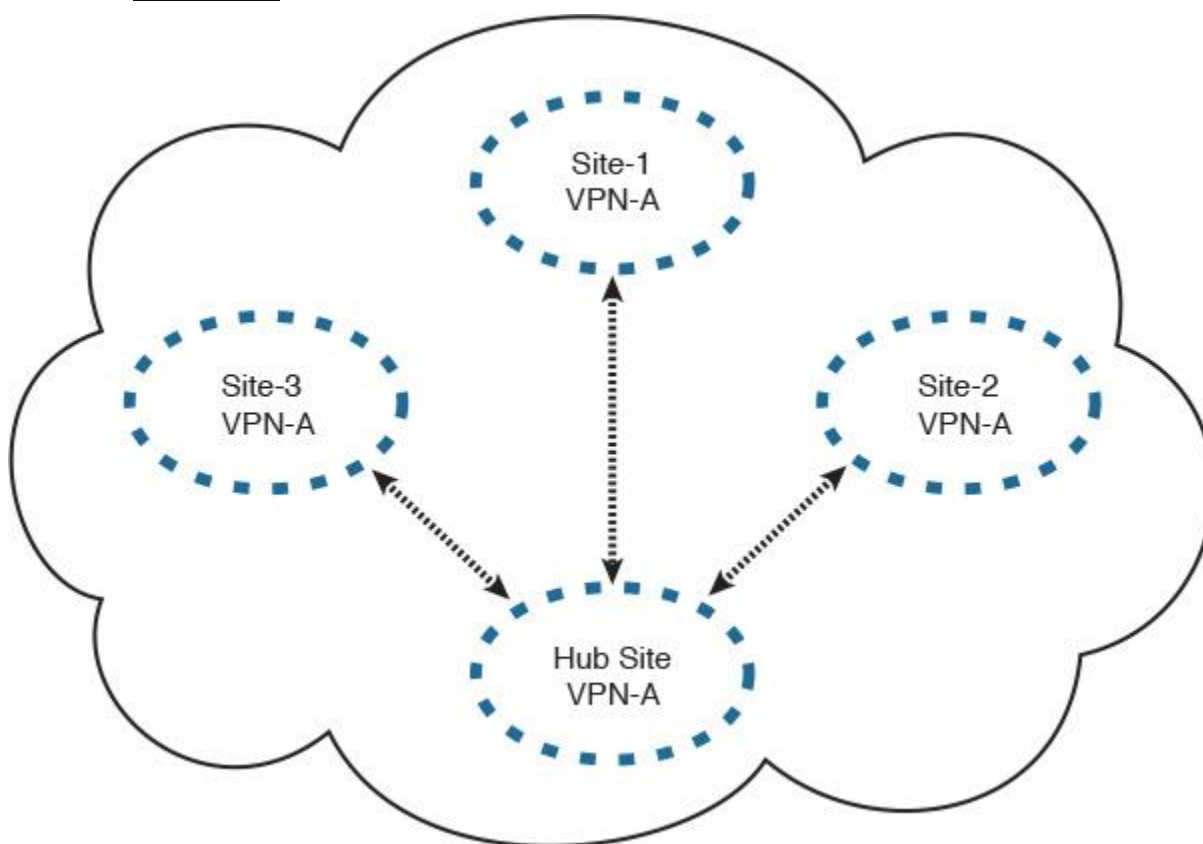


Figure 6-10 *MPLS L3VPN Hub-and-Spoke Topology*

The most common and proven way to achieve a hub-and-spoke topology over L3VPN network is to deploy two links with one VRF per link between the PE and the directly connected CE hub, as shown in [Figure 6-11](#).

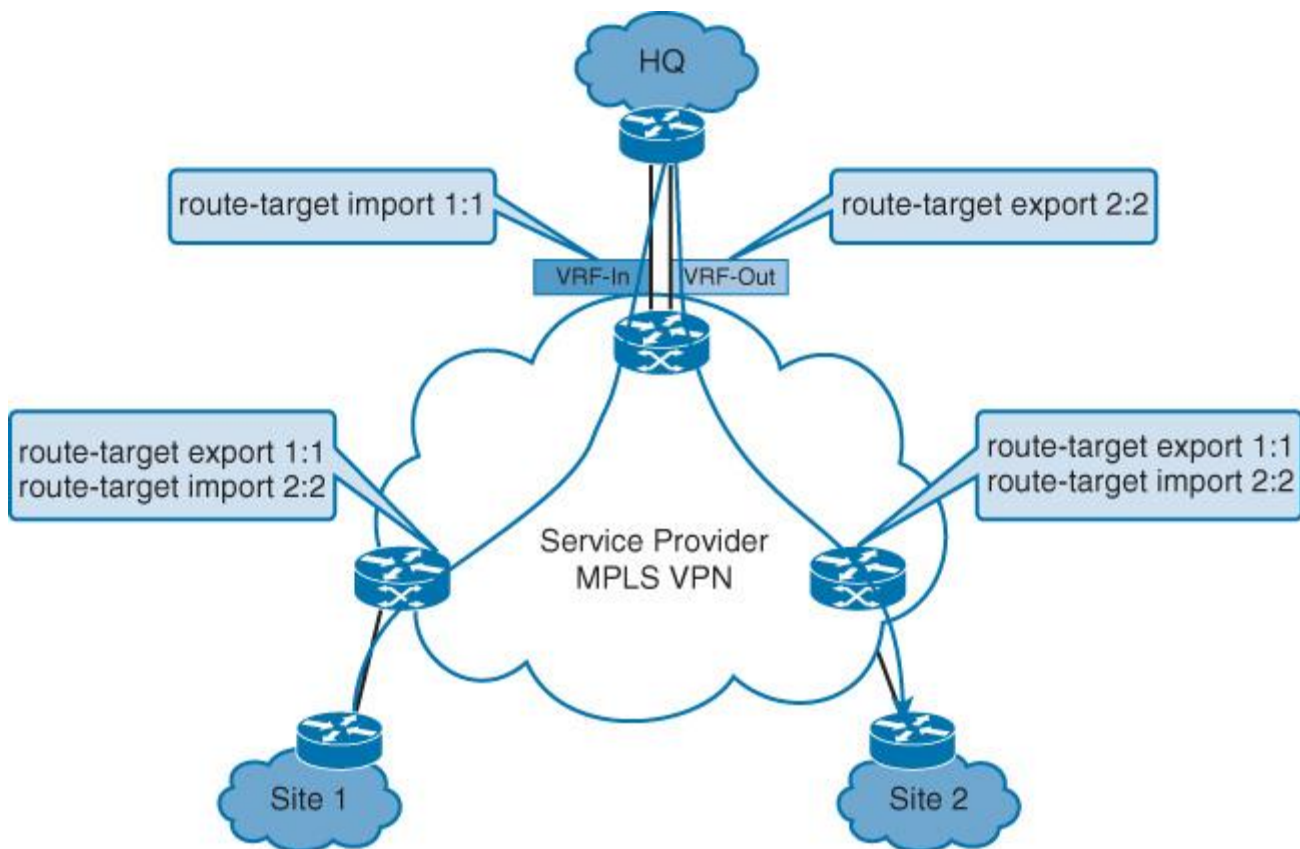


Figure 6-11 MPLS L3VPN Design

Achieving a hub-and-spoke topology in an MPLS L3VPN environment is as simple as controlling the import and export of RT values. However, network designers must be aware of the following design features to avoid breaking the communications across the overlaid hub-and-spoke topology:

- If the BGP used as the PE-CE routing protocol across the hub-and-spoke topology over L3VPN and each site uses the same BGP autonomous system number (ASN), BGP AS-override should be used by the PE connected to the hub-and-spoke sites. This avoids blocking communication among the sites as a result of BGP loop-prevention behavior. Although BGP allows the AS-in feature to be used for the same purpose from the CE side, it must be planned carefully to avoid any unexpected BGP AS_PATH looping.
- If more than one spoke is connected to the same PE half duplex, VRF is required to avoid traffic bypassing the hub site.
- If the hub site has two edge CE routers connected to the MPLS L3VPN cloud, in this case each CE must (ideally) be assigned the role of handling route/traffic in one direction; one hub CE is connected to the receiving link, and the other hub CE is connected to the sending link.

Multilevel Hub and Spoke

In this model, large enterprise customers (usually with distributed sites in multiple geographic areas) can take advantage of multitiered hub-and-spoke topology, as shown in the logical layout in [Figure 6-11](#). For instance, remote sites distributed across different regions can be aggregated into level 1 hub sites per region, while the level 1 hub sites connect to each other in a hub-and-spoke topology and to a second level hub site, such as a centralized data center.

In [Figure 6-12](#), each group of hub and spoke is allocated its own MPLS VPN, which can represent the grouping of sites based on geographic location. With an architecture like this, global enterprises can achieve a more structured network design and traffic flow between different sites and geographic regions. At the same time, service providers can easily provision this type of architecture by controlling routing information propagation among the different VPNs by controlling the import and export of RT values between the VPNs.

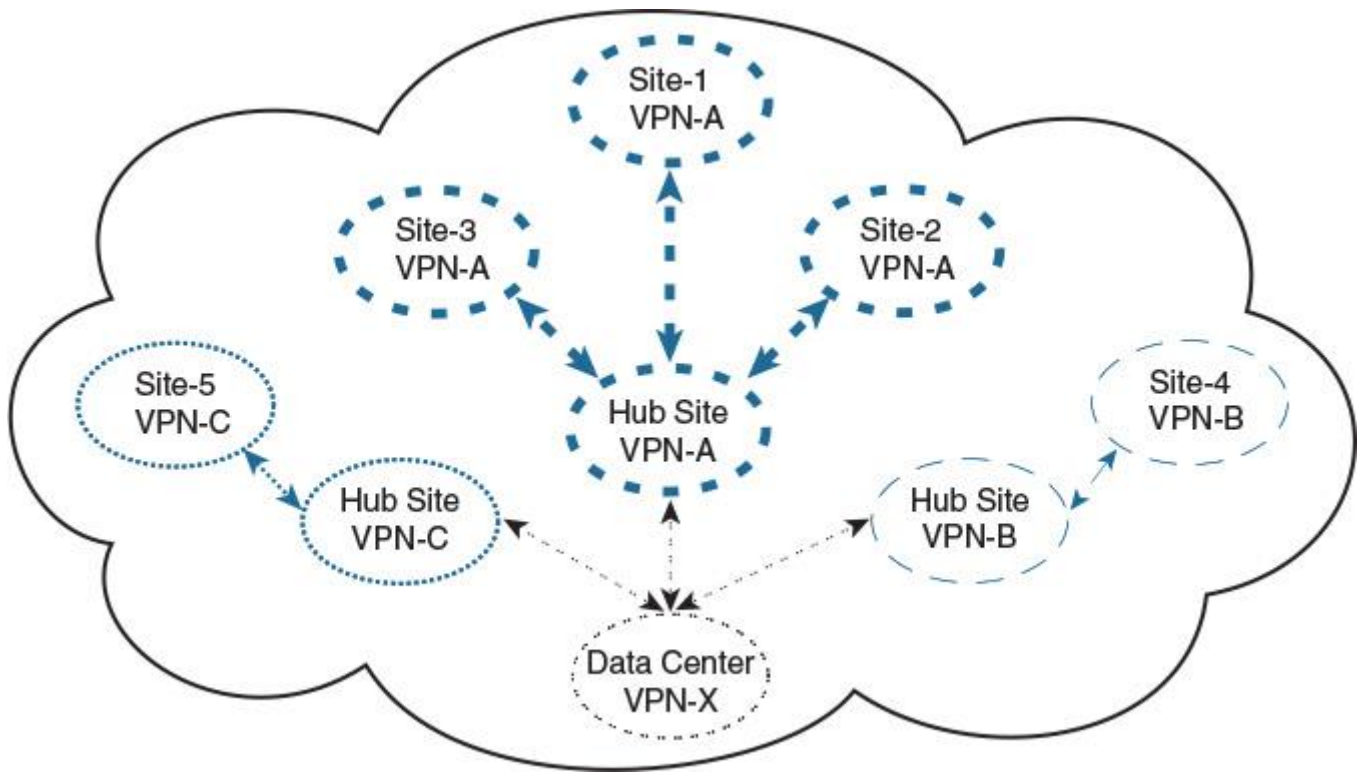


Figure 6-12 *MPLS Multilevel Hub and Spoke*

Extranet and Shared Services

In this particular design model, communication between one or more different VPN networks and a centralized VPN network is required. This is achieved in the same manner as the previous models: by controlling the import and export RT values. In this particular model, the central VPN must have import RT values matching the different export RT values of the VPNs that require access, as shown in [Figure 6-13](#).

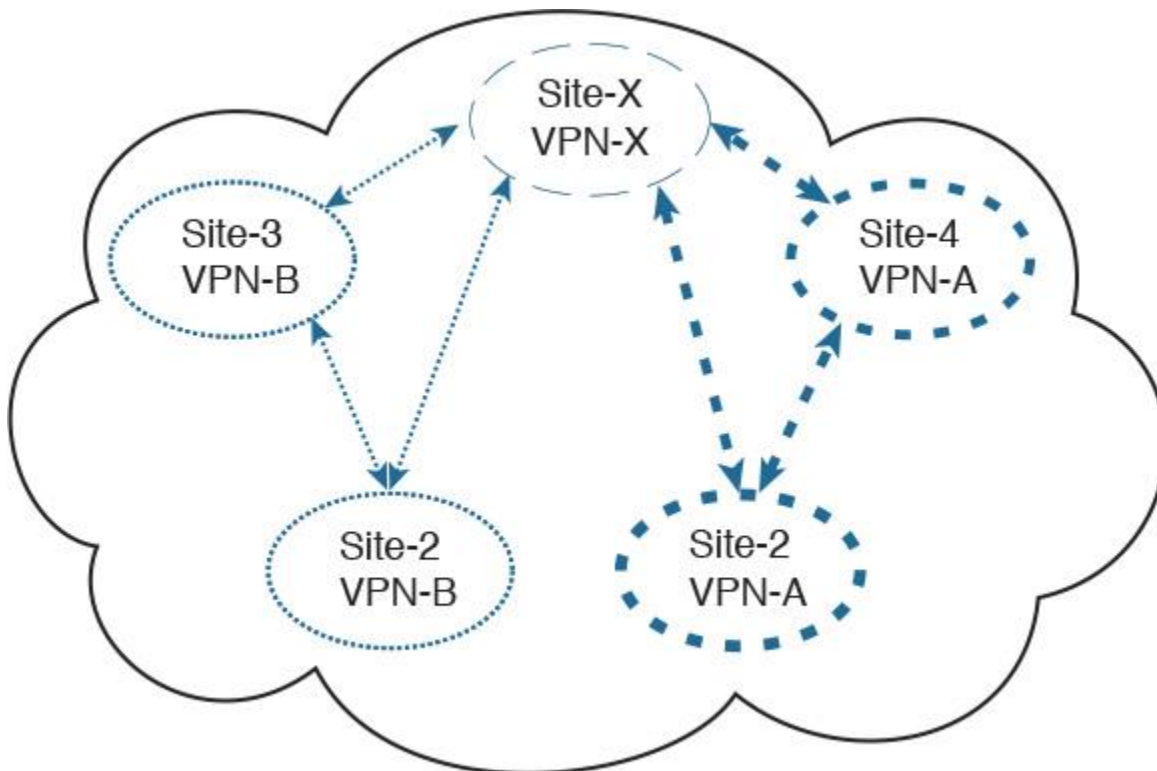


Figure 6-13 *MPLS L3VPN Extranet Topology*

[Figure 6-14](#) illustrates a detailed view of how the import and export routes happen in a shared services VPN architecture.

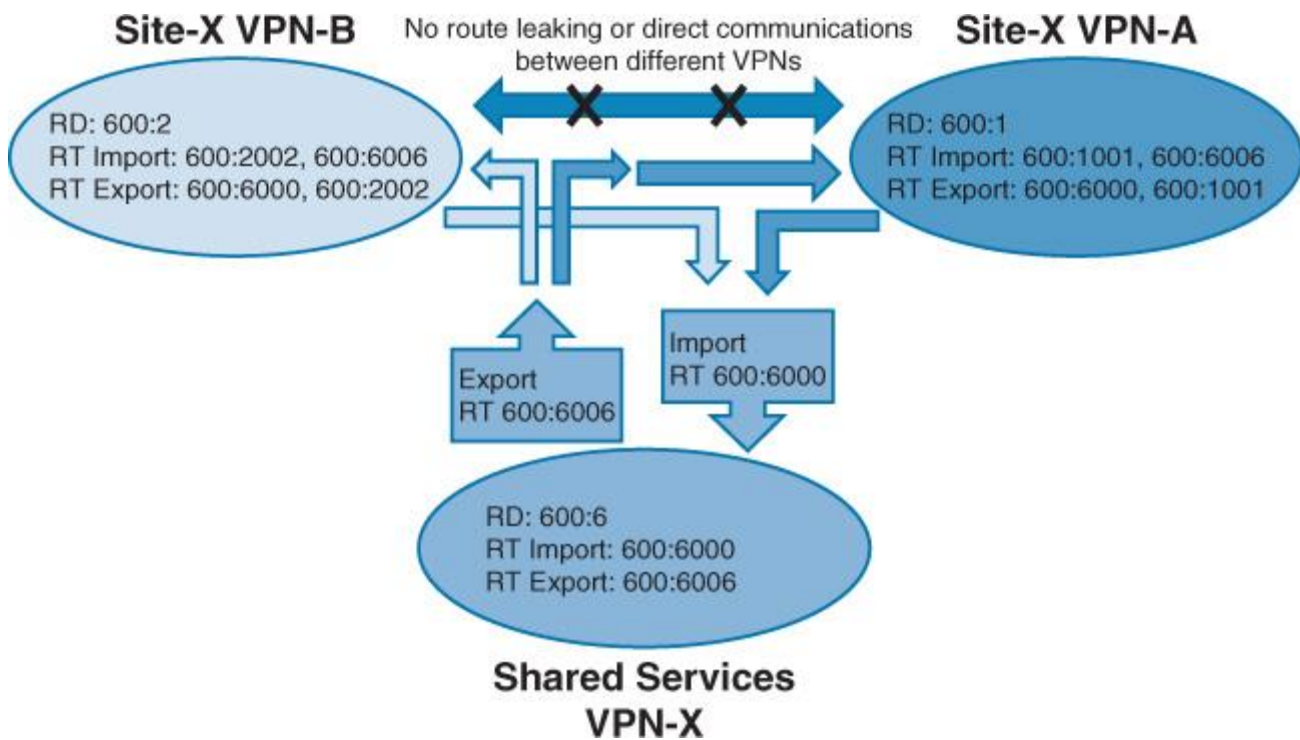


Figure 6-14 MPLS L3VPN Shared Services Connectivity Model

With this design, the shared services VPN will be accessed by the both VPN A and VPN B without compromising routing and reachability separation requirements between these VPNs (no communication between VPN A and VPN B). However, because the shared services VPN will have visibility on both VPNs routes in this example, the IP prefixes of VPN A and B must be unique; otherwise, VRF-aware Network Address Translation (NAT) should be deployed.

The following are the most common scenarios used with this design model:

- **Management:** Network operations center (NOC) management access. For example, the service provider can offer managed CE services to its clients. Accordingly, the NOC or management VPN of the service provider requires access to the relevant customer's VPN, to manage or monitor its CE routers.
- **Shared services:** In general, shared service access refers to several services that have to be accessible from different MPLS L3VPN networks, such as file services, Voice over IP (VoIP) gateways, and hosted applications (software as a service [SaaS]) or Internet connection.
- **Extranet or business-to-business (B2B) communication:** This scenario is common in modern enterprises where vendors and partners, for example, can share limited reachability between their networks to facilitate different types of communications, such as business-to-business telepresence.
- **Community communication:** In this model, a centralized entity provides central services access via a common MPLS VPN cloud. An example of this is an educational system, where commonly a centralized entity provides shared services access to the different schools across different locations while maintaining traffic separation between the schools.

Note

Depending on the network environment, the number of prefixes to be exported and imported can be limited in a controlled manner. For example, in the case of managed services, only the loopback IP of each CE is exported from the customer VPN to the NOC/management VPN for monitoring and remote-access purposes. At the same time, controlling the number of exported and imported prefixes prevents the leaking of extra prefixes, which can lead to other issues such as exposing customer internal routes and unnecessary extra overhead on the PE nodes.

MPLS VPN Internet Access Design Options

The flexibility provided by MPLS-based infrastructures, specifically MPLS VPN, allows operators to offer more than basic private IP WAN connectivity to their customers. One of the primary services that today's MPLS

VPN service providers offer to their customers is Internet access as a value-added service, taking advantage of the flexible MPLS VPN architecture that uses the same service provider infrastructure to provide various connectivity models and services. In addition, it is an additional source of revenue, from the operator's business perspective.

However, from a network design perspective, Internet access for MPLS VPN customers can be provided using multiple design options. Typically, each option has its strengths and weaknesses. This section discusses and compares the most common design options used to provide Internet service from the MPLS VPN service provider's perspective. The design models are categorized as follows:

- Non-MP-BGP VPN Internet routing
- MP-BGP VPN-based Internet routing

Note

Although this section targets service provider networks with these design models, all the concept and design options still apply to any enterprise network that has MPLS VPN enabled (commonly across the campus or WAN core "service provider style").

Non-MP-BGP VPN Internet Routing

The primary concept of this design model is that Internet routes are carried across the carrier network using the global routing table. The following are the two common design options of the model:

■ **Option 1:** VRF specific default route. This design option is based on using the static default route to redirect traffic from a customer VRF to the Internet gateway (Autonomous System Boundary Router [ASBR]) using the PE's global routing table, as shown in [Figure 6-15](#). For the traffic in the other direction (Internet to VPN customers), the same concept is used (static route to move Internet traffic from the global routing table to the corresponding VRF/VPN per customer).

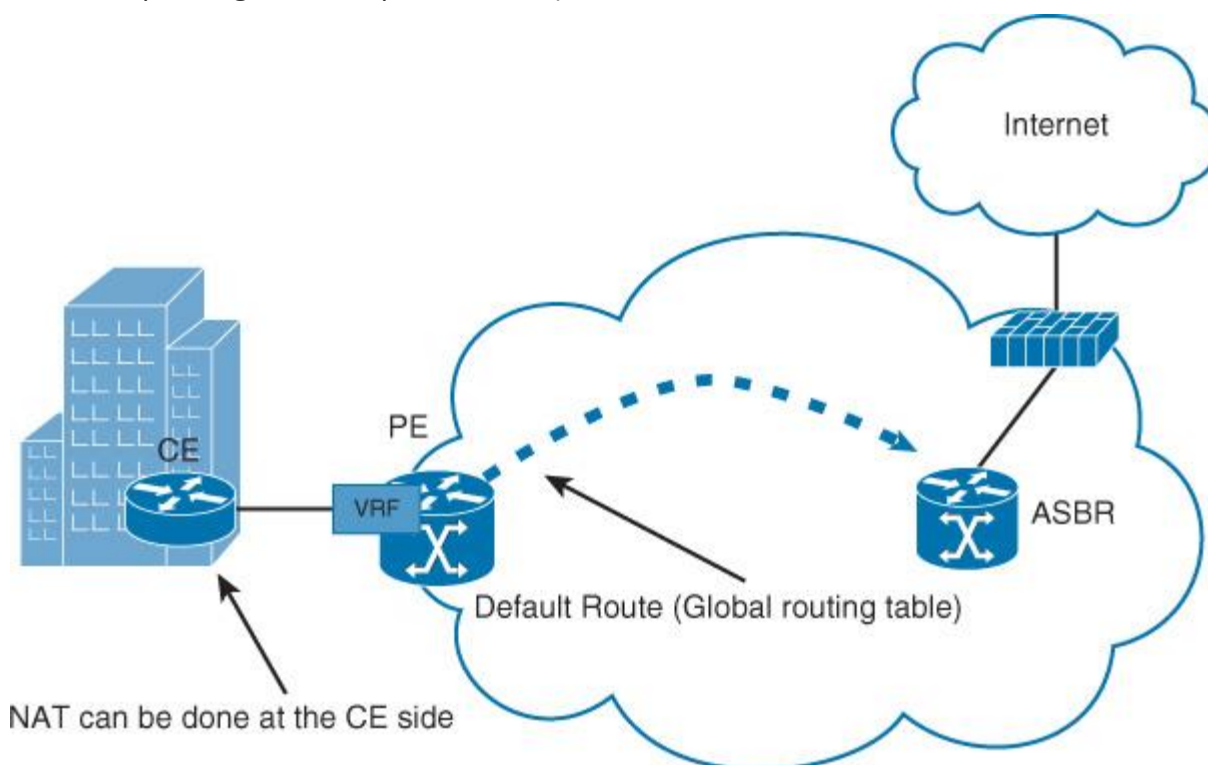


Figure 6-15 MPLS L3VPN Internet Access Option 1

■ **Option 2:** Separate PE-CE subinterface. This design option uses the conceptual model of Option 1, but with the ability to separate Internet traffic and IP VPN traffic over two physical links or subinterfaces (such as 802.1Q based, or interface or data link connection identifier [DLCI] based in case of FR CE-PE physical connectivity). For the operators to propagate Internet routes (or default only) between a PE and the directly

attached CE, either a static route or BGP is used over the Internet link (the non-VRF interface), as shown in [Figure 6-16](#).

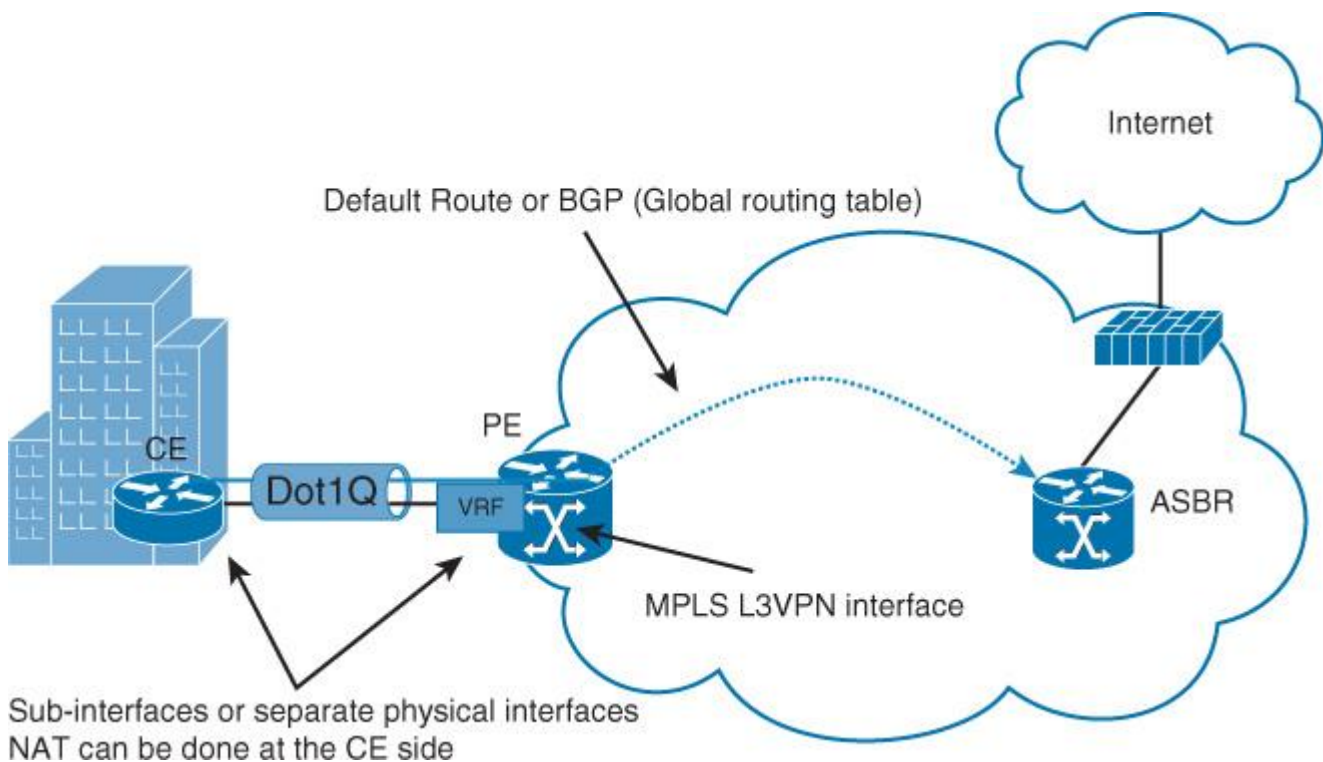


Figure 6-16 MPLS L3VPN Internet Access Option 2

MP-BGP VPN Internet Routing

This design model relies primarily on MP-BGP VPN to carry Internet routes across the carrier network. The following are the two common design options of the model:

Note

The sequence of numbers of the following design options continues from the two other design options previously discussed, to facilitate the reference of each model in [Table 6-2](#).

■ **Option 3:** Extranet with Internet-VRF. This design option is based on the principle discussed earlier in this chapter, which is the extranet or shared VPN where the Internet gateway (ASBR) installs Internet routes in its own VPN. This VPN has bidirectional communication with the customer's VPN, a customer that requires Internet access over the same link/VPN (operators control the propagation of the Internet routes or default route into a customer's VRF routing table via the import and export RT values), as shown in [Figure 6-17](#). In addition, if the full Internet routes are injected into the customer VPN, it is highly recommended to change MPLS VPN label allocation mode to either per CE or per VRF. This prevents the default behavior per-prefix allocation mode, which will usually allocate a label per prefix. Based on the size of the Internet routes, there can be noticeable performance inefficiencies in the case of the per-prefix label allocation model.

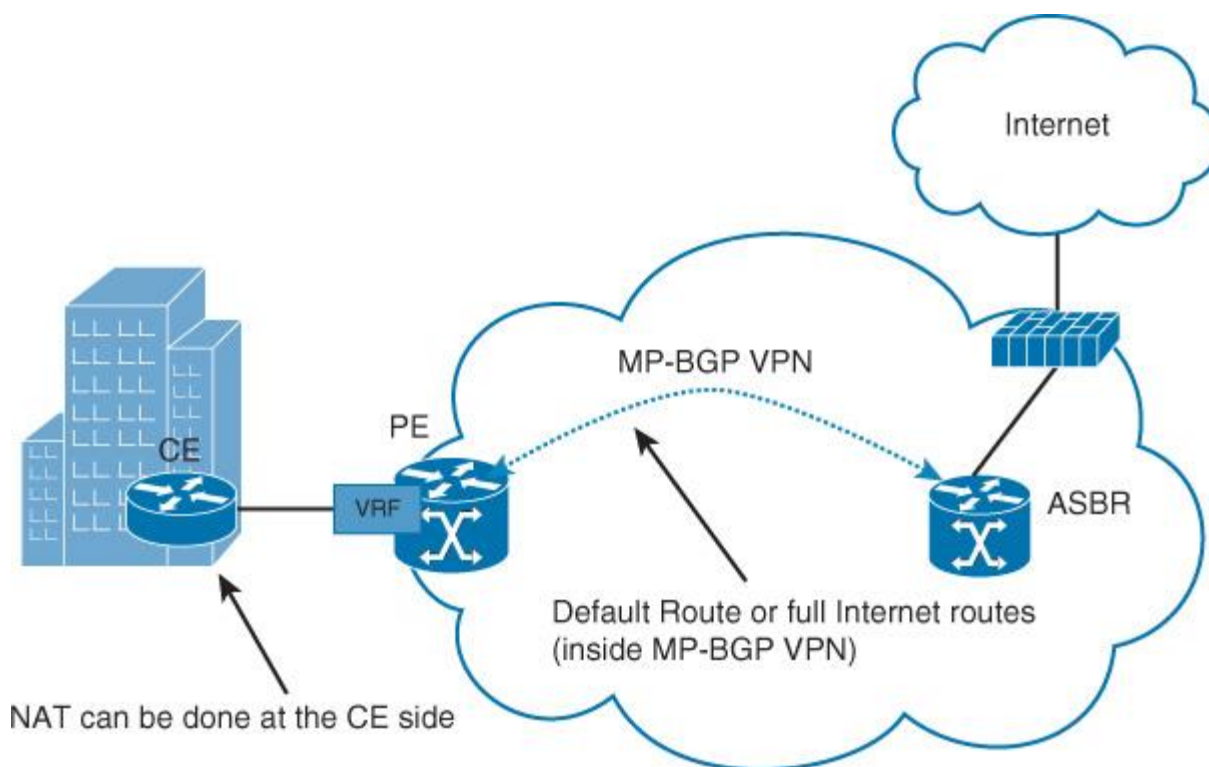


Figure 6-17 MPLS L3VPN Internet Access Option 3

■ **Option 4:** VRF-aware NAT. This design option relies on MP-BGP VPN to customer Internet traffic, like Option 3. However, in this design option, the Internet gateway (ASBR) is not placed in a shared VPN. Instead, each VRF/VPN is created at the ASRB and each VRF is injected with a default route to be propagated to the corresponding VPN across the MPLS VPN network (toward the ASBR inside interface), while performing VRF-aware NATing on the outside interface (Internet facing) for both ingress and egress Internet traffic. As result, with this design option, customers may retain their private IP addressing even if there is overlap with other customer IP addressing ranges, as shown in [Figure 6-18](#).

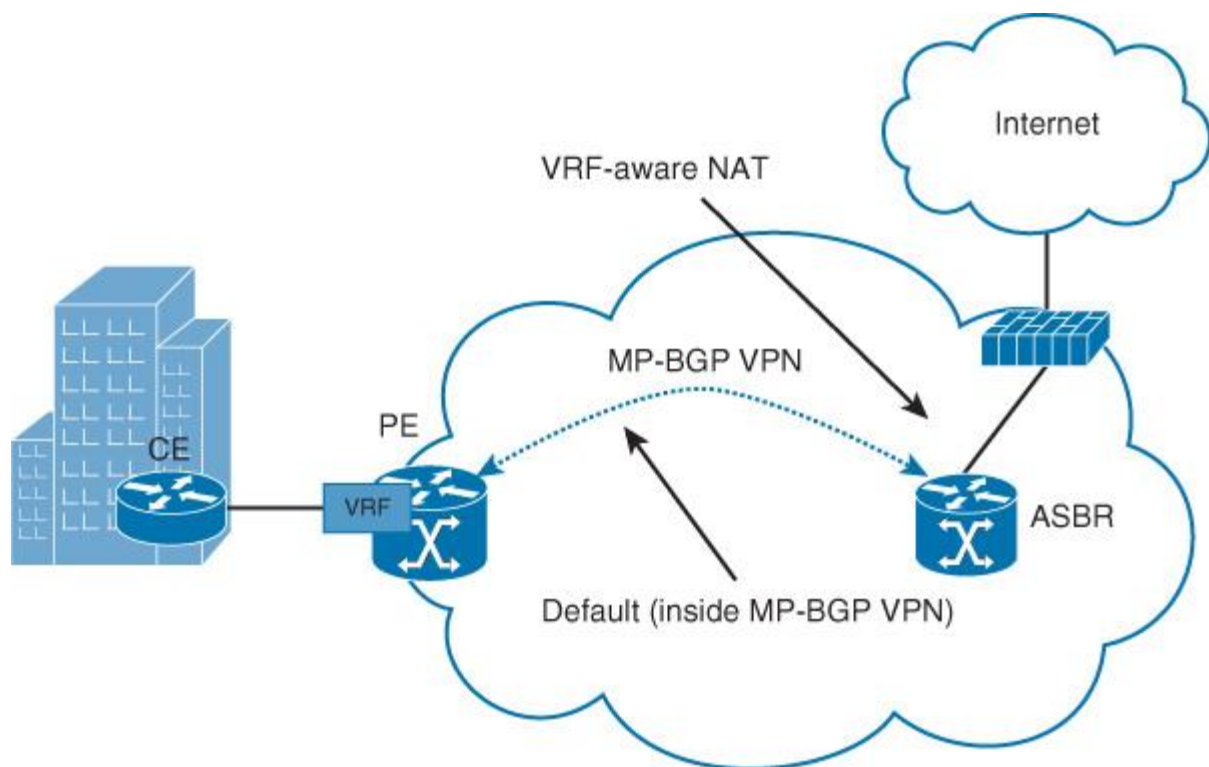


Figure 6-18 MPLS L3VPN Internet Access Option 4

[Table 6-2](#) compares these design options from different design aspects.

Design Consideration	Option 1	Option 2	Option 3	Option 4
PE manageability	Moderate.	Simple.	Simple.	Simple.
ASBR manageability	Simple.	Simple.	Moderate.	Complex.
CE simplicity	Simple.	Simple.	Moderate.	Moderate.
Scalability	High.	High.	Moderate.*	High.**
Supports overlapped customer' IPs	No (should be direct public provider independent / provider assigned [PI/PA]).	No (should be direct public PI/PA).	Yes.	No.
Full Internet routes at the PE	Yes.	May.	May.	No.
Design concerns	Static routing may lead to configuration errors and operational complexity.	May add routing/BGP configurations to the CE side.	Several VRFs on the same PE, each with full Internet routes, can add significant load on the PE.	VPN customers do not receive full Internet route.

* If the full Internet route is installed per VRF, there can be scalability limitations at the PE level.

** There might be limitations at the ASBR level (VRF routes or NATing entries/sessions).

Table 6-2 MPLS L3VPN Internet Access Design Options Comparison

The comparison in [Table 6-2](#) might make it seem like one option is better than others for certain design requirements, but usually the network environment and the situation drive the design choices (for instance, if an Internet service provider [ISP] wants to start providing MPLS VPN to their customers, but at the same time, they cannot afford any service interruption to their existing Internet customers). In this situation, keeping the Internet routing at the BGP global routing table can be a viable solution. Similarly, if an MPLS L3VPN provider wants to offer Internet as a value-added service for their customers, it may be less interruptive to add the Internet as a new VPN (extranet VPN connectivity model). Therefore, always consider the other factors such as business priorities, design constraints, and the targeted environment, in addition to the technical aspects, before making any design decision.

PE-CE L3VPN Routing Design

Designing routing between PE and CE in an MPLS L3VPN environment can sometimes be challenging. This is applicable for any MPLS L3VPN environment, whether it is a service provider or an enterprise with MPLS L3VPN core. Although the CE and PE sides are most commonly managed and deployed by different teams (usually the CE controlled by the enterprise and the PE controlled by the service provider side), the routing design of the CE to PE should (ideally) *not* be performed in isolation. It should be coordinated and aligned to

achieve a successful design, because of dependencies in some scenarios that can impact the overall design and end-to-end communication. This section discusses the design considerations from both the CE side (usually the enterprise) and MPLS L3VPN side (either a service provider or self-deployed enterprise MPLS L3VPN).

PE-CE Routing Design Considerations

For network designers to achieve a successful PE-CE routing design, they must follow the top-down approach in a structured manner to identify the goals and direction of the design during the planning phase. To achieve that, network designers must have a good understanding of the business, functional, and business-critical application requirements. The following questions can be considered the foundation, during the information-gathering and planning phases, for making a suitable design choice that can deliver value to the business:

■ Business requirements:

- Is it to reduce the cost of existing expensive links (for example, as an alternative connectivity to dark-fiber)?
- Is it to reduce the time to expand the business (for example, to add new remote sites and integrate them into the network very quickly)?
- Is it to increase the reliability of business applications and services with minimal cost (to be used as backup path)?
- Is it to optimize the ROI of the links by using multiple links in a load-balancing/sharing manner?

■ PE-CE functional and application requirements:

- Provide primary WAN connectivity?
- Provide backup WAN connectivity?
- Provide WAN and Internet access?
- Provide a primary path for applications that need high bandwidth or strict end-to-end quality of service (QoS) requirements?
- Provide efficient bandwidth utilization over multiple paths (multihoming)?
- Provide connectivity to remote areas with limited optical coverage (for example, remote sites connect over 3G)?

■ Technical requirements (connectivity characteristics):

- Is the CE single-homed or multihomed to the MPLS L3VPN network (PE) ?
- Is there any backdoor link from the CE side?
- Is there any limitation to run any specific routing protocol (such as lack of staff knowledge, software limitations, or is it not supported by the SP side)?

After identifying the different requirements, network designers can have a good understanding of the design goal and direction, so as to achieve a successful design. For example, it is always common for the MPLS L3VPN to be used as the primary path; however, in some scenarios, the backdoor link between any given sites might have a high-bandwidth capacity and offer lower latency and more control from the enterprise point of view. Therefore, in this case, the enterprise may prefer to use the MPLS L3VPN as a backup path only, which has to be reflected in the routing design.

PE-CE Routing Protocol Selection

One of the critical parts in designing PE-CE is the routing protocol selection, because normally each protocol has its strengths and weaknesses. In particular, in the PE-CE type of scenarios, the alignment of routing design between the service provider and the enterprise (CE side and PE side) is important. Otherwise, the lack of coordination and alignment between these two networks can lead to serious issues, such as suboptimal routing and routing loops. As shown in [Figure 6-19](#), in the PE-CE type of scenario, what normally

happens is a route is received from one CE via a routing protocol. Next, this route is converted into a MP-BGP route, along with some protocol related information that is normally transformed into BGP extended community values. These values can be used at the remote egress PE when the route is reconverted back into the original routing protocol, such as Open Shortest Path First (OSPF) or Enhanced Interior Gateway Routing Protocol (EIGRP) (assuming both CEs are using the same routing protocol with the PEs; otherwise, this specific routing protocol information most probably will be invaluable). Nonetheless, each routing protocol behaves differently in these types of scenarios.

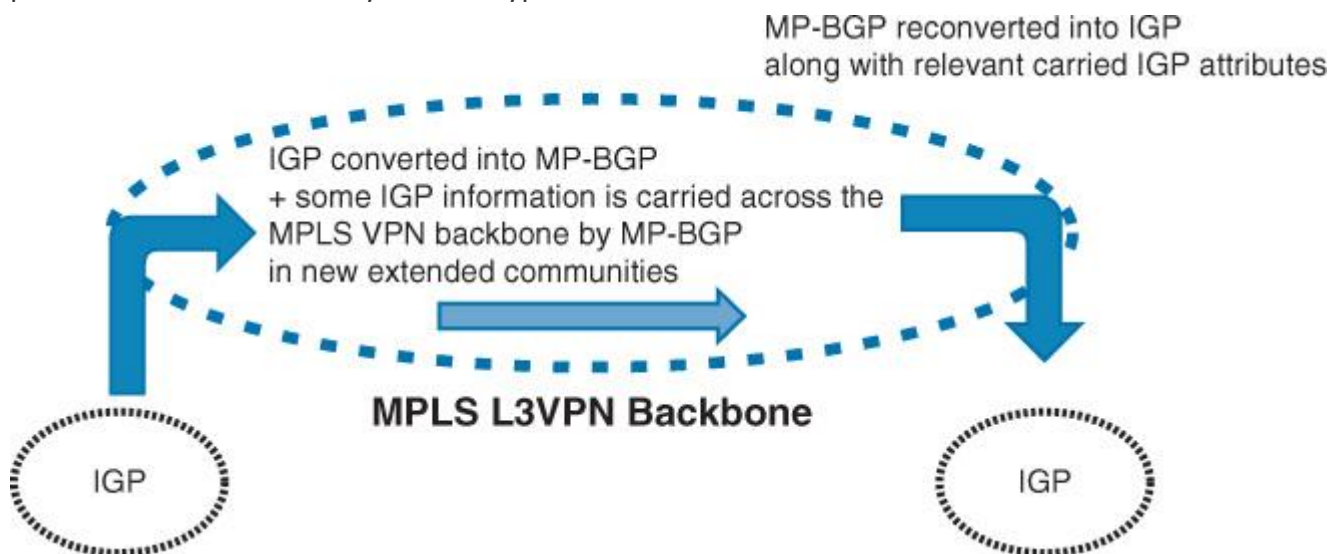


Figure 6-19 PE-CE Routing Principle

Therefore, understanding the business and functional requirements and the targeted topology (single-homed versus multihomed versus sites with backdoor link) can help you select the right routing protocol to a large extent, taking into consideration the specific protocol attributes and behaviors with regard to the underlying network topology, WAN connectivity, and path selection requirements.

PE-CE Design Options and Recommendations

As already mentioned, each routing protocol has its own strengths and weaknesses with regard to PE-CE routing design. This section discusses the technical design considerations and recommendations per routing protocol. [Figure 6-20](#) shows the reference network architecture exemplifying various PE-CE connectivity models that will form the foundation of the design considerations and recommendations in this section.

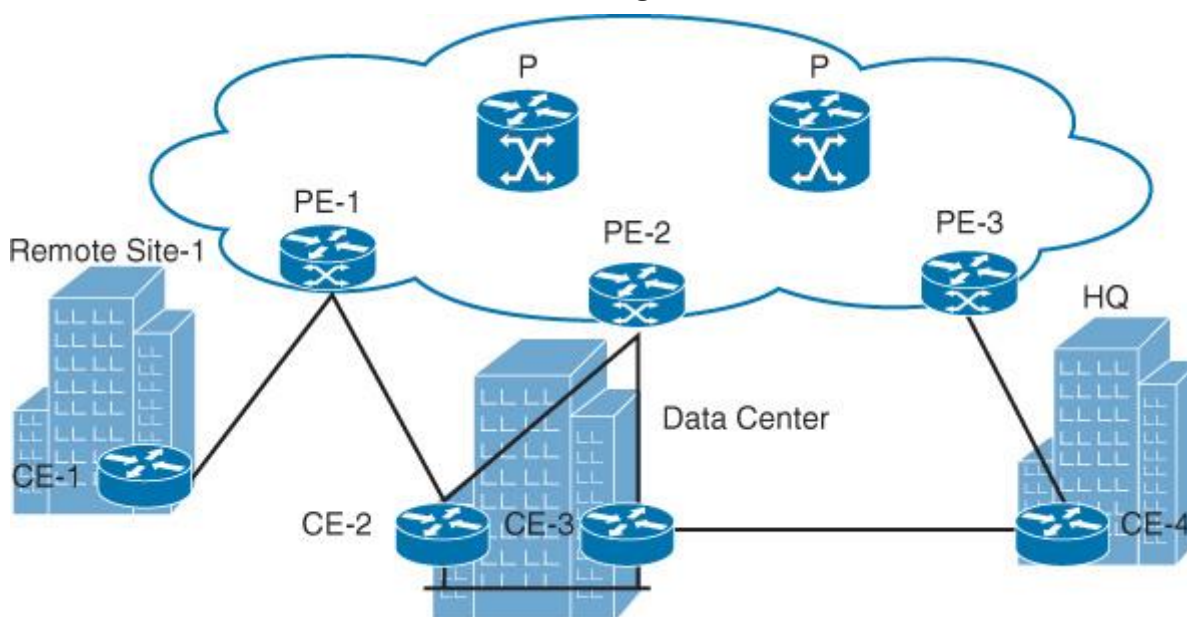


Figure 6-20 Reference Architecture

Note

This section covers the primary and most commonly used PE-CE routing protocols (static, OSPF, EIGRP, and BGP).

Static Route PE-CE

Applying a static route to the reference architecture network shown in [Figure 6-20](#) can lead to some design limitations and operational complexities. For example, CE-2, CE-3, and CE-4 have multiple links (direct PE-CE and backdoor links) using a static route, which can lead to limited design scalability and flexibility. In addition, managing these multiple edge devices with multiple links can be complicated and is associated with the high possibility of human (configuration) errors. However, the static route between CE-1 and PE-1 is a feasible design option because it is a single-homed site; even a single default route in CE-1 can be used, assuming the Internet is through the same link.

Link State: OSPF as a PE-CE Routing Protocol

OSPF is not commonly used for PE-CE routing by service providers. As stated earlier in this section, however, this design concept is not only applicable to typical enterprise to service provider MPLS WAN connectivity, it is also applicable to self-deployed MPLS L3VPN by enterprises (and OSPF is one of the most common routing protocols used by enterprises). The reason why OSPF is not commonly used as the PE-CE routing protocol is because the design can be tricky and complicated with OSPF, especially when there are sites with backdoor links or multihomed. As shown in [Figure 6-21](#), by using the same reference architecture network in [Figure 6-20](#) and applying different OSPF area designs, the implications of each differs significantly.

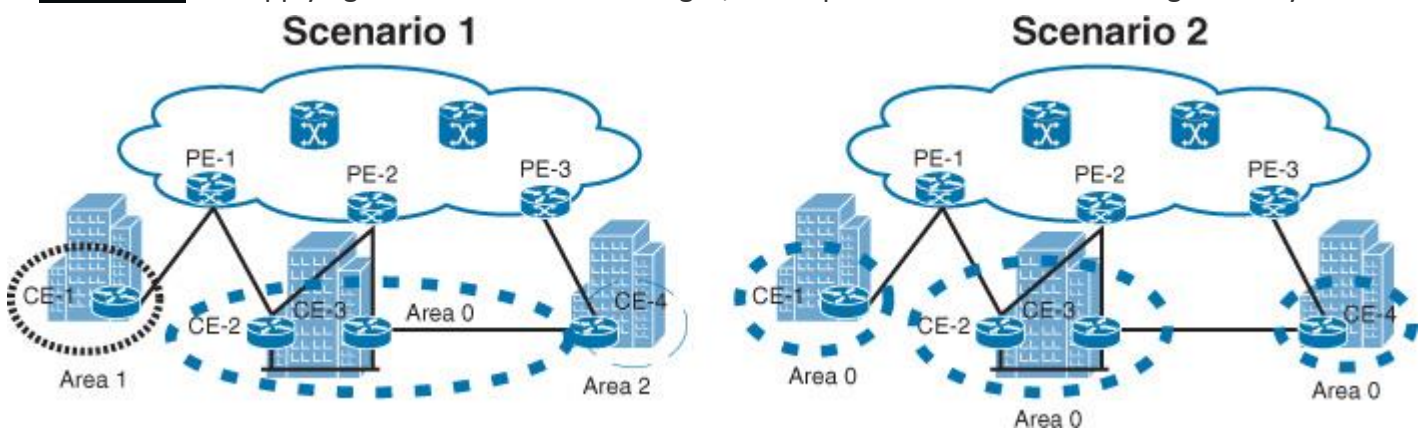


Figure 6-21 PE-CE: OSPF

Note

If you are unfamiliar with the OSPF terms used in this section such as OSPF DN bit, OSPF domain identifier and OSPF sham link, it is recommended that you refer to IETF RFC 4577 to build a foundational knowledge about these terms before reading this section.

Scenario 1 in [Figure 6-21](#) represents a multi-area OSPF design, where each site or branch is deployed with its own OSPF area. The service provider side is usually part of the super backbone area (area 0). Although this design scenario may look simple and straightforward, any lack of coordination between the service provider side and the enterprise (CE) side can break the functional requirements. For instance, if the CE side is configured with a different OSPF process ID than the PE side, traffic between the data center and the HQ will always prefer the backdoor link, as shown in [Figure 6-22](#). If the functional requirements dictate that the traffic should always prefer the MPLS cloud as the primary path, this will lead to a design failure.

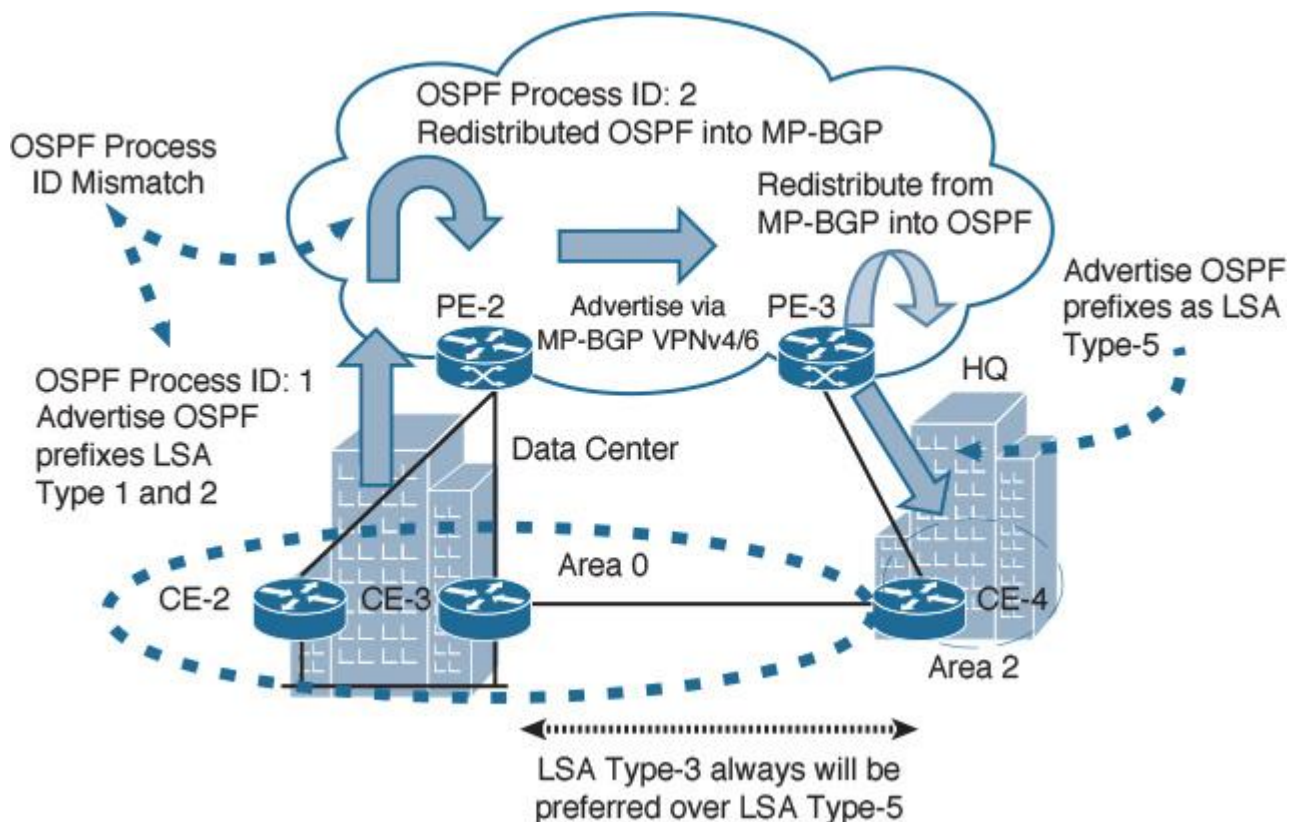


Figure 6-22 PE-CE: OSPF with Backdoor Link

This issue can be avoided by considering a single (matching) OSPF process ID between the CEs and the PEs. Nevertheless, practically speaking, this option can add operational complexity from the service provider point of view. Alternatively, service providers (PE side) can overcome this issue by deploying the same OSPF identifier on both ingress and egress PEs, along with OSPF tuning and the OSPF cost metric (to ensure that the WAN link has a lower OSPF cost metric).

In fact, OSPF incorporates multiple attributes, such as DN bit and route tag (for loop prevention) and the OSPF domain identifier. The technical reason behind these attributes is to help BGP carry this route across the MP-BGP backbone and convert the route back to OSPF appropriately, in a transparent way to the OSPF peers at the edge of the network (CEs). For instance, in [Figure 6-23](#), when PE-1 sets the DN bit to prefixes of remote site 1 (CE-1), when advertising it to CE-2 it will help PE-2 to stop re-advertising the same prefix back to the MPLS VPN super backbone when it is sourced from CE-2 or CE-3. In other words, when any of the PEs receives a type 3, 5, or 7 LSA from any CE with the DN bit set, the routing information from this LSA will not be considered in the OSPF route computation. As a result, this LSA will not be converted into a BGP route. This will ultimately help to avoid the potential loop in this scenario.

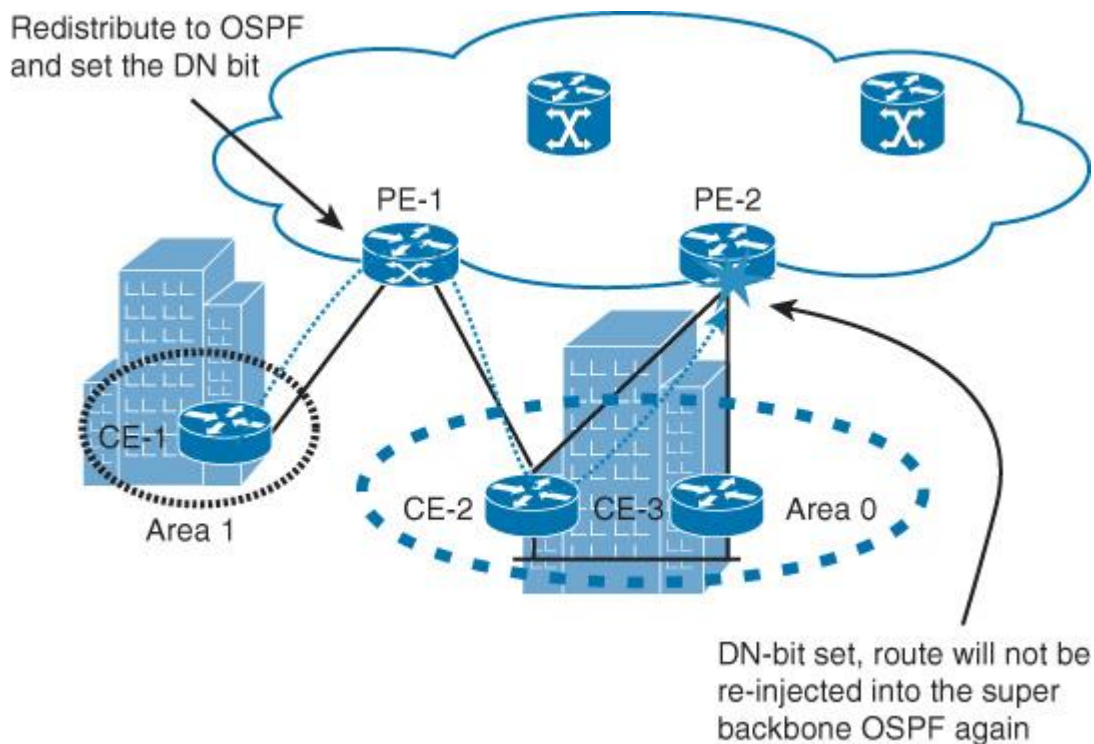


Figure 6-23 OSPF PE-CE Loop Prevention

Note

In some scenarios, such as a the multi-VRF CE (covered in [Chapter 4](#)) and the hub-&-spoke topology over MPLS L3VPN covered earlier in this chapter (see [Figure 6-12](#)), when OSPF is used as the PE-CE routing protocol with the hub-and-spoke over MPLS L3VPN topology, there will usually be multiple PE nodes communicating with a central/hub PE router that connects to the hub CE router over dual interfaces/subinterfaces (each in a separate VRF). Technically in this scenario, when two remote sites (spokes) need to communicate, the OSPF link state advertisements (LSAs) from each remote site will reach the central/hub PE, then the hub CE router, where the traffic flow loops and comes back into a different VRF (typical hub-and-spoke model). The issue here is that when these LSAs are type 3, 5, or 7, the LSAs will not be considered by the central/hub PE because OSPF will not consider them because they have the DN bit set. Therefore, the “DN bit ignore” feature is required in this scenario to “disable DN bit checking” at the hub PE node, in order to meet traffic flow and routing information distribution requirements (in which the route/LSA must be considered when it loops and comes back into a different VRF on the hub/central PE in order to reach other spokes). This OSPF feature is also known as capability VRF-lite. Ideally, before considering this feature a careful analysis is required to avoid introducing any potential routing information loop.

However, these attributes (DN bit or route tag) will be stripped from the prefixes if the route is redistributed into another routing domain (such as EIGRP) and then redistributed back into OSPF and then to MPLS L3VPN from another PE. As discussed earlier in this book, in discussions about route redistribution, route redistribution can cause metric transformation. This is a good example of how multiple redistribution may lead to a routing loop, as shown in [Figure 6-24](#).

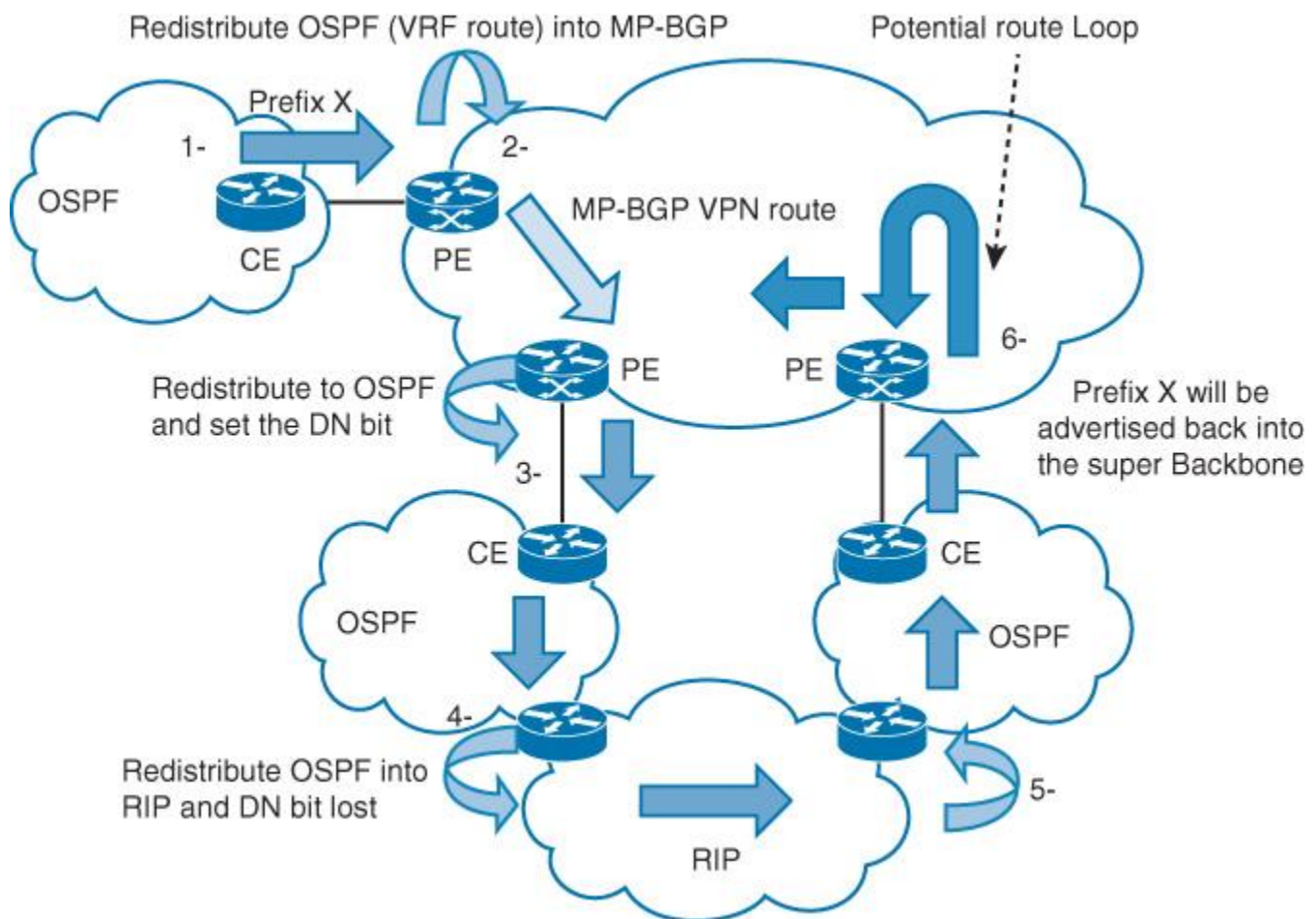


Figure 6-24 PE-CE OSPF with Multiple Redistributions

Consequently, the network designer must be careful when there is a possibility of multiple redistribution points across multiple routing domains, because it can break down the communication between OSPF islands across the MPLS L3VPN backbone.

In contrast, in Scenario 2 in [Figure 6-25](#), all the sites are deployed in OSPF area 0. Although in high level this design might look simpler, the most significant issue here is that all the routes between the data center and HQ site will be seen as OSPF intra-area routes. In other words, no matter what the WAN link cost metric is, the backdoor will always be the preferred path between the data center and the HQ (because the route from the MPLS L3VPN will be seen as either an interarea route or external route). This might not always be a desirable design. To resolve this issue, based on the OSPF area design, network designers must make sure that the route coming from the MPLS L3VPN backbone is received as an intra-area route as well. To achieve this, the service provider must coordinate and set up the OSPF sham link between the relevant PEs (in this scenario, PE-2 and PE-3) to create a logical intra-area link between the ingress and egress PEs (area 0 in this example).

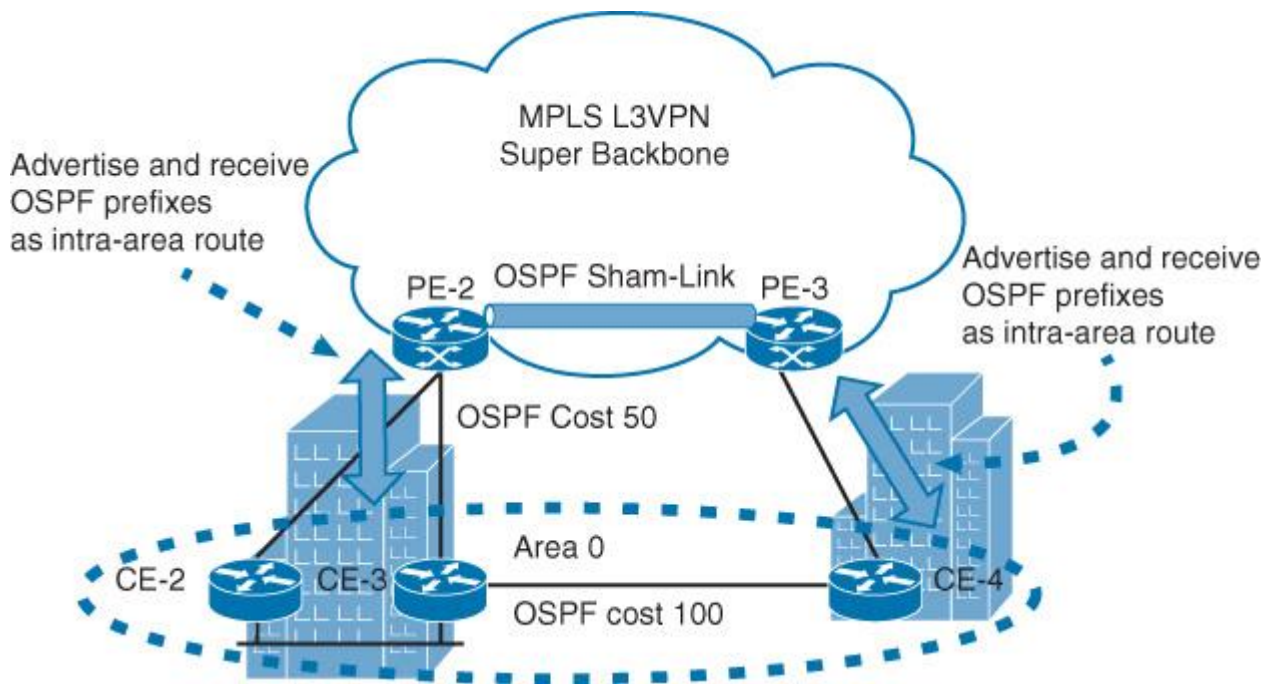


Figure 6-25 *OSPF Sham Link*

After the OSPF sham link is set up and the OSPF adjacency is established over this logical link, the network can manipulate the OSPF cost metric on the relevant interface to make the MPLS L3VPN link the preferred path.

Note

For the enterprise (CE side) to avoid the reliance on the SP side to set up a sham link, OSPF areas design can be migrated to use a unique OSPF area per site (for the sites connected with a backdoor link), if this option is available.

Consequently, using OSPF as the PE-CE routing protocol can be challenging if there is any lack of coordination between the PE side (super backbone) and the CE side. Furthermore, even if the design is performed with good coordination, OSPF can still impose some design limitations in certain scenarios, including the following:

- If any CE needs to send a summary route to other CEs or any given CE, this has to be deployed by the provider or PE side.
- In scenarios like the one discussed in this chapter, the data center has multiple paths (links and routes) to the MPLS L3VPN. OSPF can offer network designers very limited control to achieve a detailed load sharing over these paths. In addition, there is no such mechanism to influence the service provider's route selection from the CE, like BGP AS-PATH prepending.

Therefore, the protocol selection has to be aligned with all the different design requirements (business, functional, and application) to achieve a successful design.

Note

Intermediate System-to-Intermediate System (IS-IS) acts similarly to OSPF when there is a backdoor link, as the redistributed prefixes within the MPLS cloud (from MP-BGP into IS-IS) will be seen as an external route by the receiving CE while the same route over the backdoor will be received as internal route. This usually leads to always preferring the backdoor link. To overcome this issue, BGP support carries some of the critical IS-IS information as part of BGP extended communities, which can be converted back into IS-IS link-state packet (LSP). For example, if the original route was received as level 1, it will be reconverted into an IS-IS level 1 route at the other end (PE). (For more detail, refer to this IETF draft [draft-sheng-isis-bgp-mpls-vpn](#).)

EIGRP as a PE-CE Routing Protocol

Using EIGRP as a PE-CE routing protocol may offer a simpler design compared to OSPF if it is designed properly. This is because there are certain scenarios, especially when there is a backdoor link between the

CE sites, where any design mistake can lead to serious connectivity issues. Therefore, network designers must be aware of EIGRP behavior and the possible techniques and mechanisms to optimize the design.

As discussed earlier in this section, the typical routing behavior in PE-CE scenarios is that the CE routing protocols (IGP) are converted at the ingress PE router to MP-BGP route, to be carried across the MPLS VPN backbone to the egress PE router, and to be reconverted back to the relevant PE-CE routing protocol on that VRF. As with link-state protocols, MP-BGP carries specific EIGRP information in new BGP extended communities set by the PEs (usually the ingress PE), which carries EIGRP ASNs and multiple EIGRP attributes across the MPLS L3VPN backbone, such as delay, bandwidth, hop count, and reliability. This can help the receiving PE EIGRP instance to have usable information.

However, as mentioned earlier, PE-CE EIGRP design requires special considerations from network designers in some scenarios to avoid undesirable behaviors. [Figure 6-26](#) illustrates different EIGRP application scenarios using the same reference network architecture of this section.

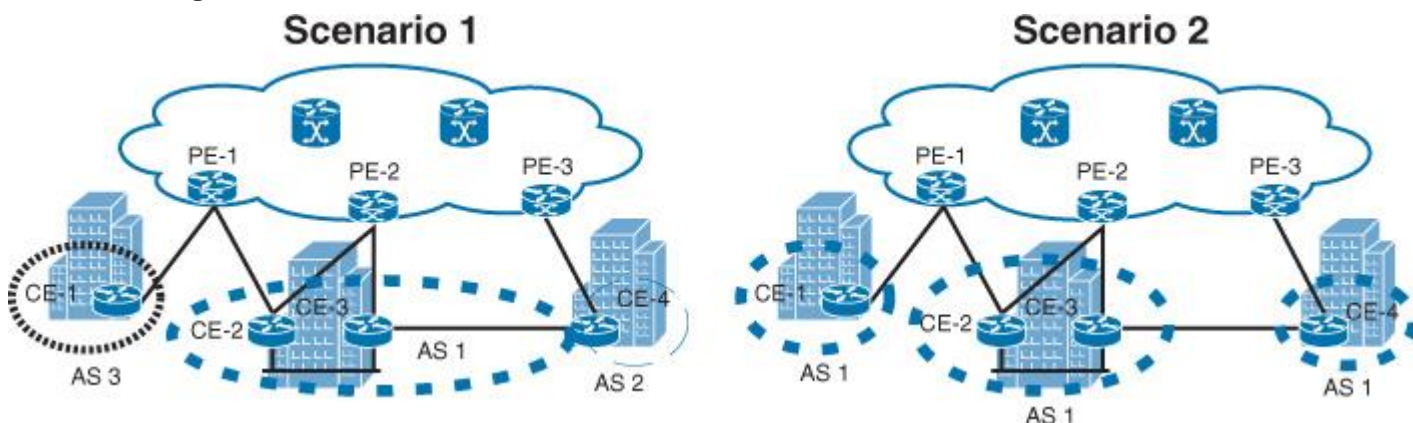


Figure 6-26 PE-CE EIGRP

In Scenario 1 in [Figure 6-26](#), EIGRP is applied with different ASNs per site. Each site will receive the EIGRP route of the other site as an external route. This is because the ASN is the carrier in the new BGP extended community. When it is redistributed from MP-BGP into EIGRP at the egress PE with ASN mismatch, EIGRP will install the route as an external route. However, the concern is always with where the backdoor link exists between different sites, because the EIGRP does not have a built-in mechanism like OSPF does to detect a loop. Therefore, there is potential that the EIGRP information will circulate in this looped topology. It is also possible as well that the EIGRP will introduce unpredictable behavior because of what is known as *race condition*, where the route is accepted based on the timing of the EIGRP and BGP updates with the local router decision.

Route racing in Scenario 1 can be mitigated to some extent by limiting the maximum number of EIGRP hop counts. However, with this approach, the network will continue to experience the undesirable (looping) behavior because of the route racing condition until the EIGRP reaches its maximum configured hop counts. In addition, it is difficult to determine the right number of hops to be configured (and there is a high degree of human configuration errors and added operational complexities). Alternatively, you can use a route tagging-based solution to offer a more deterministic and simplified solution known as *EIGRP site of origin (SoO)*, which is technically an extended BGP community associated with the route. When redistributed from EIGRP into MP-BGP, any route assigned an SoO value will not be advertised over links (interfaces) deployed with the same SoO value, which helps to avoid or mitigate the impact of routing loops in complex topologies, such as sites with EIGRP used as a PE-CE routing protocol that contain both MPLS VPN and backdoor links, as shown in [Figure 6-27](#).

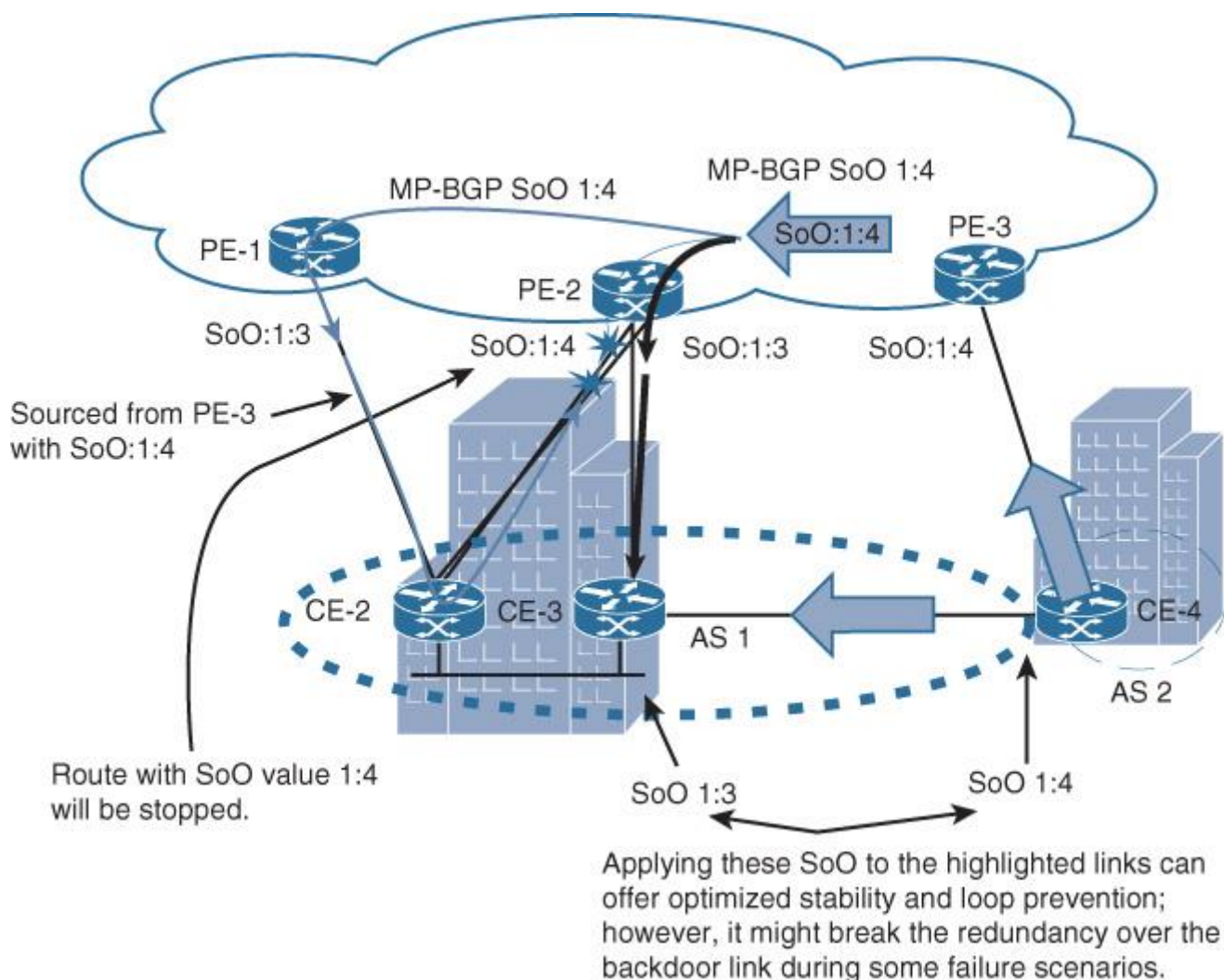


Figure 6-27 PE-CE EIGRP SoO

Table 6-3 summarizes the behavior of EIGRP SoO in different situations.

Received Route SoO Details	Action
SoO value matches the SoO value on the sending or receiving interface.	Route is will be filtered out.
CE deployed with SoO value that does not match.	Route is added to the EIGRP topology table so that it can be redistributed into BGP and the SoO value preserved.
Does not contain an SoO value.	Route is accepted into the EIGRP topology table, and the SoO value from the interface that is used to reach the next-hop CE router is appended to the route before it is redistributed into BGP.

Table 6-3 EIGRP SoO Actions

In Figure 6-27, SoO helps to optimize the EIGRP looping (race condition) by preventing the route from being re-injected back to the network based on the attached SoO value to the route and the deployed SoO value on the interface. For instance, traffic sourced from the HQ LAN (CE-4) passing through PE-3 will have an SoO

value of 1:4 assigned to it. Then, any interface in the scenario shown in [Figure 6-27](#) that has an SoO value of 1:4 will not pass this route information through.

Although in this type of scenario SoO can help to mitigate route looping and racing issues to a certain extent, it might sometimes be necessary to introduce other limitations, such as reduced redundancy. For example, if the SoO values are applied on the backdoor link (as covered in [Figure 6-27](#)), and the PE-3-CE-4 link goes down, any traffic with SoO value of 1:3 or 1:4 destined for the HQ (behind CE-4) will be isolated (because of the SoO filtering at backdoor link), even though the backdoor link is available. Therefore, as a network designer, you must understand the design goals and priorities and what the impact is of applying SoO on the different interfaces/path (for example, redundancy + suboptimal routing versus stability + optimal routing). In other words, if the time required for the EIGRP to stabilize following a failure event is acceptable, a simple SoO design should be sufficient, like the one shown in [Figure 6-28](#), in which SoO stops the information feedback looping faster than relying on the hop count for EIGRP to stabilize after CE-2 failure.

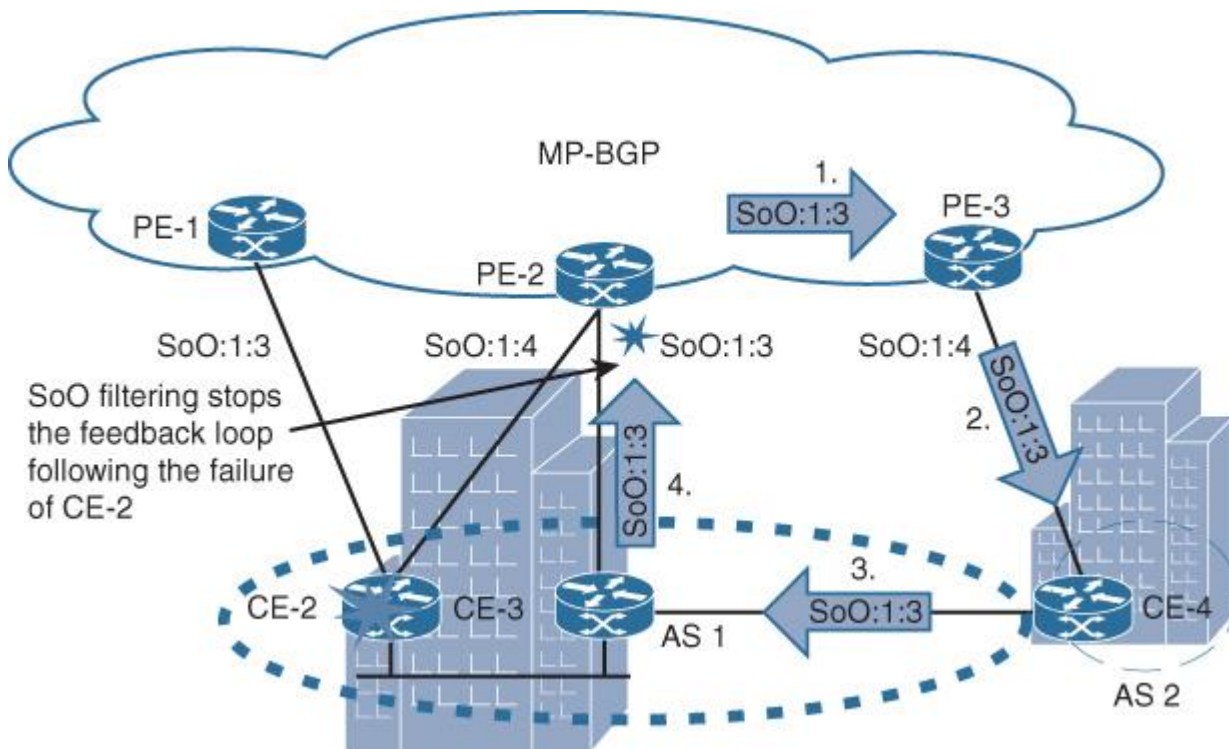


Figure 6-28 EIGRP PE-CE Loop Prevention with SoO: Failure Scenario

However, Scenario 2 shown in [Figure 6-26](#) is designed with the same EIGRP ASN on all sites, which is typical in that all the routes learned over MPLS L3VPN and the backdoor links will be an internal EIGRP route.

One of the common design concerns with this setup is when the backdoor link is intended to be used only as a backup path (because with this design there is a possibility that some remote sites will use the backdoor link to reach either the DC LAN or the HQ LAN). For instance, in [Figure 6-29](#), the HQ LAN prefix is advertised in EIGRP to the MPLS VPN PE-3 and to CE-2 and CE-3 over the backdoor link. Likewise, CE-2 and CE-3 advertise this route to PE-1 and PE-2, respectively. Therefore, PE-1 in this case has two BGP paths available for the HQ LAN: iBGP path via PE-2 and PE-3, and the locally redistributed BGP route from EIGRP advertisement via CE-2 EIGRP.

Scenario 2

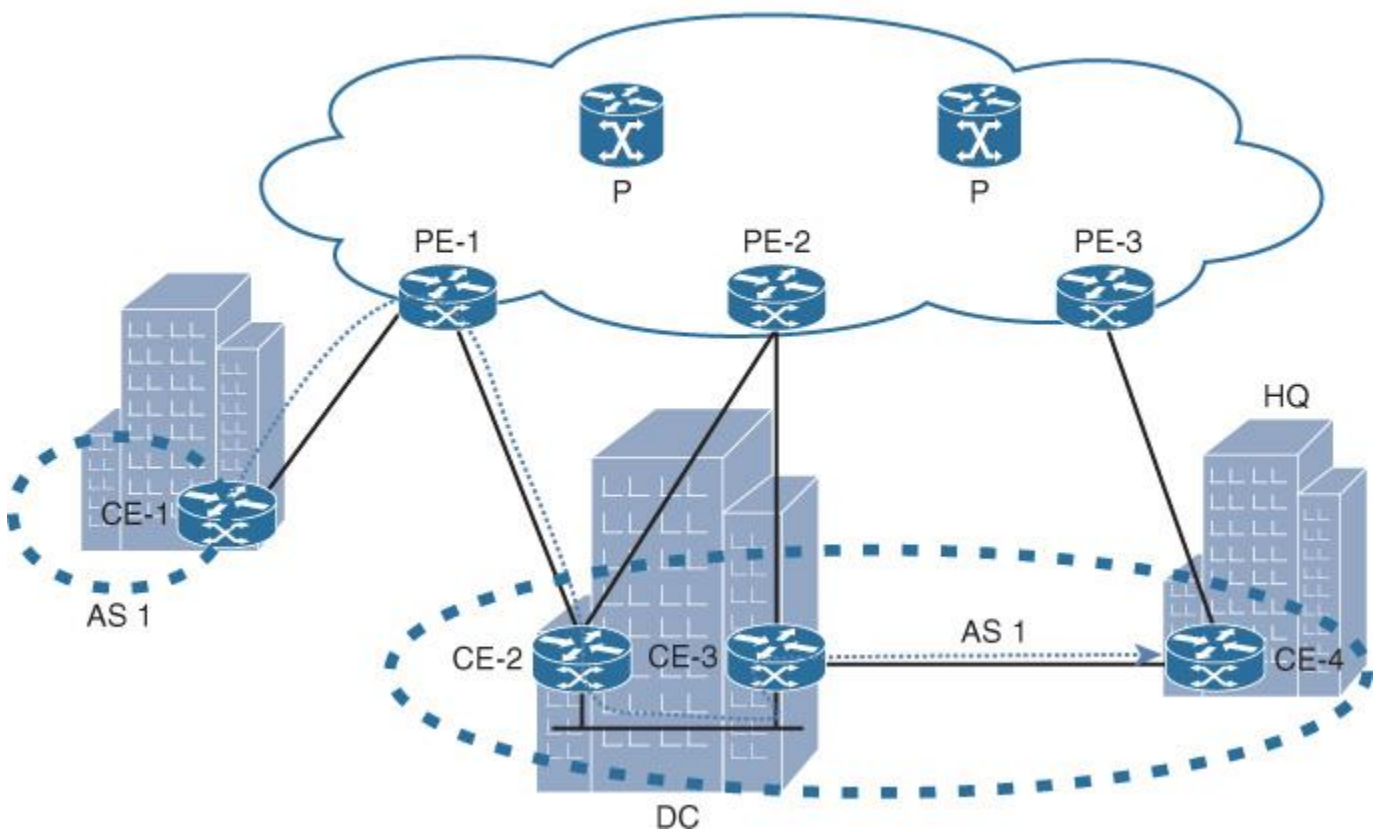


Figure 6-29 EIGRP PE-CE Suboptimal Routing

Consequently, from the PE-1 point of view, the locally originated route will be preferred based on the BGP best path selection. As a result, traffic from CE-1 destined for the HQ site (CE-4) will use the backdoor link as a primary path, as shown in [Figure 6-29](#).

There must be an attribute to influence PE-1 path selection in which the internal BGP (iBGP) learned routes is to be preferred over other routes. One possible and common solution that can be used in this scenario is “BGP cost community” to influence BGP path selection.

EIGRP routes injected into MP-BGP with a point of insertion (POI) value of 128, along with the cost set to the EIGRP composite metric (in this case, the community’s value), must be considered before the typical BGP path selection algorithm (such as local_preference and AS-PATH). Consequently, when the PEs redistribute the EIGRP routes into BGP, the BGP cost community attribute will be populated along with the accumulated EIGRP metrics. This means that each PE will have a level of visibility of the EIGRP path cost. As a result, PE-1 will prefer the HQ LAN route advertised by PE-3, because the associated EIGRP metric of this LAN route advertised by either CE-2 or CE-3 contains the accumulated path cost, including the backdoor link, while the BGP cost community of the iBGP (from PE-3) has a lower cost. Therefore, it will be the preferred path, as shown in [Figure 6-30](#).

Scenario 2

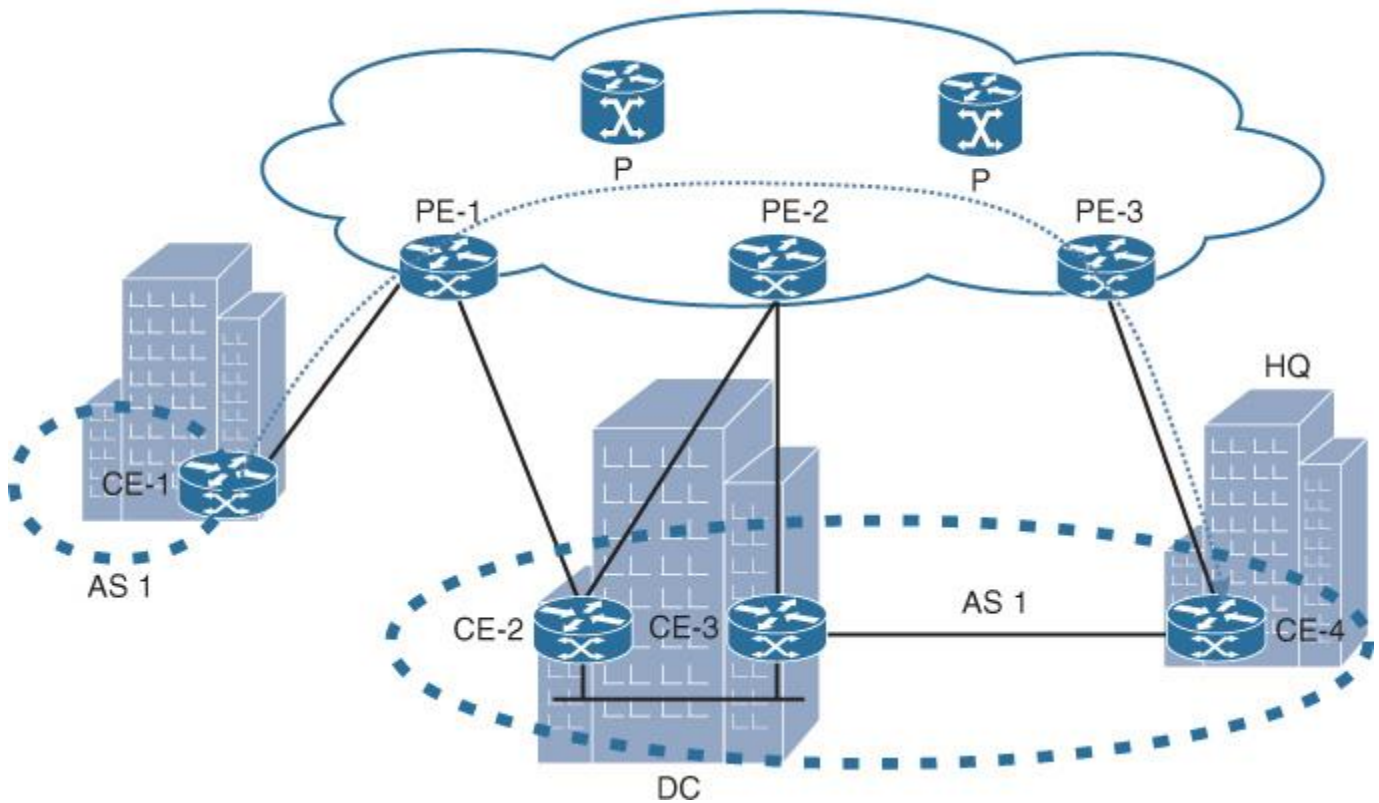


Figure 6-30 EIGRP PE-CE Optimal Routing

Furthermore, BGP cost community helps to optimize optimal routing in scenarios like this (and scenarios like Scenario 1 shown in [Figure 6-27](#)). For example, if the route from site 1 (CE-1) is assigned a BGP cost community value at the ingress PE (PE-1), it will always be preferred by PE-2 and PE-3 over any other BGP advertisement that has no cost community attached to it, regardless of what BGP attributes it has. As a result, the site 1 route will always be preferred via PE-1 (directly connected PE to site 1), as shown in [Figure 6-31](#).

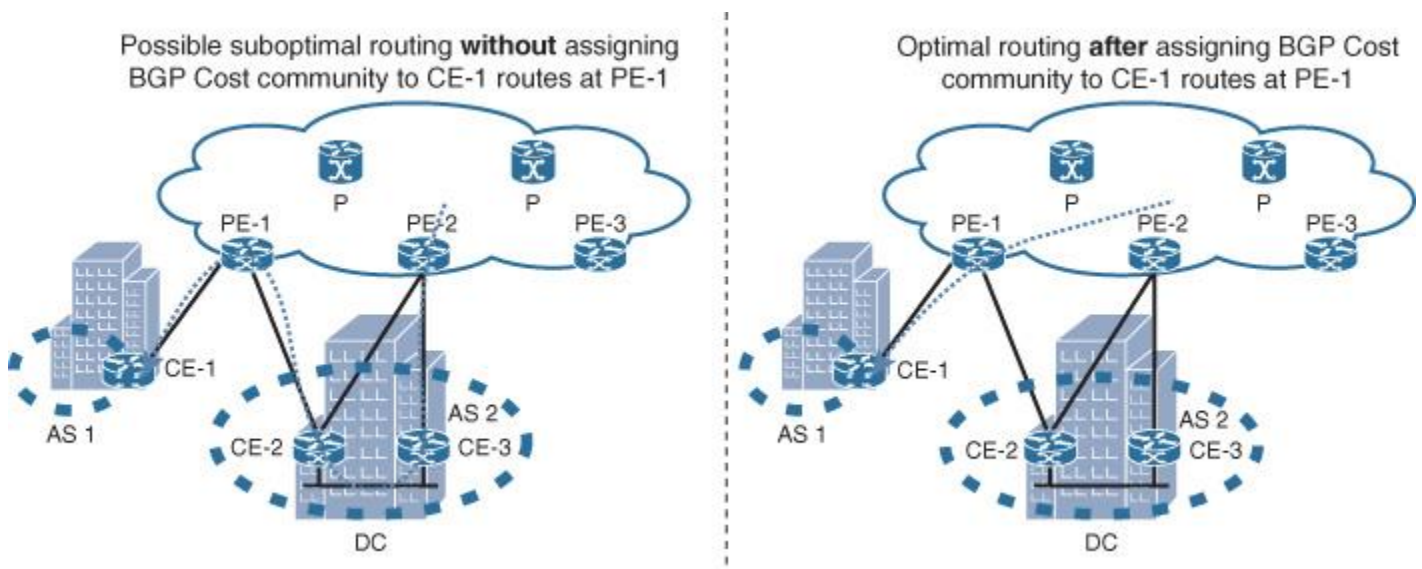


Figure 6-31 EIGRP Suboptimal Routing-2

Note

This cost community may transform BGP to act in way it is not designed for (like IGP), which may lead to undesirable behaviors in some scenarios.

Consider, however, what happens when a new remote site is added (for example, in a new country where the current service provider does not have presence and requires inter-AS communication to extend the MPLS-L3VPN reachability). In this scenario, the BGP cost community will not be a valid solution because it does not support propagation over external BGP (eBGP) sessions, as shown in [Figure 6-32](#).

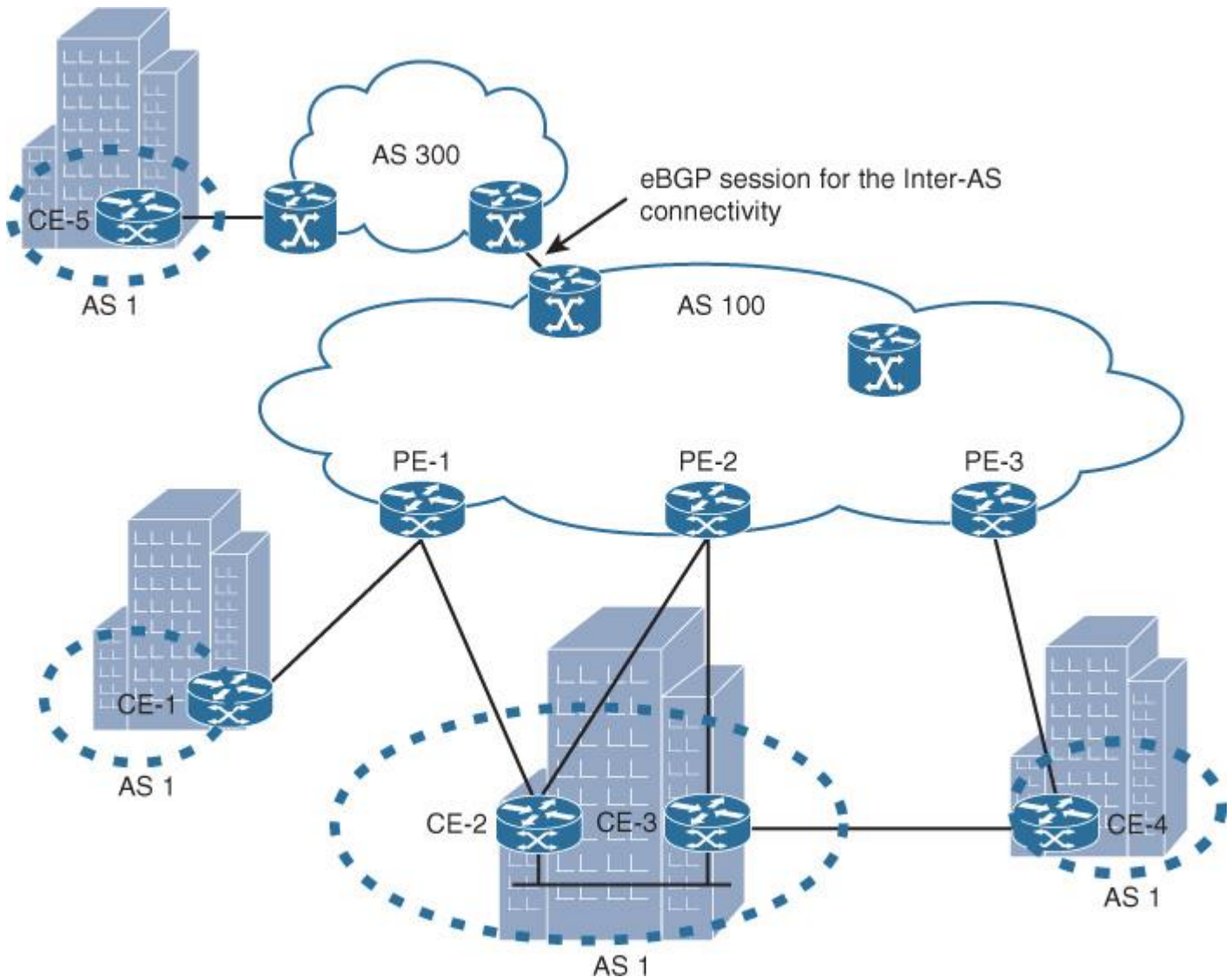


Figure 6-32 PE-CE EIGRP: BGP Cost Community Limitation Scenario

Consequently, using EIGRP as PE-CE routing protocol may add simplicity to the enterprises that already use EIGRP as the enterprise routing protocol. However, when there are multihomed sites to the MPLS provider or sites with backdoor links, the design may prove too complicated, and overall flexibility and stability may be reduced. EIGRP Over the Top (OTP), however, can offer more flexible PE-CE design and is independent of the service provider routing control (as covered in [Chapter 4](#)).

BGP as a PE-CE Routing Protocol

BGP, however, as a PE-CE routing protocol can achieve the optimal routing and traffic control over the most complex connectivity layouts because it has multiple powerful attributes that can influence inbound and outbound path selection. All the design considerations and options discussed earlier in this book (especially in the “[Enterprise Multihoming](#)” section in [Chapter 4](#)) are applicable to the design of BGP as a PE-CE routing protocol. In addition, the BGP SoO attribute uses the same concept of the EIGRP SoO to control route propagation when there is a backdoor link between the customer sites.

In other words, when possible, it is always desirable to consider BGP as the PE-CE routing protocol for multihomed sites to single- or multiple-provider networks. For network designers, this can facilitate the achievement of a very advanced level of traffic engineering with flexible BGP policies. This, in turn, offers the business adequate flexibility to use the available paths in a way that aligns with the business, functional, and application requirements in a more dynamic manner regardless of the extent of the physical connectivity's complexity. The flexibility of BGP can obviously provide an optimized ROI of the available paths and lower operational complexities when there is complicated connectivity with multiple links.

Normally, MPLS VPN providers consider two BGP ASN allocation models when BGP is used as a PE-CE routing protocol as following (see [Figure 6-33](#)):

■ **Same ASN (ASN) per site:** With this model, the MPLS provider allocates the same ASN to all the customer sites. One of the main advantages of this model is reduced BGP ASN collisions.

■ **Single ASN per site:** With this model, the MPLS provider allocates each of the customer sites a separate BGP ASN. This model offers network designers and operators the ability to identify the source of prefixes (from which site) in simple way (based on BGP ASN in the AS PATH attribute of each prefix). However, it may introduce scalability limitation with regard to the available ASNs.

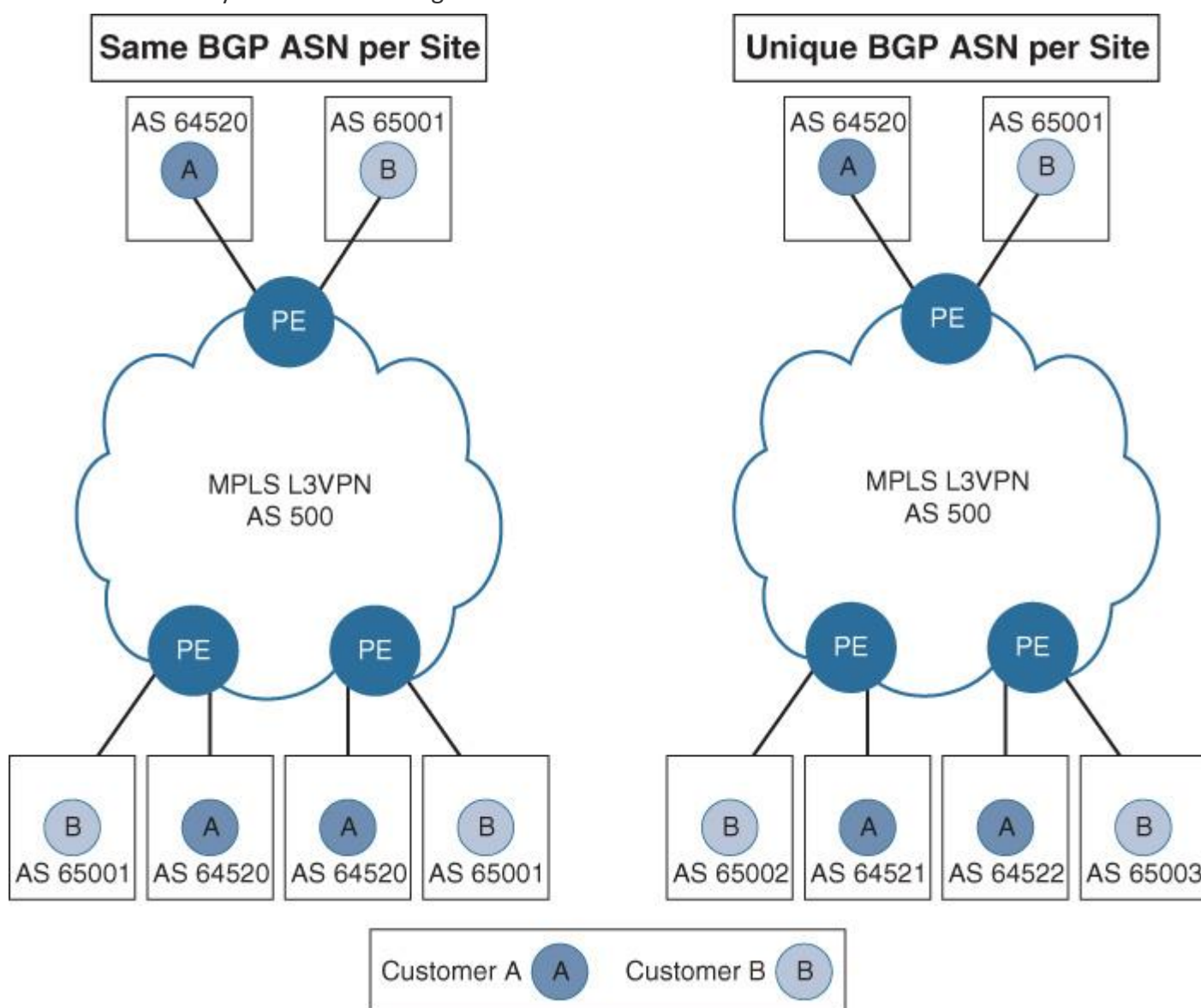


Figure 6-33 BGP ASN Allocation Models as an MPLS VPN PE-CE Routing Protocol

Although allocating the same BGP ASN offers better scalability with regard to the number of ASNs/sites and reduced ASN collisions, this model introduces some design concerns for multihomed sites. For example, in [Figure 6-34](#), the service provider must rewrite the customer ASN (AS override) to overcome the default

Routing Protocol	Strengths	Weaknesses
Static	Simple and reliable “when combined with IP SLA”. Low operational complexity in small environments with a small number of prefixes.	Nonscalable. High operational complexity in large environments. Limited flexibility in multihomed scenarios with automatic failover limitation.
Link state*	Reliable to certain extent (for example, supports built-in loop prevention). Supports multiple connectivity and flooding domain design scenarios.	High design and operational complexity. Limited flexibility in large environments with backdoor links and multihoming scenarios.
EIGRP*	Reliable to a certain extent (topology dependent) EIGRP OTP can simplify and optimize CE-PE design to a large extent.	High design and operational complexity and limited flexibility in large environments with multihoming scenarios. EIGRP SoO may lead to inefficient use of available paths or lack of redundancy in multihoming scenarios with backdoor links.
BGP	Most powerful and flexible protocol that can support all types of connectivity.	For multihomed sites with complex policies, it requires advanced operational staff expertise. May not be supported by some low-end routers (for very small remote sites).

* Not commonly used as a PE-CE routing protocol

Table 6-4 Comparison of PE-CE Routing Protocols

Layer 2 MPLS VPN (L2VPN)

Operators understand the importance and the real need for an infrastructure that offers flexible and responsive network architecture, which supports current and future requirements without requiring any major change or upgrade to the network in the future. Therefore, in today’s market, almost all service providers have either partially or fully migrated their core infrastructure to be MPLS enabled. This enables them to take advantage of the MPLS infrastructure and offer the same services to their customers (Layer 2 over legacy and modern WAN protocols) over one unified MPLS core infrastructure. This is a big incentive for those service providers that already offer MPLS L3VPN, because it will significantly reduce capital expenditures and operational expenses where the same core infrastructure and control plane can be used to transport both L2 and L3 services. Commonly, modern service providers that offer Layer 2 Ethernet

services (for example, Metro Ethernet) are referred to as *Carrier Ethernet providers*. Based on that, the following are the primary business drivers for service providers to adopt and invest in modern Layer 2 services (L2VPN for example, Metro Ethernet):

■ **Flexible offerings:** Customized solutions with a flexible mix of services, data rates, and revenue (Ethernet private line [EPL], Ethernet virtual private line [EVPL], Ethernet LAN [E-LAN]).

■ **Flexible bandwidth:** For example, service providers can provision 1G or 10G Ethernet physical links and offer fractional line access rates based on the agreed SLA. This provides operators and customers seamless roadmap upgrades in the future without changing any link or device (as compared to legacy services, such as a time-division multiplexing (TDM)-based WAN). Also, with today's flexible Ethernet nodes, an access port upgrade from 1G to 10G can be as simple as port shortest path first [SFP] replacement.

■ **Flexible media access:** A variety of access networks that leverage Ethernet, such as WiMAX, IP digital subscriber line access multiplexers (DSLAMs), Ethernet over Fiber, Ethernet over Copper, and so on. Also supports interworking between legacy access media, such as ATM and Frame Relay, which might need to be retained for some remote areas where Metro Ethernet access coverage is not available.

■ **Cost savings:** Moving to Ethernet infrastructure offers lower capex compared to legacy hardware such as ATM switches. In addition, bandwidth flexibility, discussed earlier in this list, offers an optimized capex and investment protection for both service providers and customers.

■ **Scalable services that can increase revenue:** Support a large number of customers over one common infrastructure.

Note

This covers the technologies and designs from a service provider point of view. Nevertheless, these technologies and designs can be applicable to enterprises that deploy any of these services across their networks, such as self-deployed L2VPN to interconnect enterprise data centers over a Layer 3 WAN core network.

Figure 6-35 illustrates the different Layer 2 media access methods and topologies that can be transported over an MPLS-enabled backbone (L2VPN).

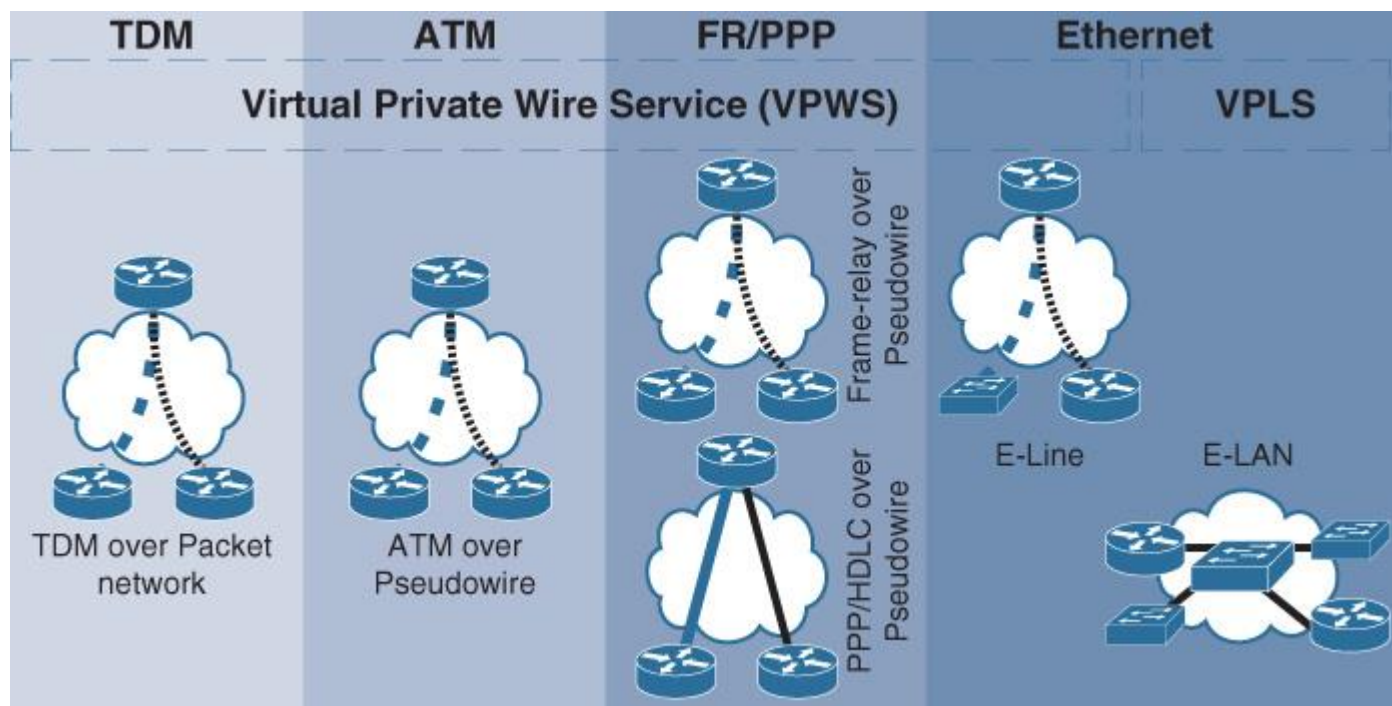


Figure 6-35 MPLS L2VPN Connectivity Models

This section focuses on the L2VPN MPLS (Ethernet-based) models of modern service provider networks, as covered in [Figure 6-36](#).

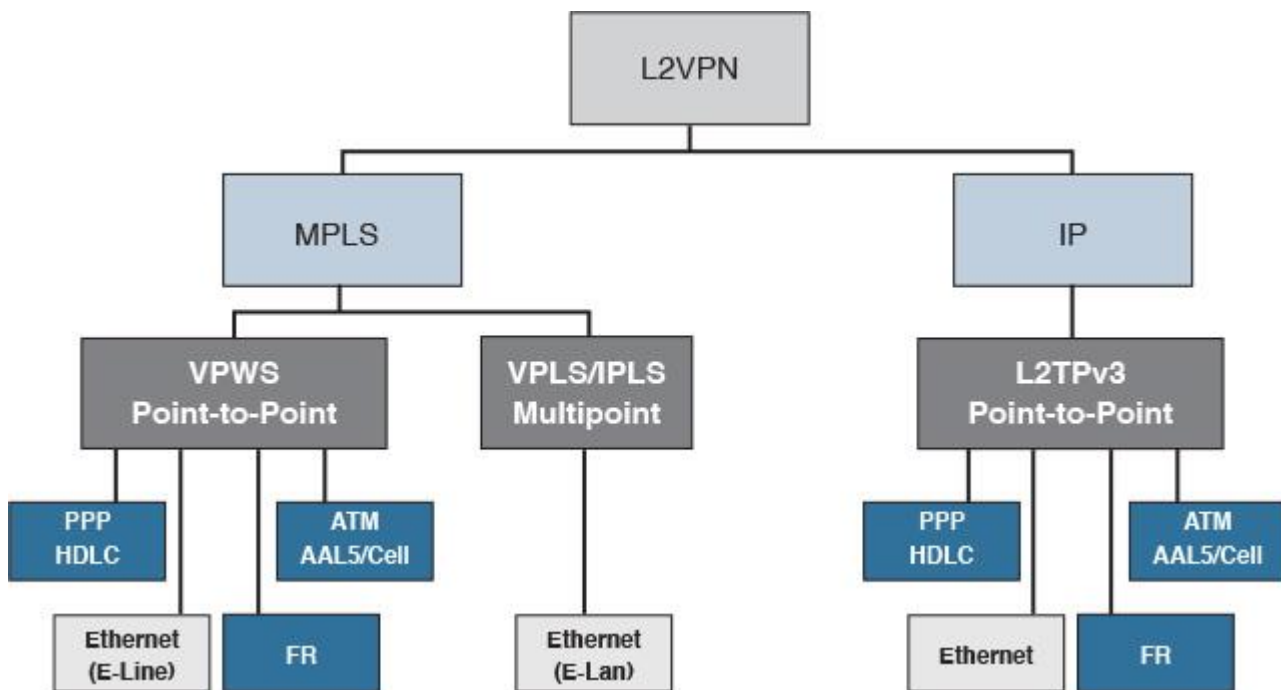


Figure 6-36 L2VPN Models

IP NGN Carrier Ethernet

The IP NGN Carrier Ethernet refers to the modern service provider architectures that offer a cost-effective converged network infrastructure capable of delivering new and future services by providing scalable and flexible designs and technologies that can meet the different bandwidth and network requirements. As a result, IP NGN Carrier Ethernet offers a wide range of modern services, including the following:

- Business services such as Metro Ethernet enterprise connectivity services (E-Line, E-LAN)
- Consumer and residential services (VoD, IPTV)
- Business services over DOCSIS
- Broadband mobility services
- Wholesale services

Figure 6-37 shows a high level view of the IP NGN Carrier Ethernet architecture.

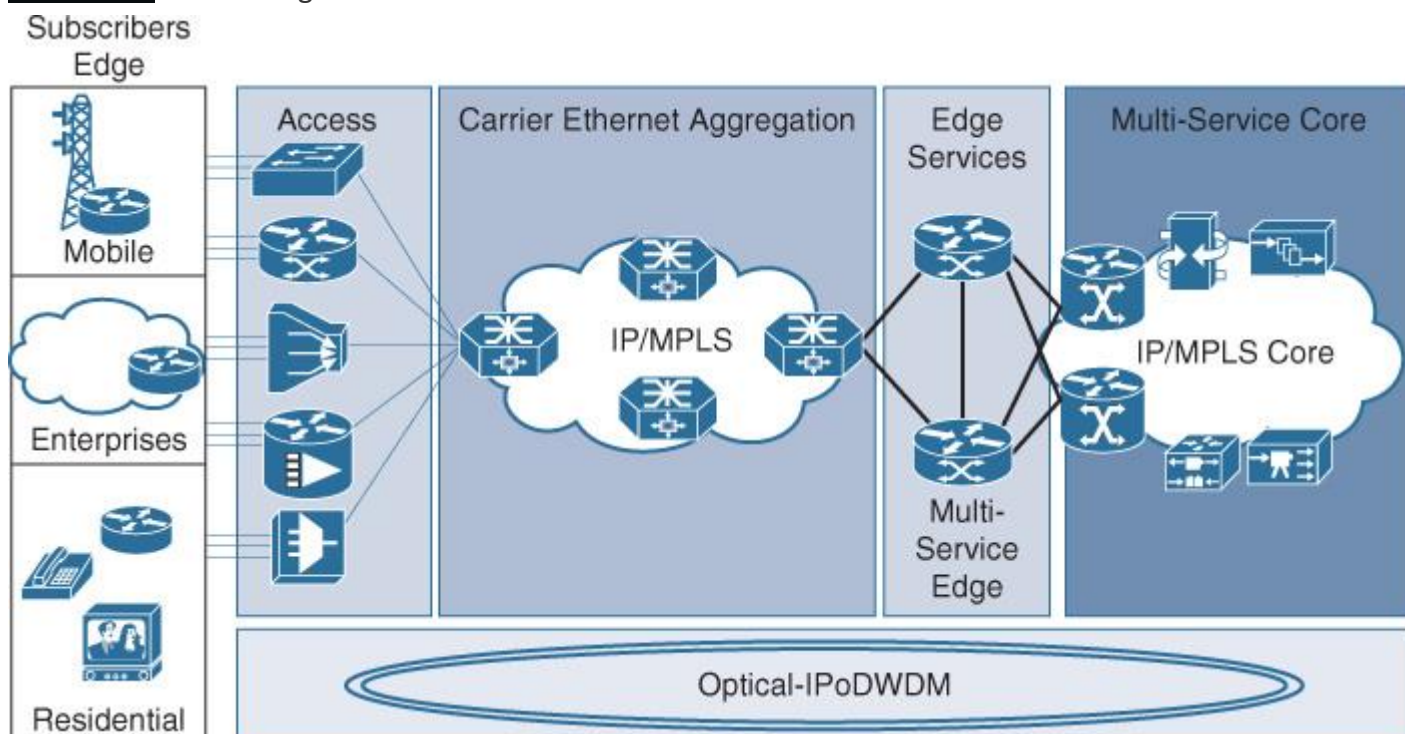


Figure 6-37 IP NGN Carrier Ethernet Architecture

The following components are as primary elements that comprise the architecture shown in [Figure 6-37](#):

- **Access:** Provides access to residential and business customers over digital subscriber line (DSL), fiber, cable, or wireless.
- **Carrier Ethernet aggregation:** Aggregates the access network across a Carrier Ethernet network and provides interconnectivity to the IP/MPLS edge and IP/MPLS core.
- **IPoDWDM optical network:** Enables optical aggregation services with intelligent Ethernet multiplexing using MPLS/IP over dense wavelength-division multiplexing (IPoDWDM).
- **Multiservice edge:** Interface services with the IP/MPLS core; this is the provider edge for both residential and business subscriber services.
- **IP/MPLS core:** Provides scalable IP/MPLS routing in the core network.

Metro Ethernet (ME) services can be classified into the following two general categories:

- **Point to Point (P2P):** In which a P2P Ethernet circuit can be provisioned between two user network interfaces (UNIs).
- **Multipoint to Multipoint (MP2MP):** In which a multipoint-to-multipoint Ethernet circuit can be provisioned between two or more UNIs. If there are only two UNIs in the circuit, more UNIs can be provisioned to the same Ethernet virtual connection if required, which distinguishes this from the point-to-point type.

Metro Ethernet Forum (MEF), however, defines the two categories just discussed as two main types of Layer 2 Ethernet services:

- **Ethernet line service (E-Line):** Point to point, such as Virtual Private Wire Service (VPWS)
- **Ethernet LAN service (E-LAN):** Multipoint to multipoint such as Virtual Private LAN Services (VPLS)

Note

There is also a P2P Metro Ethernet service known as E-Tree, as an abstraction E-LAN, in which the spoke “leaves” can communicate with the hub or “root” location but not with each other, as shown in [Figure 6-38](#). The typical application for E-Tree is in franchise operations [\[43\]](#).

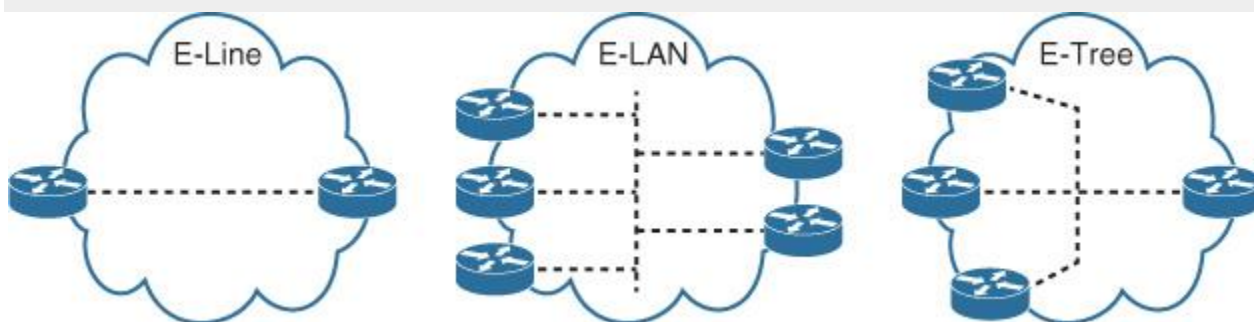


Figure 6-38 ME models

Metro Ethernet services can be created by assigning values to a set of attributes grouped according to the following [\[44\]](#):

- **User network interface (UNI):** Represents a physical demarcation point between the connection and the subscriber. Also known as *port based*.
- **Ethernet virtual connection (EVC):** Represents the association of two or more UNIs, which limits the exchange of service frames to UNIs within the EVC in Ethernet. When multiple EVCs could exist per single UNI and each EVC is distinguished by 802.1Q VLAN tag identification, this is known as *VLAN based* (ERS or EVPL).

[Table 6-5](#) summarizes the relationship between the transport model and the MEF Ethernet service definitions.

Service Type	Port Based	VLAN Based
E-Line	Ethernet private line (EPL)	Ethernet virtual private line (EVPL)
E-LAN	Ethernet private LAN (EPLAN)	Ethernet virtual private LAN (EVPLAN)

Table 6-5 MEF Transport Models

In addition to E-Line and E-LAN services, additional services are available for Layer 2 that are mainly to facilitate carrying legacy WAN transport over MPLS networks, such as the following:

■ Frame Relay over MPLS (FRoMLS)

■ ATM over MPLS (ATMoMPLS)

Note

Cisco's implementation of VPWS is known as *Any Transport over MPLS* (AToM) and delivers what is known as *Ethernet over MPLS* (EoMPLS). L2TPv3, however, can be used as an analogous service to AToM over any IP transport.

Virtual Private Wire Service Design Considerations

VPWS is based on the concept of pseudo wire (PW) described in (RFC 3916), which can be defined as a connection between two edge nodes connecting two attachment circuits (ACs) to emulate a direct AC connection of a packet-switched network (PSN), as shown in [Figure 6-39 \[45\]](#). The typical implementations of PWs are carried either over a native IP, such as L2TP, or over an MPLS network, such as EoMPLS. Within an MPLS-enabled network, the core visibility is limited to the transport MPLS labels (LSP) that each PW uses to communicate with the remote end, while the remote PE relies on the virtual circuit (VC) label to identify the intended AC per pseudo wire (PW-to-AC mapping), as shown in [Figure 6-39\[46\]](#).

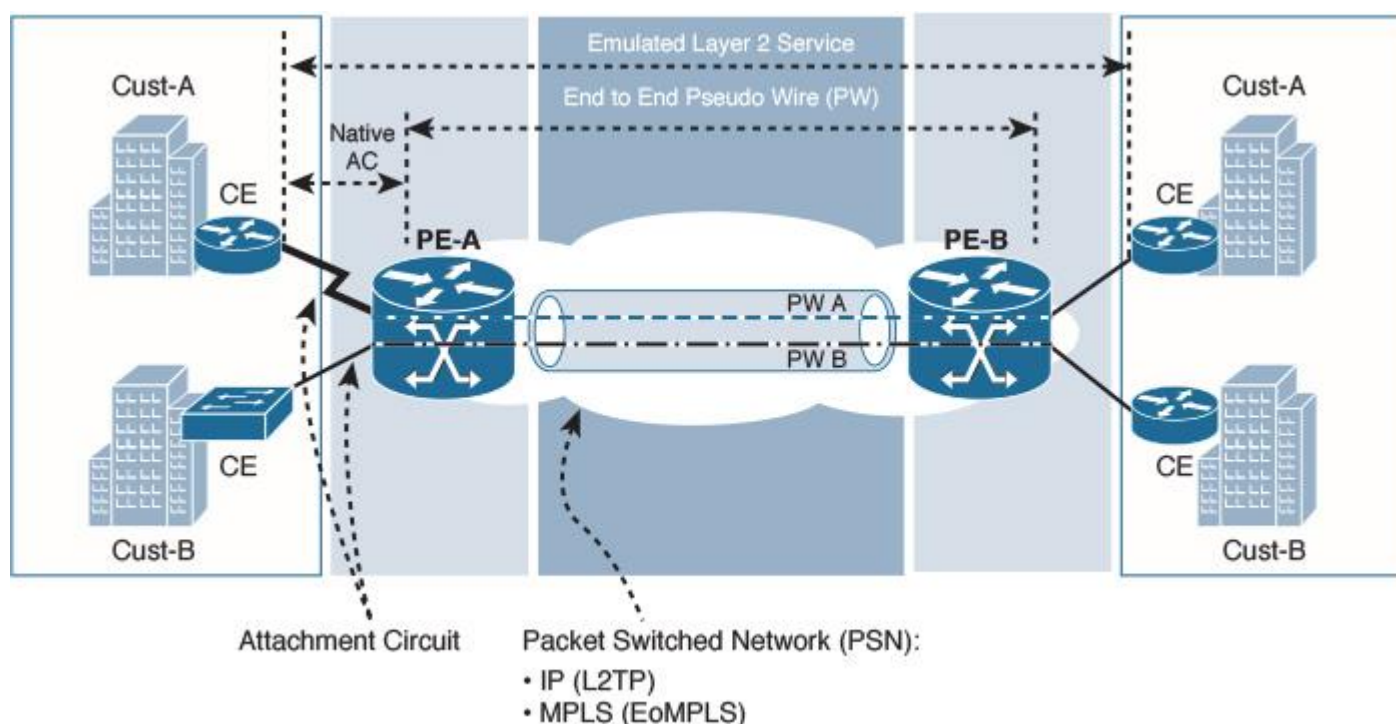


Figure 6-39 Virtual Private Wire Service

Transport Models

As covered earlier in this chapter, the primary goal of service providers is to offer services that can satisfy their customer requirements; therefore, their core public services network (PSN) is the main business

enabler. Pseudo wire over PSN gives the service provider flexibility to handle customer connectivity using primary connectivity models, each of which can serve different set of requirements:

■ **VLAN-based model:** As its name implies, it is a VLAN-based transport model. VLANs defined on the physical port can be switched to different remote PEs (for example, the scenario in [Figure 6-40](#) where enterprise customers are seeking a service from the service provider to be analogous to their current WAN service [traditional Frame Relay or ATM], in which no routing and IP addressing changes are required).

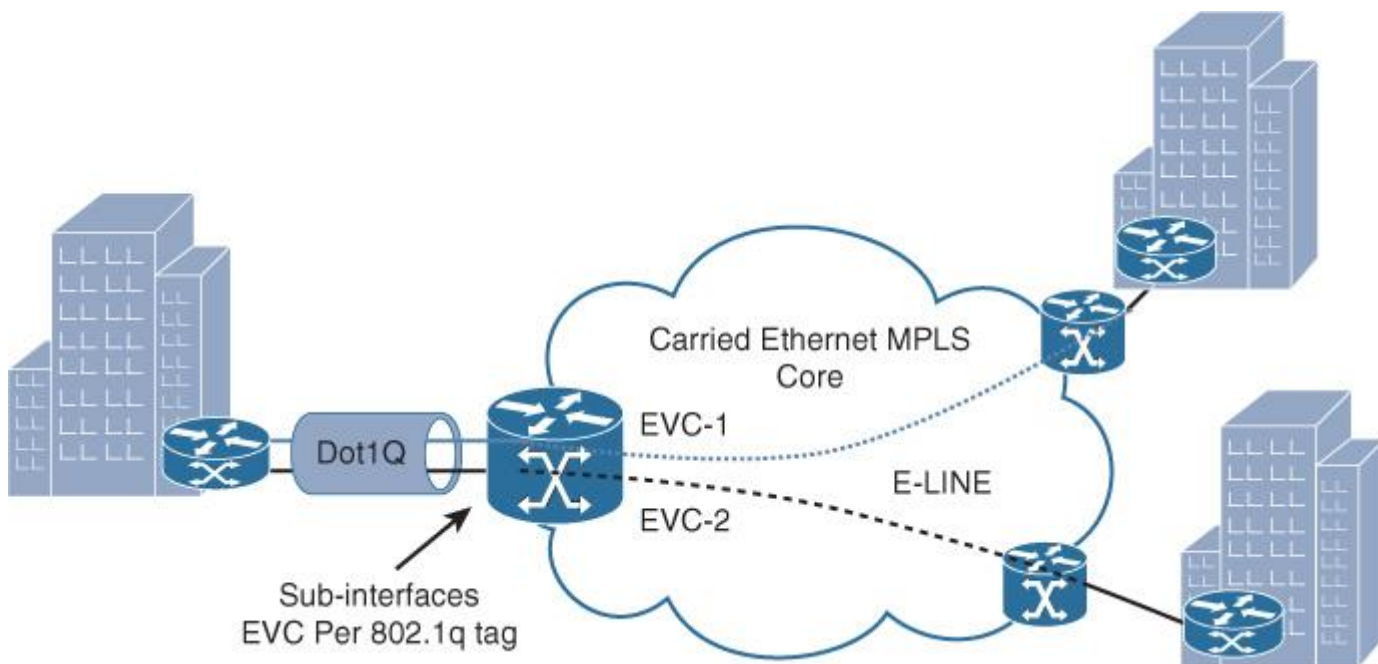


Figure 6-40 PW VLAN-Based Model

■ **Port-based model:** This transport model tunnels all traffic entering the AC physical port and transports it to the targeted remote PE without any change. This model can be applicable in scenarios where the Layer 3 customer network is traditional WAN serial links and needs to be replaced with higher-speed P2P links. Another possible usage of this model is the case of a customer that needs to interconnect two LAN islands over the service provider's Layer 2 P2P PW (whether it is native IP core or MPLS enabled). Furthermore, this model is commonly used to provide Layer 2 data center interconnection to support the extending of multiple Layer 2 VLANs over the same link, as shown in [Figure 6-41](#).

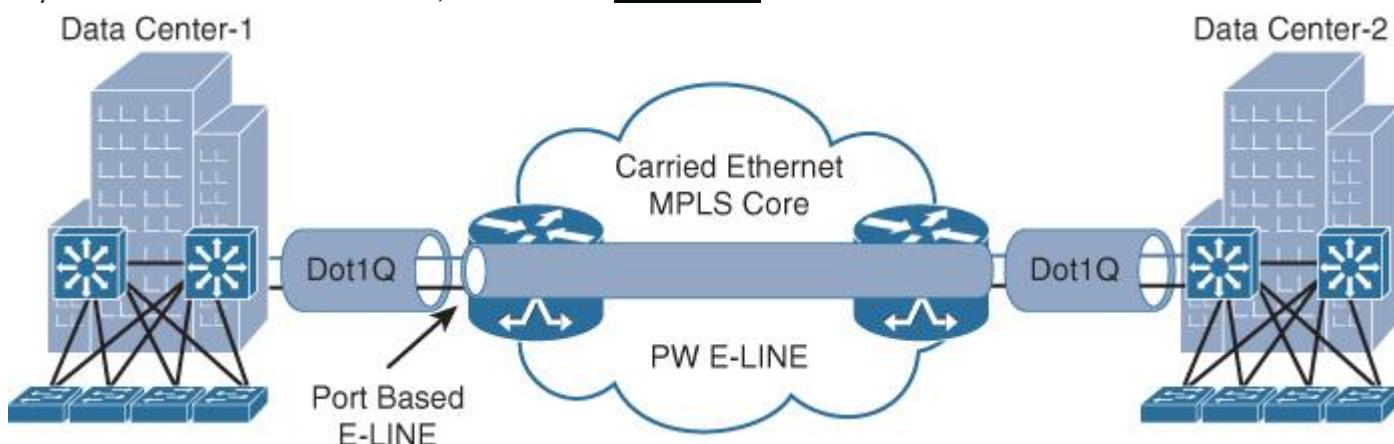


Figure 6-41 PW Port-Based Model

Furthermore, in some situations, the user access side may not always be provisioned as Ethernet. Instead, customers with legacy WAN media access such as ATM and Frame Relay might to be provisioned with L2VPN services. With the VPWS (L2VPN MPLS based), carriers still can maintain customer access as it is and offer the same service capability. In addition, service providers can even maintain the connectivity between sites that have different media access methods, such as Frame Relay, ATM, and standard Ethernet [\[45,46\]](#). [Table](#)

6-6 compares and summarizes the primary L2 WAN media access technologies (AC) along with its transport models over PWs.

L2 Technology	Transport Mode over PW
Frame Relay	Port based Per DLCI
ATM	AAL5 protocol data units (PDUs) over PW Cell relay over PW
Ethernet	Protocol based VLAN based

Table 6-6 PWs L2 Media Access

VPWS Control Plane

In general, the control plane in L2VPN is responsible for two primary functions:

- Identify the remote end (PE) with which the PW session needs to be established
- Negotiate the setup of the PW session with the relevant remote node (PE)

For the PEs to set up PWs for the VPWS, a control plane mechanism should exist to establish the session. The following are the two primary control plane models used for this purpose:

- LDP based (RFC 4447)
- BGP based (RFC 6624)

In fact, LDP based and BGP based have different attributes with regard to L2VPN signaling. However, both mechanisms have the same operational duties in L2VPN environments. For instance, with both signaling models, each PE needs to know the VPN label of the remote PE where the remote CE is attached, in addition to the consistency check via the exchange of PW attributes such as the PW maximum transmission unit (MTU) value. In addition, both signaling models support the ability to inform local and remote PEs during connectivity failure scenarios, such as PE-CE link failure. Table 6-7 compares the characteristics of each control protocol [46, 47, 48].

	LDP Based	BGP Based
Control plane sessions	Fully meshed	Can use route reflectors or confederations to avoid full mesh
Autodiscovery	No	Yes
Complexity of full mesh	Complex	Simple
Inter-AS support	Supported but can be complex with a large number of PWs	Supported, using analogous concept to L3VPN Inter-AS
Scalability	Not scalable	Scalable
P2P deployments (EPL) between 2 sites (virtual leased line)	Recommended	Not required if BGP is not used as a control plane anywhere in the network

Table 6-7 L2VPN PW Control Plane

Note

Typically, as part of the control plane signaling, PEs exchange PW payload MTUs and must agree to use the same MTU value. The PW MTU is usually derived from the directly attached circuit (AC) MTU. Therefore, the MTU within the SP core must be large enough to carry PW payload and MPLS stack to avoid any failure because of payload fragmentation. In contrast, L2TPv3 does support MTU path fragmentation.

From a design perspective, the nature of the L2VPN solution and the environment in general influence to a large extent the selection of the L2VPN control plane (LDP versus BGP). For example, the typical use of VPWS is as a P2P L2VPN solution, and mostly used to connect either two sites or a very small number of remote sites, in which control plane scalability is less of a concern, whereas the number of PWs supported by any PE is more of a concern here. Therefore, LDP is still a feasible choice as a control plane in this case. Nonetheless, if the same L2VPN carrier offers L3VPN to its customers over the same infrastructure, then having BGP as the control plane for L2VPN (VPWS) will help to simplify the overall solution manageability because there will be a unified control plane for both L2VPN and L3VPN (and so the same operation folks with the same expertise can manage both MPLS VPN services). This translates into lower operational complexity and cost from the business point of view. If VPWS is used to interconnect many remote sites with increased LDP-based control plane complexity, however this complexity here is as a result of selecting the inappropriate L2VPN solution. Consequently, the network designer may consider introducing VPLS in this case to provide a more scalable L2VPN solution for those customers with several dispersed remote sites.

Virtual Private LAN Service Design Considerations

The preceding section discussed how today's service providers can take the advantage of their MPLS-based infrastructure and provide a P2P MPLS L2VPN network, also known as a *virtual leased line* (VLL). The VLL can support sites with different WAN media access methods, which offers increased flexible and reliable solutions in terms of bandwidth and satisfying tight SLAs.

However, in today's market, where technology has become the primary key driver and facilitator for many businesses to achieve their goals, there is a significantly increasing demand for interconnected multiple sites over one common Layer 2 network. Along with that, there is an increasing demand to extend multiple modern data centers over Layer 2 transport. This supports the requirements of the next generation of data centers and the mobility of applications, such as virtual machine mobility and distributed workload (geoclustering), which supports the overall business continuity plan of the customer's business. Taking the aforementioned into consideration, this section covers how service providers can leverage the P2P PWs L2VPN architecture to offer any-to-any LAN service to their customers (usually enterprise customers). This architecture is known as *Virtual Private LAN Service* (VPLS), or in Metro Ethernet Forum (MEF) terms, *E-LAN*.

With the VPLS architecture, end users see their network nodes are interconnected directly over a shared Ethernet LAN segment; this shared segment is an emulated LAN created by the VPLS domain (emulated switched network), as shown in [Figure 6-42](#). In addition, it does support legacy media access methods that support Ethernet encapsulation, such as Frame Relay.

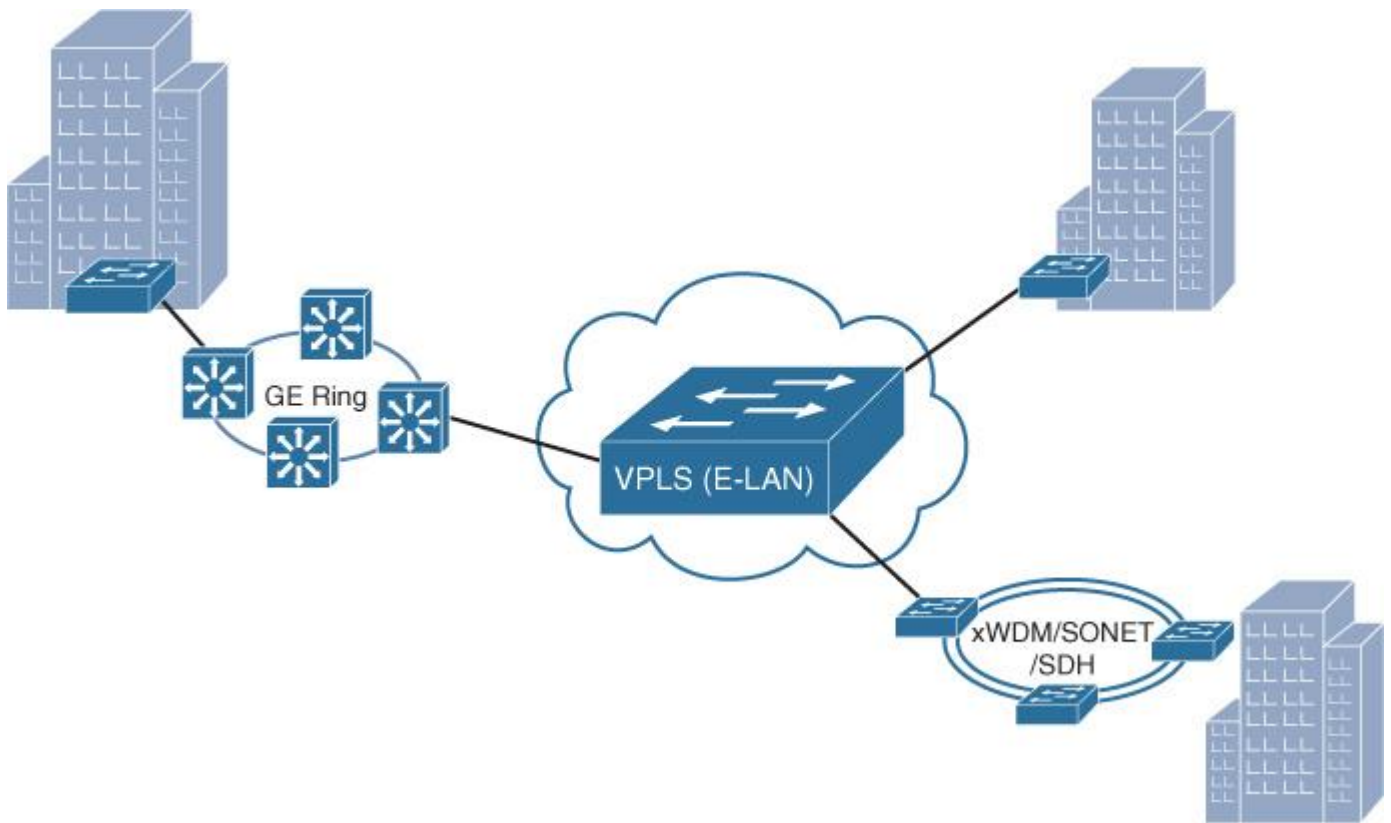


Figure 6-42 VPLS Conceptual View

As with L3VPN, in a typical VPLS architecture service providers are normally required to update the PE or PEs, to which the customer needs to be connected to the same VPLS domain of a certain customer (equivalent to the concept of adding a new L3VPN customer to an existing VPN customer). In contrast, with the P2P Layer 2 VPN solutions, the service provider must reconfigure each of the peering PEs every time a change is required (typical scalability limitation of a P2P architecture). This optimized Layer 2 VPN architecture, along with its simplified operation compared to legacy Layer 2 WAN services, is a key business driver that attracts many Carrier Ethernet providers to adopt it and offer it to their customers quickly. As a result, VPLS in today's market is one of the most common reliable and mature Layer 2 WAN services.

Note

Although this section always refers to the network as a service provider network, in fact all the concepts are applicable, to certain extent, to large-scale enterprise networks that have self-deployed Layer 2 VPN services, such as VPLS.

VPLS Architecture Building Blocks

In general, VPLS brings the standard IEEE 802.1 MAC address learning, flooding, and forwarding concept over the MPLS-based packet-switched network extended by the pseudo wires (PWs), which commonly interconnect different customer LANs. In other words, VPLS is like the glue that interconnects the multiple LANs across one common packet-switched network over MPLS. Like any system architecture, multiple components construct the overall VPLS architecture. Each component performs specific functions to achieve the desired end-to-end result. In addition, as VPLS becomes a more mature architecture, there are multiple design models of the VPLS-induced architecture, with the components and their functions differing slightly based on the architecture used. This section discusses these different design models.

VPLS Functional Components

Figure 6-43 illustrates the common functional components that construct a typical VPLS architecture. However, these components may vary slightly based on the used VPLS model, as covered in the following section:

- **N-PE:** Provides VPLS termination/L3 services

■ **U-PE:** Provides customer UNI

■ **CE:** The customer device

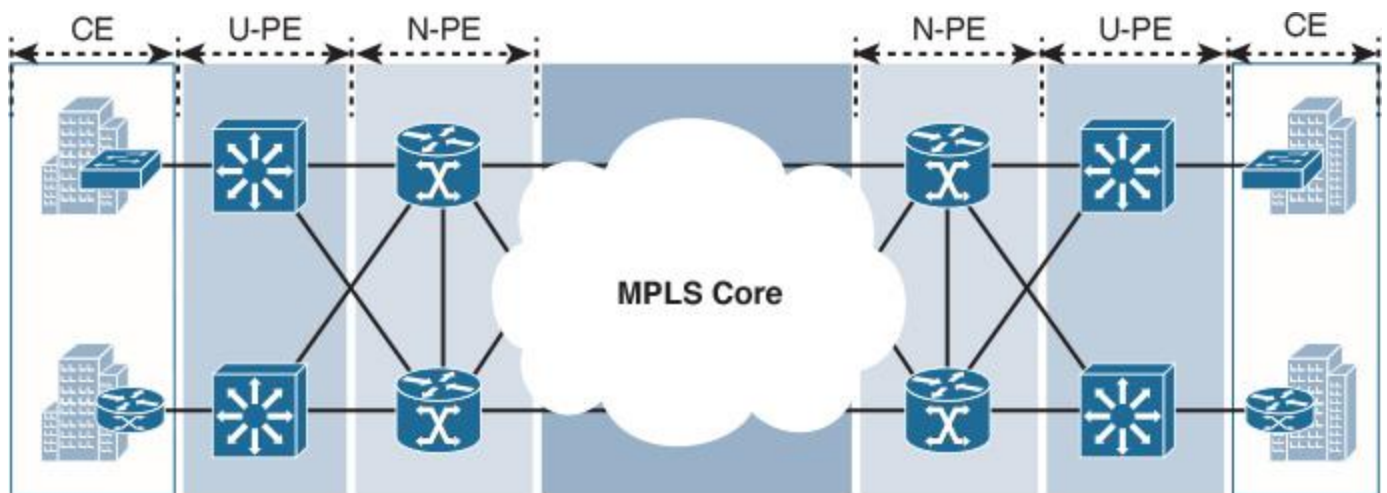


Figure 6-43 VPLS Architecture Functional Component

Virtual Switching Instance

The virtual switching instance (VSI) performs standard LAN (that is, Ethernet) bridging functions, including the participation in the learning and forwarding process based on MAC addresses and VLAN tags. Each VSI defined at each provider edge node usually handles the forwarding decisions of a single VPLS instance.

VPLS Control Plane

To a large extent, the concept of the VPLS control plane is the same as VPWS (covered earlier in this chapter). However, the primary difference with VPLS is the need of full-mesh PWs among all the participating PEs in any given VPLS instance. With VPWS, in contrast, the typical scenario is P2P PWs between two sites or a few sites in a full-mesh or hub-and-spoke connectivity model. Consequently, with VPLS, the automation of the PE discovery and the PWs set up among the relevant PEs are important factors to consider. Otherwise, the design will encounter increased operational complexity when the network grows with a large number of customers (VPLS instances) and remote sites (PEs per VPLS instance).

Nonetheless, as covered earlier, VPLS can be deployed by enterprises across their backbone to form an overlaid L2VPN connectivity (self-deployed VPLS). In such an environment, the number of PEs and VPLS instances can be limited compared to service provider networks. Therefore, using a manual mechanism to set up the PWs, such as LDP, can be a viable solution in this type of scenario. Furthermore, LDP is accustomed to more people and is a widely deployed control plane for L2VPN, even though it may not be the optimal scalable choice. However, upgrading this control plane may not be an option for those operators because of certain constraints and concerns, such as the operator cannot afford any service interruption to their existing VPLS customers or the existing PE nodes do not support BGP as L2VPN control plane protocol. In these scenarios, network designers are forced to comply or accommodate their design suggestion with regard to these design constraints.

In general, designing a typical VPLS with automated PE discovery and signaling can offer a more scalable and simple operational solution. This can be achieved by BGP as an L2VPN control plane for signaling and autodiscovery (RFC 4761), or by a suboptimal solution that offers an optimized version of the only LDP control plane that retains the signaling to be based on LDP while automating the discovery of the participating PEs in each VPLS instance using BGP (as specified in RFC 6074).

VPLS Design Models

One of the main drivers in the selection of one VPLS design model over others is the level of scalability possible. The primary factors that influence VPLS solution scalability are the supported number of VLANs (customers) and the number of PWs established between the PE nodes. Therefore, network designers should

consider the answers to the following questions, which can help to build the foundation of the design decision during the planning phase:

- What is the goal of the solution (enterprise grade to interconnect multiple data centers of an enterprise; offer E-LAN services by a carrier Ethernet service provider grade)?
- Number of customers?
- Number of participating PEs?
- Scale of the interconnected sites in terms of the amount of MAC addresses? (For instance, hosting cloud providers usually requires support for an extremely huge number of MACs.)
- PW termination between N-PEs versus between U-PEs along with the VSI?
- Any platform limitation, such as supporting VPLS VSI configuration or hardware resource capacity?
- Is traffic load-balancing/sharing for the multihomed sites required not (active-active versus active-standby)?
- What is the targeted resiliency level? And how can be archived (for example, MPLS-TE FRR, redundant PWs)?

Flat VPLS Design Model

The flat VPLS design model is the classic VPLS design model. It is the simplest design model among other VPLS models, with the least functional components, as shown in [Figure 6-44](#).

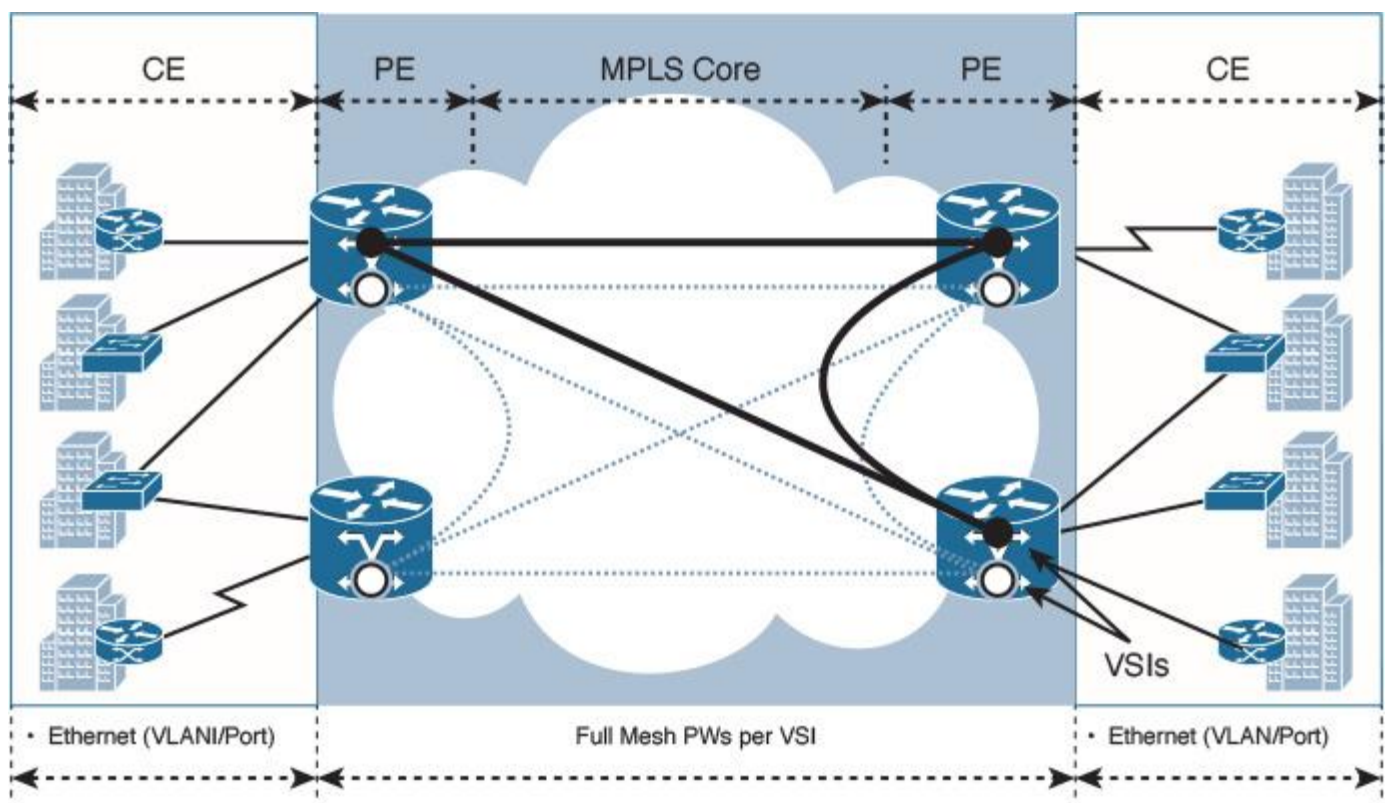


Figure 6-44 Flat VPLS

With this design model, each PE usually hosts the VSIs of VPLS domains, along with a full mesh of PWs among the participating PEs per VPLS domain. Based on this, each PE will maintain a P2MP perspective of all other PEs per VPLS domain, and each PE will be seen as the root bridge of the PW's mesh to other PEs, as shown in [Figure 6-45](#).

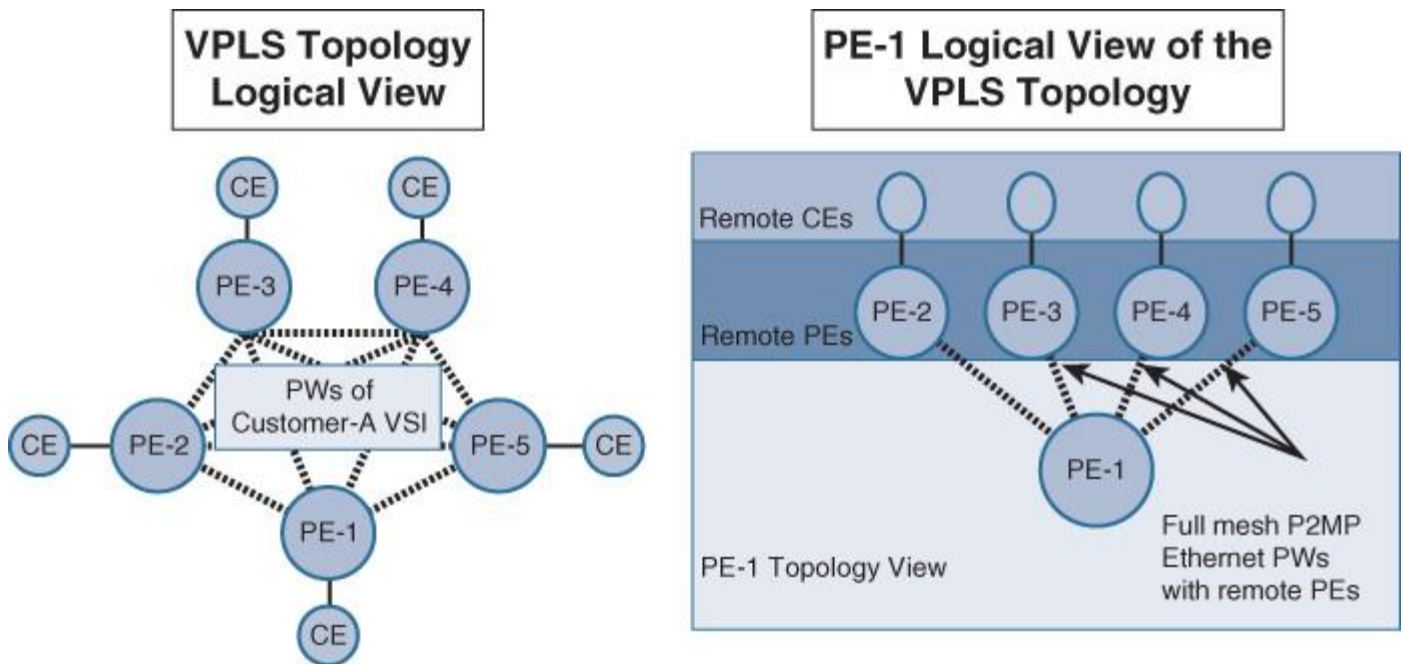


Figure 6-45 PE PW Topology View

When PEs maintain a full mesh of PW connectivity, it eliminates the need to rely on or enable Spanning Tree Protocol (STP) across the PSN MPLS network. However, from the customer point of view, their STP (bridge protocol data units [BPDUs]) can be transparently carried over the emulated VPLS LAN between the different sites without impacting the service provider network. The split-horizon concept, however, is used with the model to block any potential loop of traffic received over one core/PE PW into another core/PE PW. Based on that, the following summarizes the characteristics of this VPLS design model:

- Limited scalability. The more PEs and VSIs there are, the more network hardware resources are consumed per PE associated with operational complexity:
- A large number of PWs because of the nature of this model, where a full mesh of directed LDP sessions is required ($N * (N - 1) / 2$ PWs required)
- Potential signaling and packet replication overhead, when the number of PWs increases across multiple PEs to cover multiple remote sites (CEs) per customer VSI using LDP as the control plane protocol
- Large amount of multicast replication, which may result in inefficient network bandwidth utilization (unless some mechanisms is used to mitigate it such as Interior Gateway Management Protocol [IGMP] snooping with VPLS)
- CPU overhead for replication
- MAC distribution across the network limitations
- Support limited number of customers/VLANs (maximum 4096)
- VLAN and port-level support (no QinQ)
- Support multihomed CEs in active-standby manner
- Suitable for simple and small deployments, such as a self-deployed enterprise VPLS solution

Hierarchical VPLS Design Model

The hierarchical VPLS (H-VPLS) design model aims to optimize the high complexity of PW mesh complexity and the limited scalability of the flat VPLS design model by introducing a hierarchical structure to the VPLS design. This hierarchical structure consists of hub-and-spoke and full-mesh networks, as shown in [Figure 6-46](#).

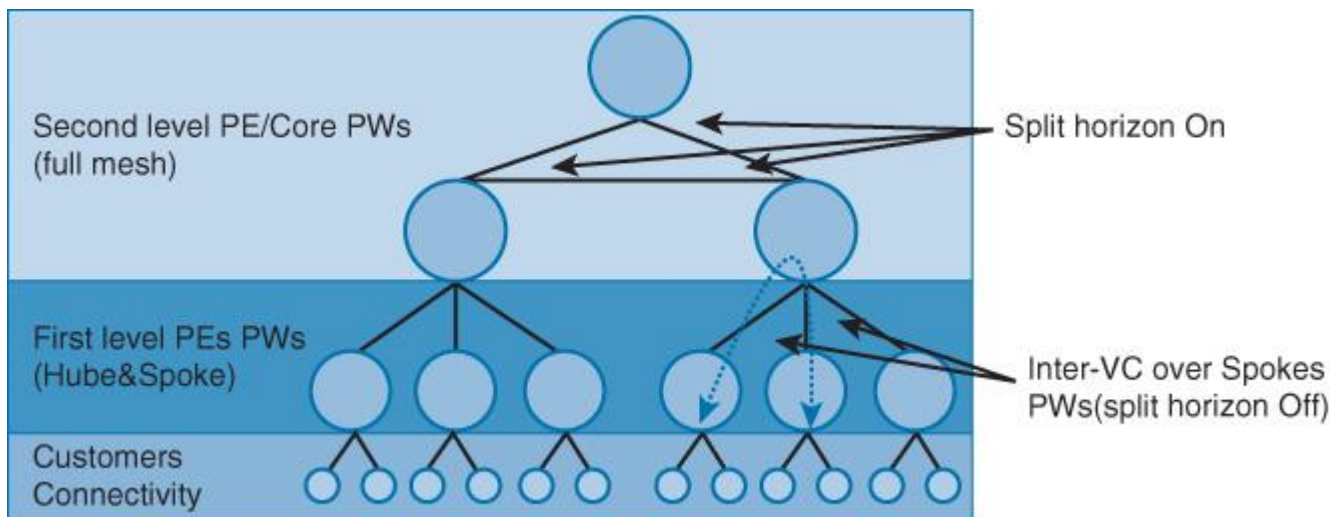


Figure 6-46 H-VPLS Structure

As shown in [Figure 6-46](#), this design model provides a more structured multistage PW connectivity model, where the hub or core of fully meshed PWs in each PE node form a multipoint-to-multipoint forwarding relationship with all other PE nodes at the same level within the VPLS domain. At the hub-and-spoke level, each PE node needs a single PW to the hub PE node and can operate in a non-split-horizon mode. This allows inter-VC connectivity between PEs at the same level connected to the same hub PE node. In other words, the hub PE nodes perform PW aggregation of the edge node PWs. Consequently, this hierarchical VPLS model offers significant reductions to the number of mesh PWs, as shown in [Figure 6-47](#). The model also overcomes the scalability and performance efficiency limitations of the flat VPLS model to a great extent.

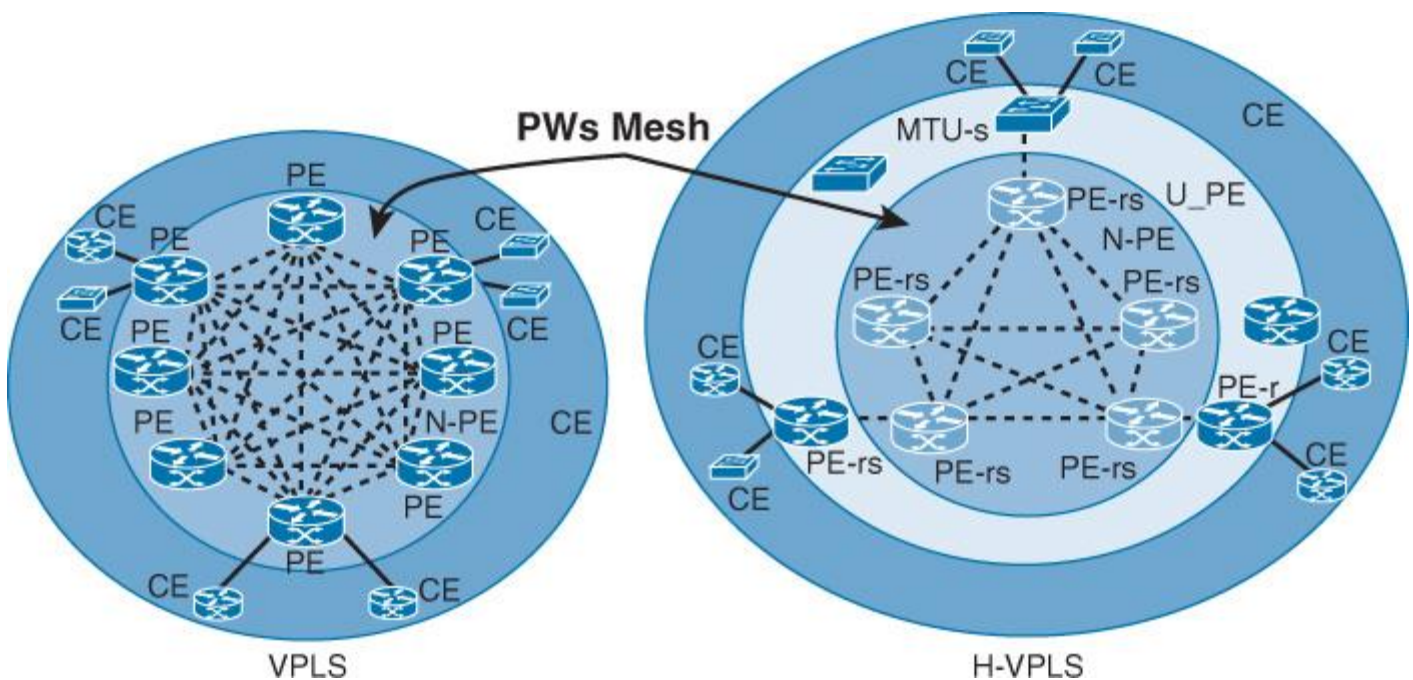


Figure 6-47 H-VPLS PW Aggregation

As shown in [Figure 6-47](#), H-VPLS has some additional terms for its functional components:

- **MTU-s:** Multitenant unit switch capable of bridging (U-PE)
- **PE-r:** Nonbridging PE router
- **PE-rs:** Bridging and routing capable PE

Moreover, H-VPLS supports two primary access models (the access tier):

- Ethernet access model
- MPLS access model

Ethernet Access Model

This model is primarily based on Ethernet and 802.1ad (QinQ), as shown in [Figure 6-48](#). With this model, service providers (Carrier Ethernet) can extend their access to different geographic areas to offer flexible Ethernet WAN/MAN access. However, this model can be limited to up to 4094 service instances per bridged access network.

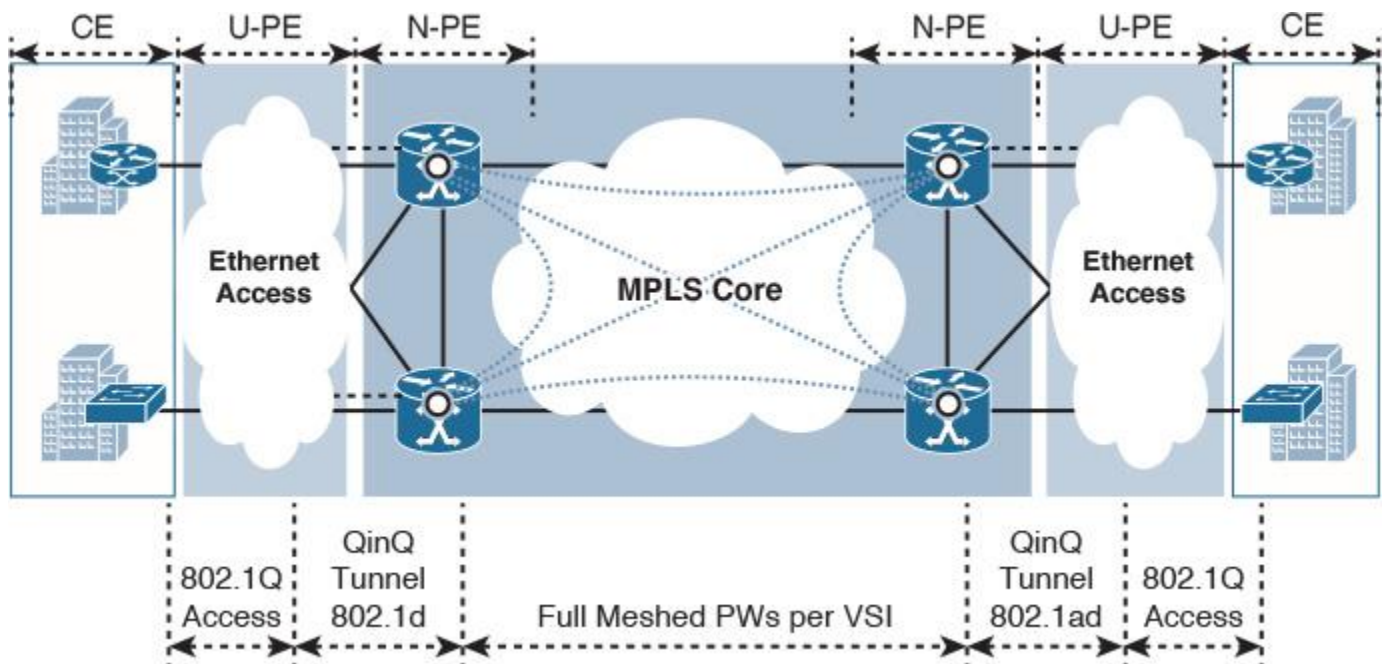


Figure 6-48 H-VPLS: Ethernet Access Model

Furthermore, with this access model, local edge traffic between ACs (within the same VLAN/VPLS domain) does not have to traverse across the N-PE. Instead the MTU-s can switch traffic locally; this can offload some of the traffic load from the core and PEs, which ultimately will lead to optimized bandwidth efficiency on circuits to the N-PE, as shown in [Figure 6-49](#).

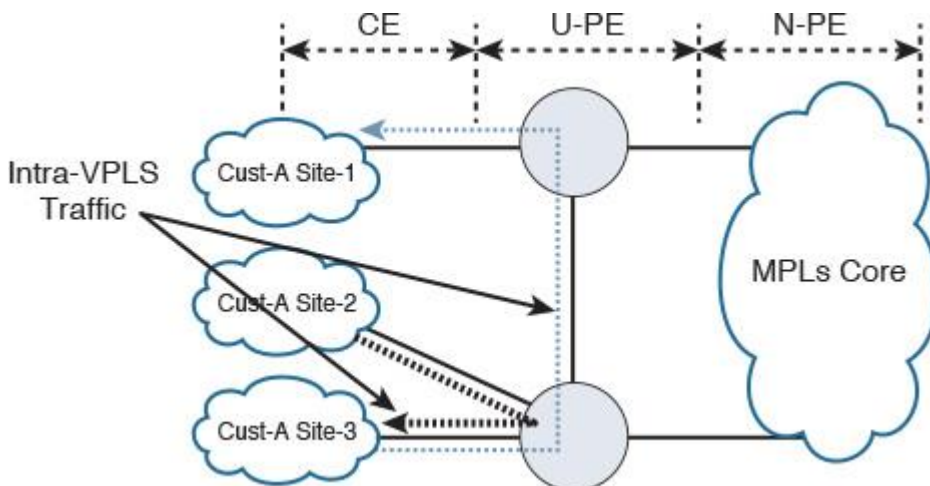


Figure 6-49 H-VPLS: U-PE Local Forwarding

MPLS Access Model

It is obvious, given the name of this model, that it is based on extended MPLS to the access model. This can offer flexibility to service providers to take advantage of certain points of presence (POPs), which indeed provide multiservice access, such as L3VPN and L2VPN (VPLS) access termination. With this model, the PWs from the U-PE to the N-PE usually connect in a hub-and-spoke fashion, and the N-PEs connect in a full mesh, as discussed earlier in this section. This can offer an optimized version of E-LAN service as compared to the flat VPLS model, as shown in [Figure 6-50](#). In addition, with regard to the optimization of the scalability and

efficiency levels of this model, in a large-scale network with a large number of PWs/VSI, the N-PE will encounter performance and scalability limitations because it will aggregate all the PWs of U-PEs connected to it.

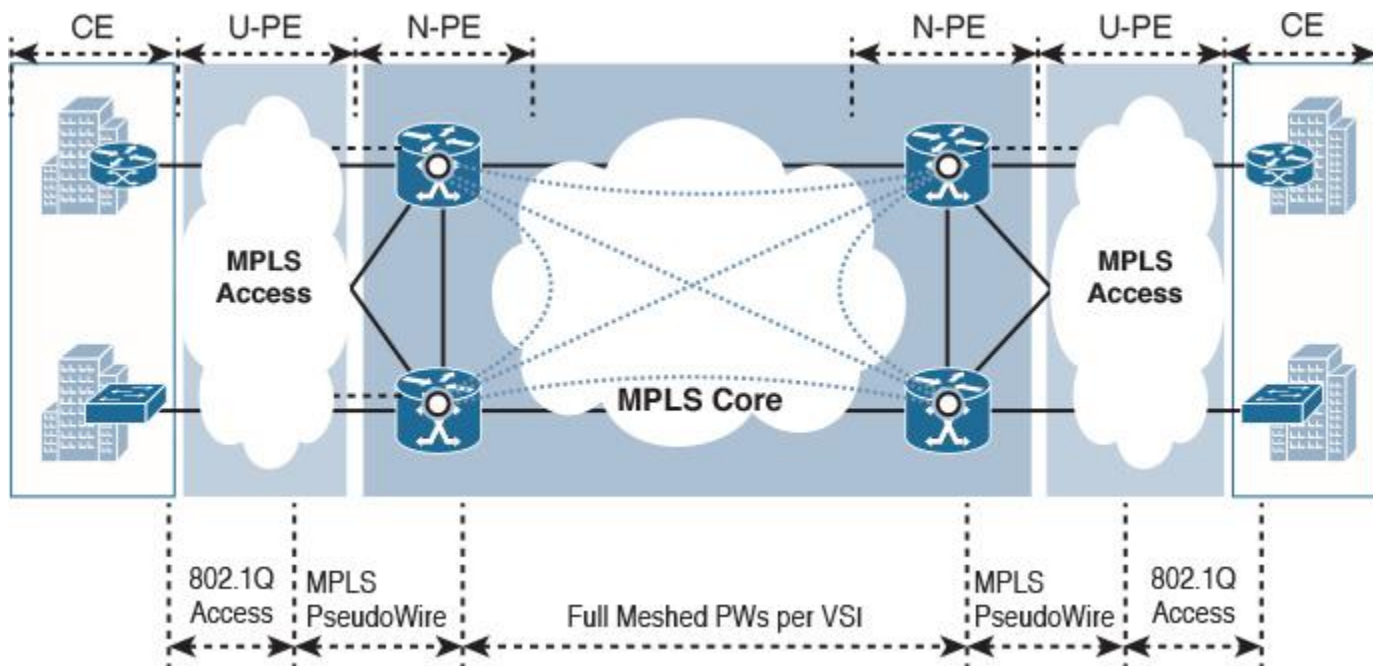


Figure 6-50 H-VPLS: Ethernet Access Model

Note

Each access model of the H-VPLS offers a slightly different level of scalability. Network designers must consider the following points, and should always ask why before selecting any given approach or design model. This ensures the selected design approach is driven by factors that collectively align with the business goals and priorities. For instance, if a POP (access network) is required to provide Ethernet access in a specific city (for example, E-LAN) and the existing infrastructure does not support MPLS, is it feasible to enable MPLS in this POP network? Yes! Why? No! Why?

■ **Ethernet access:** Limited to 4096 service instances (per access network); also STP in the Ethernet access network might slow down network convergence time

■ **MPLS access:** Increased complexity when the number of PWs between the U-PE and N-PE increases significantly

The MPLS access model can offer more flexibility if other services are required to be terminated at the same access POP/node, such as L3VPN.

With these two variant access models of the hierarchical VPLS, service providers can take advantage of the multiple access model options. For example, in one region/city, MAN access can be based on Ethernet access (QinQ), where only Metro Ethernet services are required, whereas in other regions, the access can be based on MPLS, where multiaccess services nodes are required to provide L2VPN and L3VPN client access termination, as shown in [Figure 6-51](#). In this approach, the SP core appears like the hub that connects the different access networks (spokes).

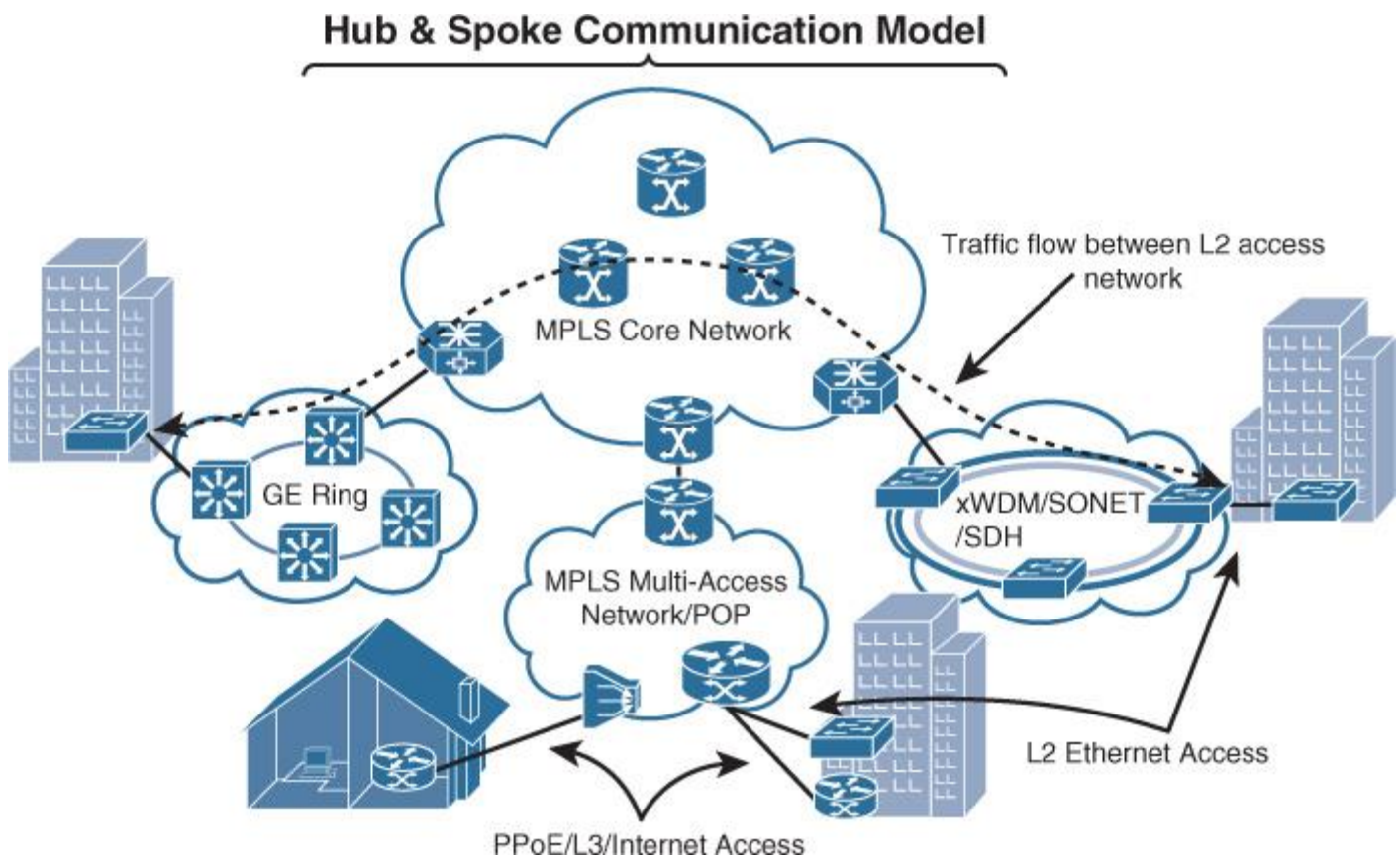


Figure 6-51 H-VPLS with Multiaccess Network

This architecture has the following characteristics:

- Offers a more structured design, where the full PW mesh is required among core nodes only, in addition to the partitions node discovery process
- Minimizes signaling overhead
- Full PW mesh among core devices only
- Packet replication done on the core only
- Provider H-VPLS flavors: H-VPLS with Ethernet access and H-VPLS with MPLS access
- Suitable for larger-scale deployment

Note

There is another variation of VPLS solution called *Advanced-VPLS (A-VPLS)*, which is the Cisco enhanced version of the standard VPLS and offers the following:¹

1. "Advanced Virtual Private LAN Service," http://www.cisco.com/c/en/us/solutions/collateral/data-center-virtualization/data-center-interconnect/design_guide_c07-647422.html

- Capability to load balance using equal-cost multipathing concept (ECMP)
- Simplified operations though the command-line interface (CLI) enhancements to facilitate configuration of the L2VPN A-VPLS feature
- Optimized DCI redundancy as its support for redundant DCI and PE switches (in active-active and active-standby modes)
- Optimized overall convergence time for a solution with multiple edge nodes and operating in active-active manner

H-VPLS with Provider Backbone Bridging

The H-VPLS model offers an optimized version of the flat VPLS in terms of scalability and performance efficiency. However, there are still scalability and performance limitations with this design model for Carrier Ethernet, which offers services to a large number of customers with large-scale networks; in particular, large-scale modern data centers (virtualized and cloud-based data centers) with layer interconnect between them,

where the service provider may need to carry millions of MAC addresses across their backbone. With H-VPLS, the following limitations are most probably going to constrain the service provider from achieving this goal:

- In both flat and hierarchical VPLS models, each PE that performs MAC switching has to learn customer MAC addresses, which often leads to performance and scalability deficiencies.

- With the H-VPLS Ethernet access model, each access network is limited to about 4094 customer instances. With the MPLS access model, the larger the number of customers, the larger the number of PWs and VSIs to be handled by the N-PE (limited performance and scalability).

Therefore, the IEEE standard was developed to provide an optimized approach that can scale to a very large number of customer instances and MAC addresses (millions) without compromising network performance efficiency. This standard is called *provider backbone edge* (PBB). This section covers the high-level architecture of the PBB and how it can be integrated with VPLS/H-VPLS to enable service providers to meet the requirements of today's large-scale enterprise customers and cloud-based data center interconnect requirements.

Provider Backbone Bridging Overview

PBB is an IEEE (802.1ah-2008) technology that defines an architecture based on a MAC tunneling mechanism, which enables service providers to build a large-scale Ethernet bridged network. The primary concept behind PBB is the hierarchical MAC address learning and forwarding in a similar manner to the MPLS label staking concept, where the typical top label used for the transport within the SP core network and the bottom labels are relevant to the customer VPN or VCID in L2VPN. Instead, PBB uses multitier MAC address learning and forwarding in which the core PEs, also known as *backbone core bridges* (BCBs), communicate based on backbone MAC addresses of the core components. Customer MAC frames (C-MAC), however, are encapsulated (tunneled) at the edge of the bridged network into the backbone MAC, which enables Carrier Ethernet to scale a large number of C-MACs without impacting core components, because the communication (including broadcast, unicast, and multicast flooding BUM [broadcast, unknown destination address, multicast]) will be based only on the backbone MACs. [Figure 6-52](#) shows the high-level architecture of the PBB.

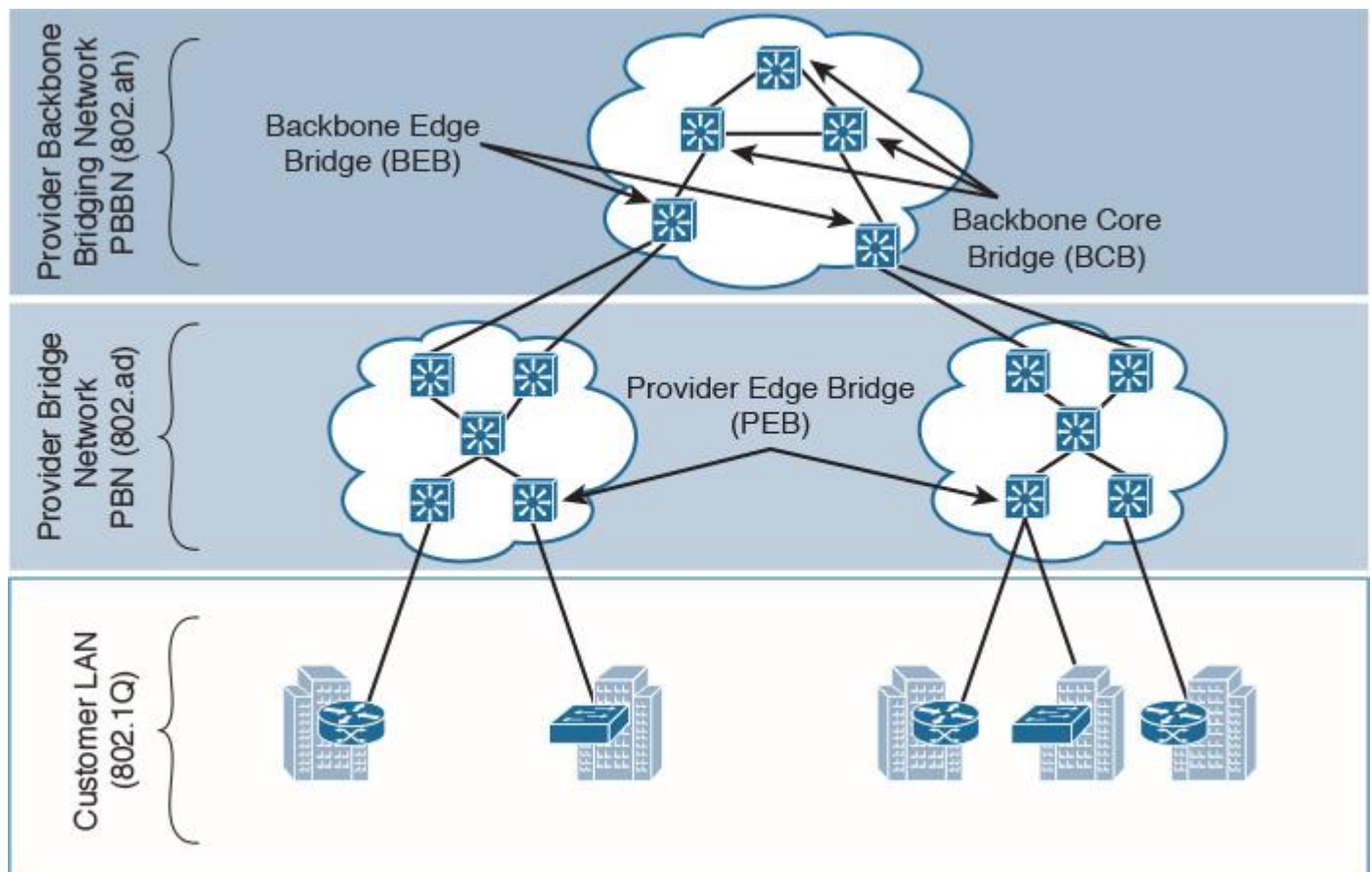


Figure 6-52 PBB Architecture

With PBB architecture, besides MAC tunneling, the network provides VLAN tunneling as well, at two stages. First, at the edge of the bridged network, where customer VLANs (C-VLAN) are tunneling using 802.1ad IEEE standard (QinQ), in this architecture referred to as *service provider tag* or *S-VLAN*. The second stage is where the S-VLANs are mapped into the PBB backbone VLAN (B-tag or B-VLAN).

Note

This concept can be applied to flat VPLS model, as well, where the BEB in [Figure 6-52](#) will be facing the CEs directly and performing forwarding based on both the backbone and customer MAC address (mapping between C-MAC and B-MAC).

Across the PBB bridged network, the end-to-end mapping and identification of each customer instance is based on the service instance ID (I-SID), which has to be unique across the entire bridged network and is usually performed at the edge of the network and not within the backbone. Based on that, PBB architecture is broken down into two primary components.

The forwarding database (FDB) within the SP backbone is selected based on B-VLAN, not the customer MAC/VLAN. PBB has two main components (as shown in [Figure 6-52](#)):

- I-Component
 - Face the 802.1Q or 802.1ad network (customer side)
 - Learns and forwards using C-MACs
 - Maintains a mapping table of C-MACs to B-MACs
 - Performs PBB encapsulation/decapsulation facing the PBB component (adding PBB header for ingress traffic)
- B-Component
 - Learns and forwards using PBB encapsulated frames (B-MAC, B-VLAN)
 - Pushes/pops B-VLAN on CBP

[Figure 6-53](#) illustrates the conceptual architecture of PBB encapsulation and forwarding.

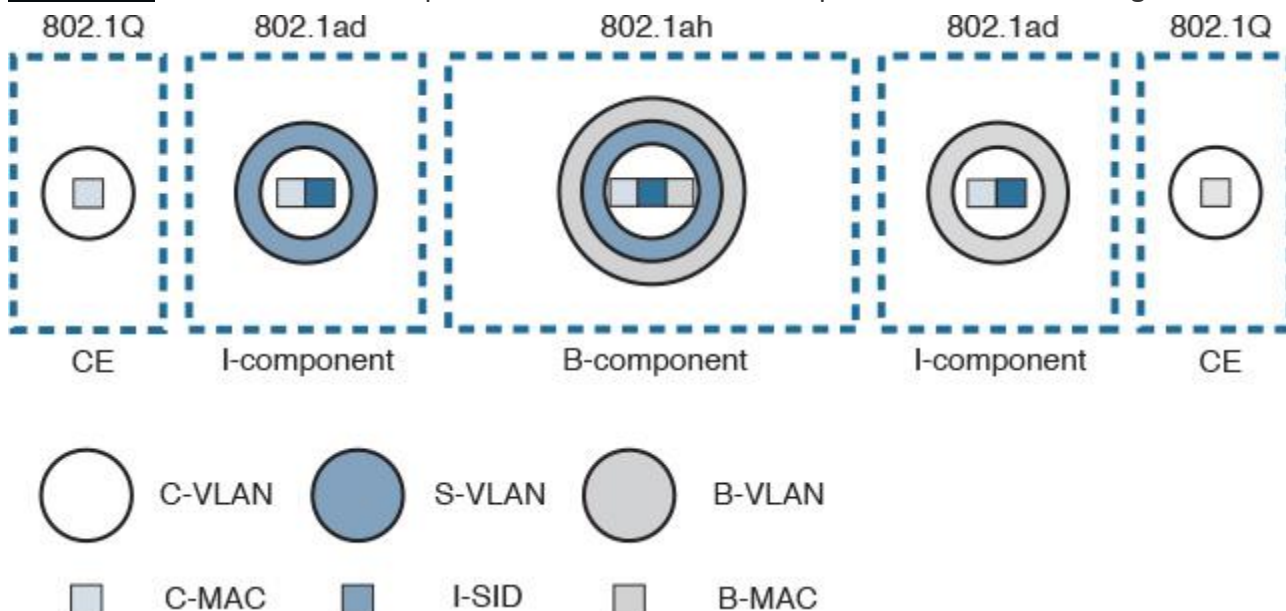


Figure 6-53 Conceptual Architecture of PBB Encapsulation

Note

In the case of broadcast of unknown unicast and multicast (BUM) traffic, the ingress BEB sets the B-MAC destination address (DA) to a B-MAC multicast group address, where one or multiple egress BEB can listen to this group address.

Furthermore, the flexibility of the PBB architecture offers multiple options to network designers to map between customer VLANs and the service interface, which is more flexible compared to the typical L2VPN VLAN- and port-based forwarding modes discussed earlier in this chapter.

PBB with this hierarchal architecture can offer the following:

- Up to 2^{24} service instances per bridge domain by defining a 24-bit service identification field (I-SID). In other words, 224 service instances per B-VLAN, also known as MAC-in-MAC.
- PBB provides the MAC address hiding (tunneling) technique from the SP core, which can significantly increase the solution scalability by confining customer MAC address learning to the edge and mapping customer MAC addresses to the backbone MAC addresses on BEBs. (In this architecture, all the intelligence is on the BEBs, because they are responsible for translating frames to/from the new PBB format.)

H-VPLS Optimization with PBB

As covered earlier in this section, PBB provides VPLS deployment a significant scalability optimization for both the flat and hierarchical models. Typically, customer MACs and VLANs will be tunneled inside the backbone MAC and VLANs along with the unique I-SID assigned to each customer instance at the network edge. B-VLAN, however, has a 1:1 mapping to VPLS VSI instance, where each B-VLAN associates with a VSI instance along with a B-MAC range.

In a backbone network, the architecture of the typical VPLS (control plane and forwarding) remains the same. However, inside each B-VLAN, there can be multiple customer VLANs, each with its I-SID encapsulated, as shown in the [Figure 6-54](#) and [Figure 6-55](#) in the next sections. In other words, each VPLS VSI in the core can carry multiple customer instances (also known as *service multiplexing*) without adding any performance inefficiencies, with the ability to scale to a large number of customer MAC addresses, because the core only learns and forwards based on the B-MAC and B-VLAN instances. The sections that follow describe the different VPLS design models combined with PBB.

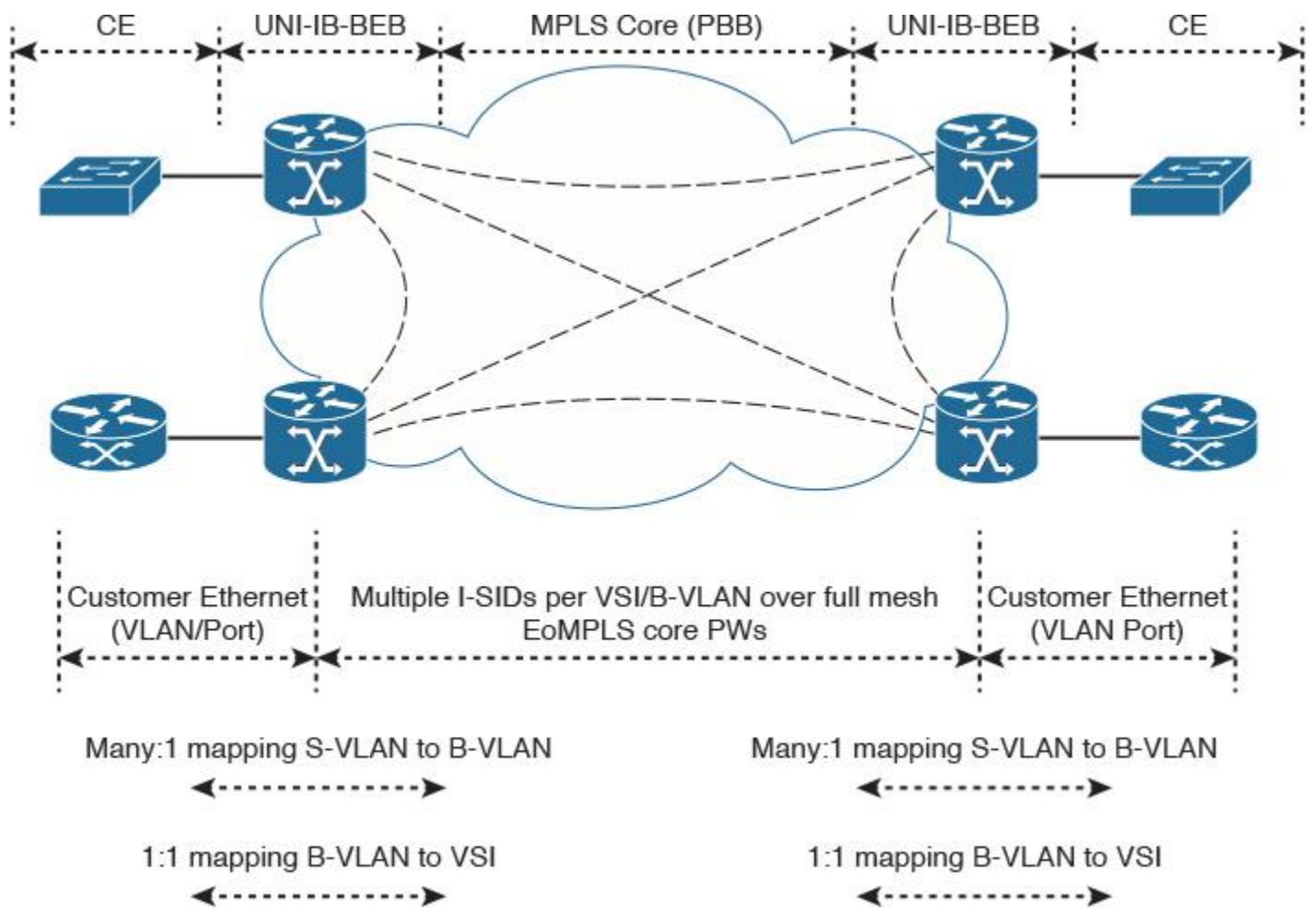


Figure 6-54 Flat VPLS: PBB

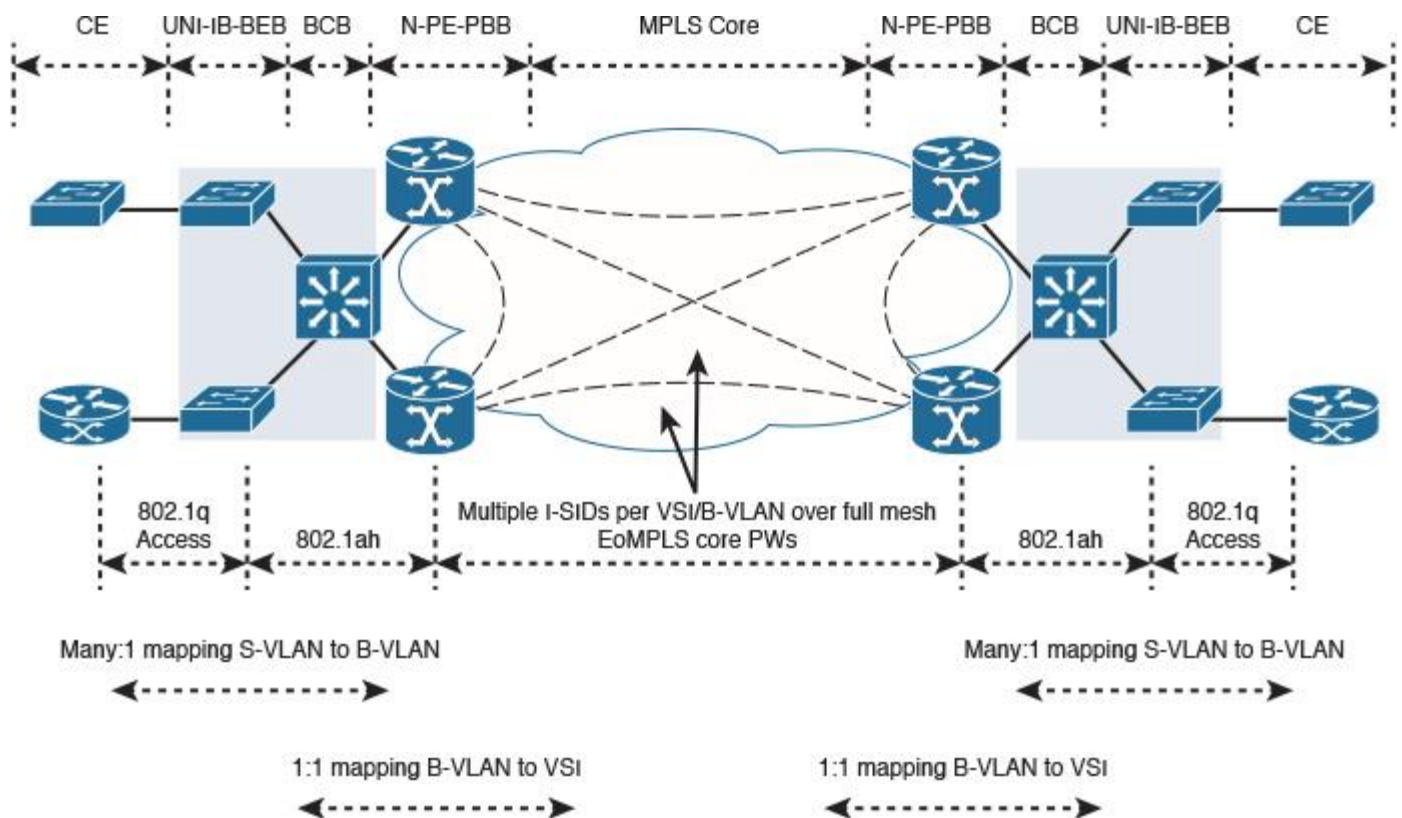


Figure 6-55 H-VPLS (B-Tagged) Ethernet Access: PBB

Flat VPLS

For the flat VPLS model, the primary optimization here is the reduction of the number of meshes of VPLS VSIs per customer. With PBB, a single VPLS (VSI) instance can multiplex multiple customer service instances, as discussed earlier and shown in [Figure 6-54](#).

Hierarchical VPLS

PBB supports both access models of the hierarchical VPLS (Ethernet based and MPLS based). With the Ethernet model, PBB transforms the access network from being based on 802.1ad (QinQ) to being based on PBB. Therefore, it is referred to as *Ethernet access B-tagged*. This approach will significantly optimize the overall scalability limitation of this model (which is limited to about 4094 customer), in addition to the multiplexing of multiple customer service instances into one VPLS instance across the core, as shown in [Figure 6-55](#).

Note

This model can also support access network with 802.1ad (QinQ), where the C-VLAN is usually encapsulated with the access network by the S-VLAN per customer; at the N-PE-PBB, one or multiple customer VLANs can be mapped to a single B-VLAN and VSI toward the core.

When PBB is combined with VPLS with the MPLS access network, the U-PE will perform the functions of encapsulation/decapsulation of the PBB header (IB-BEB) and provide the 802.1Q or 802.1ad service interface toward the CEs, as shown in [Figure 6-56](#). As with the other models, PBB provides the ability to multiplex multiple customers' service instances (I-SIDs) under a single VPLS instance (B-VLAN/VFI), which will significantly reduce the number of PWs within the access and core network, along with providing more scalable MAC address learning.

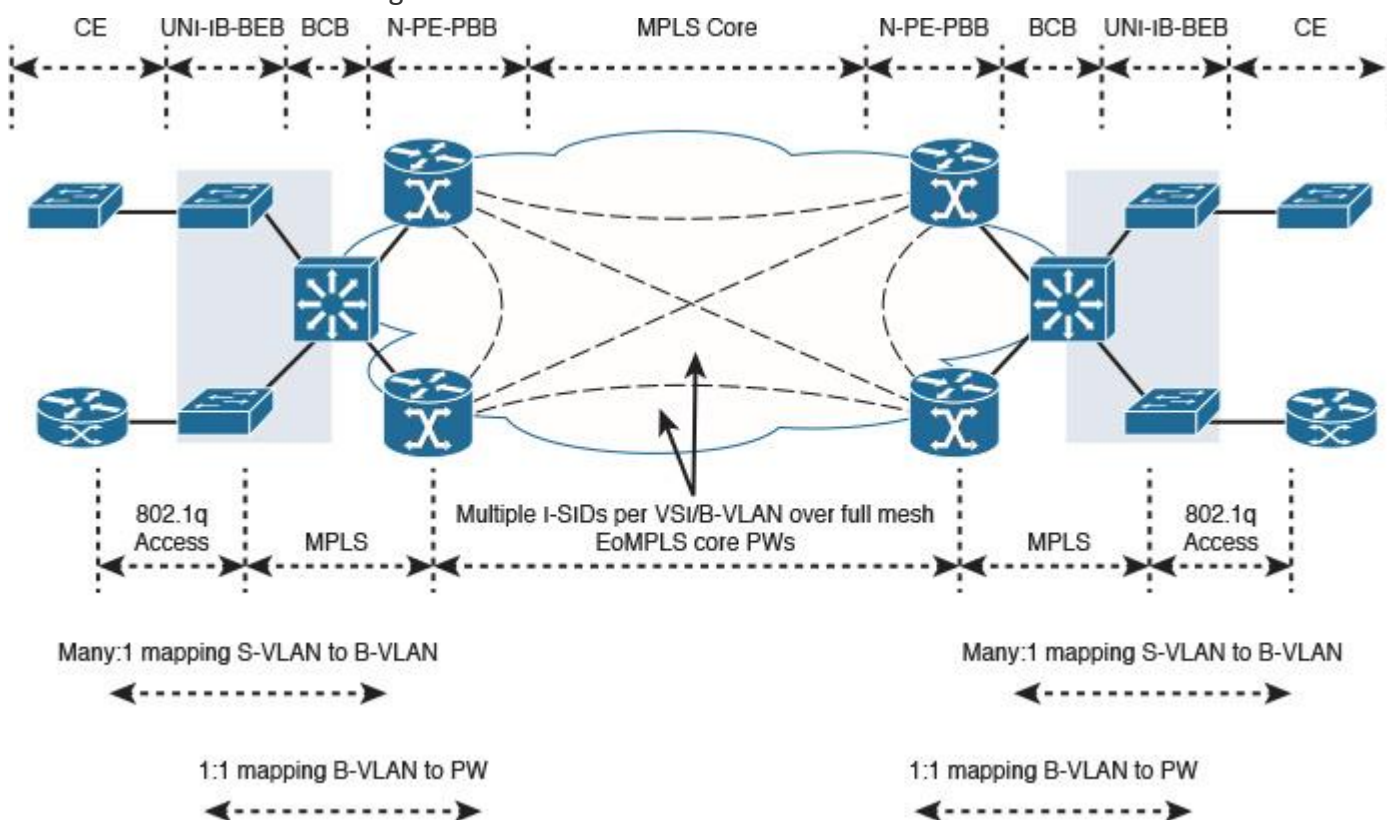


Figure 6-56 H-VPLS MPLS Access: PBB

It is obvious that PBB is not changing the actual concept of VPLS in terms of address learning, forwarding, and building the PW mesh. However, it incorporates the concept of MAC and VLAN tunneling on different stages in the bridged network, depending on the design model used, which can offer significant improvement to overall performance and scalability.

EVPN Design Model (Next-Generation MPLS L2VPN)

Ethernet VPN (EVPN) technology is a next-generation Ethernet L2VPN (defined in RFC 7432), which uses the same principle of MPLS L3VPN, in which the BGP control plane for Ethernet segment and MAC address signaling and learning over the service provider MPLS core, in addition to for access topology and VPN endpoint discovery, will eliminate the need to use PWs (thereby avoiding its operational complexities and scalability limitations). In fact, the introduction of EVPN marks a significant milestone for the industry, because it aligns the well-understood technical and operational principles of IP VPNs to Ethernet services. Operators can now leverage their experience and the scalability characteristics inherent in IP VPNs to their Ethernet offerings. In addition, EVPN supports various L2VPN Ethernet over MPLS topologies, including Ethernet multipoint (E-LAN), Ethernet P2P (E-Line), and Ethernet rooted-multipoint (E-Tree) [49].

Businesses Drivers

One of the main drivers toward EVPN is the increased demand on distributed virtualized and cloud-based data centers, which commonly require a scalable and reliable stretched layer concavity among them. Recently, DCI has become a leading application for Ethernet multipoint L2VPNs. Virtual machine (VM) mobility, storage clustering, and other data center services require nodes and servers in the same Layer 2 network to be extended across data centers over the WAN. Consequently, these trends and customer needs add new requirements for L2VPN operators to meet, such as the following:

- Flow-based load balancing (PE based and multipathing) across the PSN.
- Fast convergence (avoid C-MAC flushing).
- Support large-scale, virtualized data centers.

In addition, with the increased demand and expectations from enterprise customers or cloud hosting providers, operators also need to consider a solution that offers the following:

- Flexible forwarding policies and topologies
- A scalable solution that supports a very large number of customers and MAC addresses
- Operational simplicity

EVPN Business Strengths

Based on the market trends and new requirements and expectations of modern enterprise customers, EVPN can offer the following advantages for L2VPN service providers (carrier Ethernet):

- Ability to meet very strict customer SLA requirements for business continuity by offering the ability of all-active (per-flow) access load balancing and fast convergence (at different levels: link, node, and MAC moves), in addition to optimizing the customer ROI of their multiple links (multihomed).
- Scalable design that avoids any PW limitations and supports extremely large numbers of customers and MAC addresses.
- Simplified operations, such as EVPN. Uses the same L3VPN address learning and forwarding principle where no additional staff is required to manage or maintain the network because the same control plane (MP-BGP) is used for both. Furthermore, with the autodiscovery of PEs adding new nodes, it becomes very simplified and eliminates all the complexities of PWs that network operators need to deal with.
- Topology flexibility because EVPN supports the primary Ethernet services topologies (E-LAN, E-Line, E-Tree, and VLAN-aware bundling).
- Seamless interworking between TRILL, 802.1Qaq, and 802.1Qbp (draft-ietf-l2vpn-trill-evpn-01).
- EVPN offers investment protection to service providers because it is open standard technology and supported by multiple vendors (RFC 7432).

From the enterprise (customer) point of view, the greatest advantage of EVPN is that they can have a flexible and reliable L2 WAN/MAN service that meets their new requirements, such as large-scale next-generation data center interconnects (virtualized and cloud based).

Provider Backbone Bridging with Ethernet VPN (PBB-EVPN)

By combining the principle of MAC tunneling and hiding of PBB (802.1ah) with the principle of MP-BGP-based MAC learning and PE discovery of the EVPN, network designers can achieve the optimal L2VPN architecture, which supports an extremely large number of customers and MACs. At the same time, it can reduce control plane overhead and scale to a larger extent than EVPN alone. In addition, it will help operators to offer L2VPN with faster convergence time. As a result, combining PBB with EVPN can achieve a superior solution for modern DCIs (virtualized and cloud based) and next-generation E-LAN offerings.

In other words, both LDP and BGP are valid and proven choices. The selection of one over the other should be based on the target environment (VPWS versus VPLS) and the scale of the network (SP versus enterprise), considering design constraints.

For instance, if the SP is offering both MPLS L2 and L3VPN services and there are many clients who have full mesh or several sites to be connected using either VPLS or VPWS, the use of the BGP for L2VPN signaling with autodiscovery can be considered a scalable solution (especially if the same PEs are involved in the Layer 3 VPN service and Layer 2VPN service). Consequently, the same BGP sessions can be used for both the L3VPN network layer reachability information (NLRI) and the L2VPN NLRI, and the same BGP architecture can be used (for example, BGP RRs). This solution also simplifies the design and operations of the network because there is no need to maintain multiple separate protocols (BGP and LDP) for the control plane of L2VPN and L3VPN services.

EVPN Architecture Components

The architecture of EVPN has three primary foundational building blocks, as described in more detail in the sections that follow:

- EVPN instance (EVI)
- Ethernet segment (ES)
- EVPN BGP routes and extended communities

EVPN Instance

An EVI represents an L2VPN instance on a PE node. Similar to the VRF in MPLS L3VPN, import and export RTs are allocated to each EVI. In addition, a bridge domain (BD) is associated with each EVI. Mapping traffic to the bridge domain, however, is dependent on the multiplexing behavior of the user to network interface (UNI). Typically, any given EVI can include one or more BDs based on the PE's service interface deployment type, as summarized in [Figure 6-57](#).

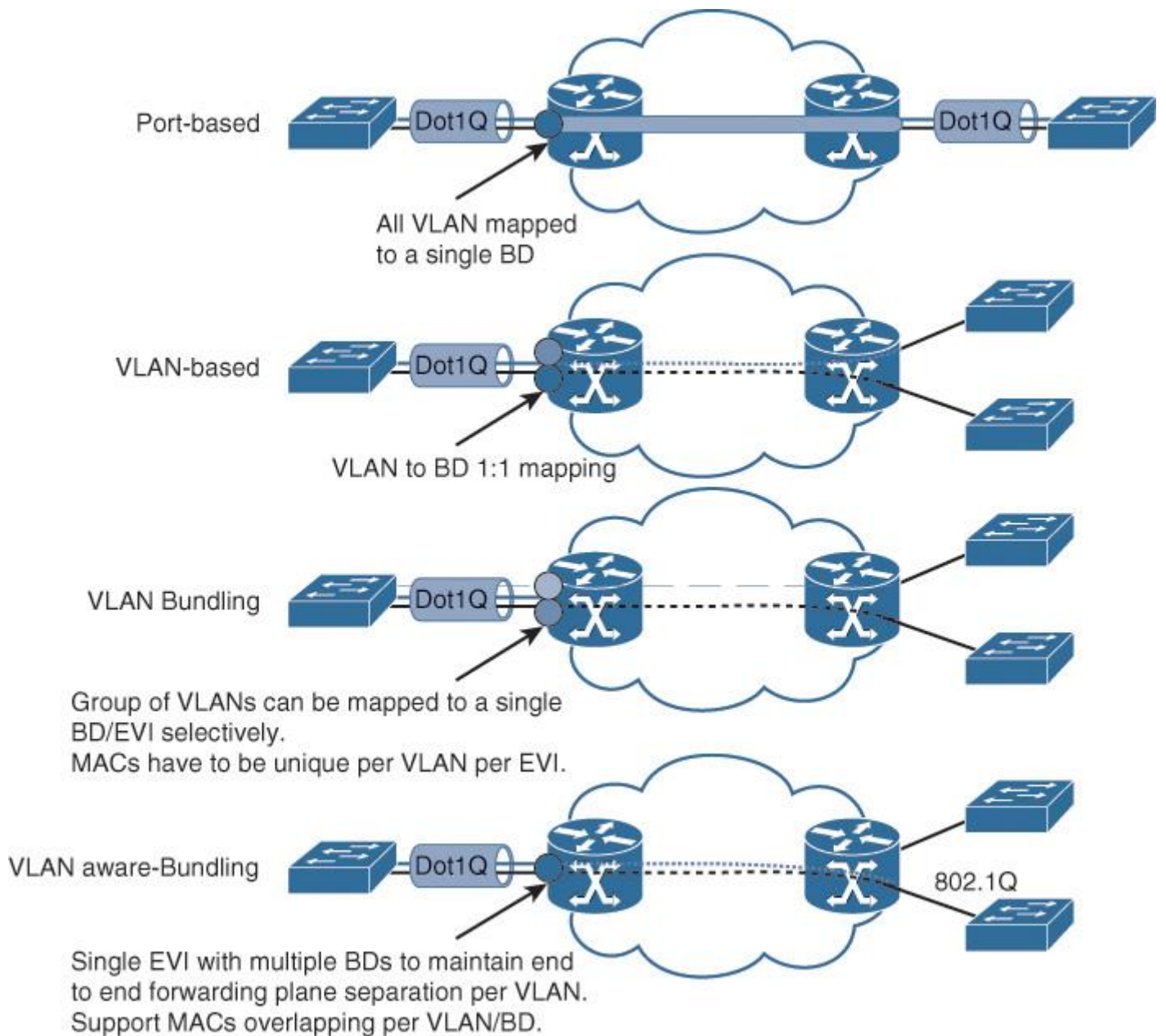


Figure 6-57 EVPN EVI Models

For instance, you can use the VLAN bundling model in environments that require multiple VLANs to be carried transparently across the EVPN cloud between two or more sites. In contrast, the VLAN-aware bundling is more feasible for multitenant data center environments where multiple VLANs have to be carried over the DCI over a single EVI with multiple DBs (VLAN to BD 1:1 mapping), because the overlapping of tenant MAC addresses across different VLANs is supported.

Ethernet Segment

Ethernet segment (ES) refers to a site that is connected to one or more PEs. (ES can be either a single CE or an entire network.) Typically, each network segment is assigned to a single unique identifier, commonly referred to as *Ethernet segment identifier* (ESI), which can eliminate any STP type of protocol used for loop prevention and normally will add limitations to the design, especially for the multihomed CE scenarios. Based on this identifier, EVPN can provide access redundancy that offers georedundancy and multihoming, where a site (CE or entire network with multiple CEs) can be attached to one or multiple PEs that connect to the same provider core using multiple combinations of the CE-PE connectivity. [Figure 6-58](#) illustrates the various EVPN supported access connectivity models:

- Single-homed device (SHD)

- Multihomed device (MHD) with mLAG
- Single-homed network (SHN)
- Multihomed network (MHN)

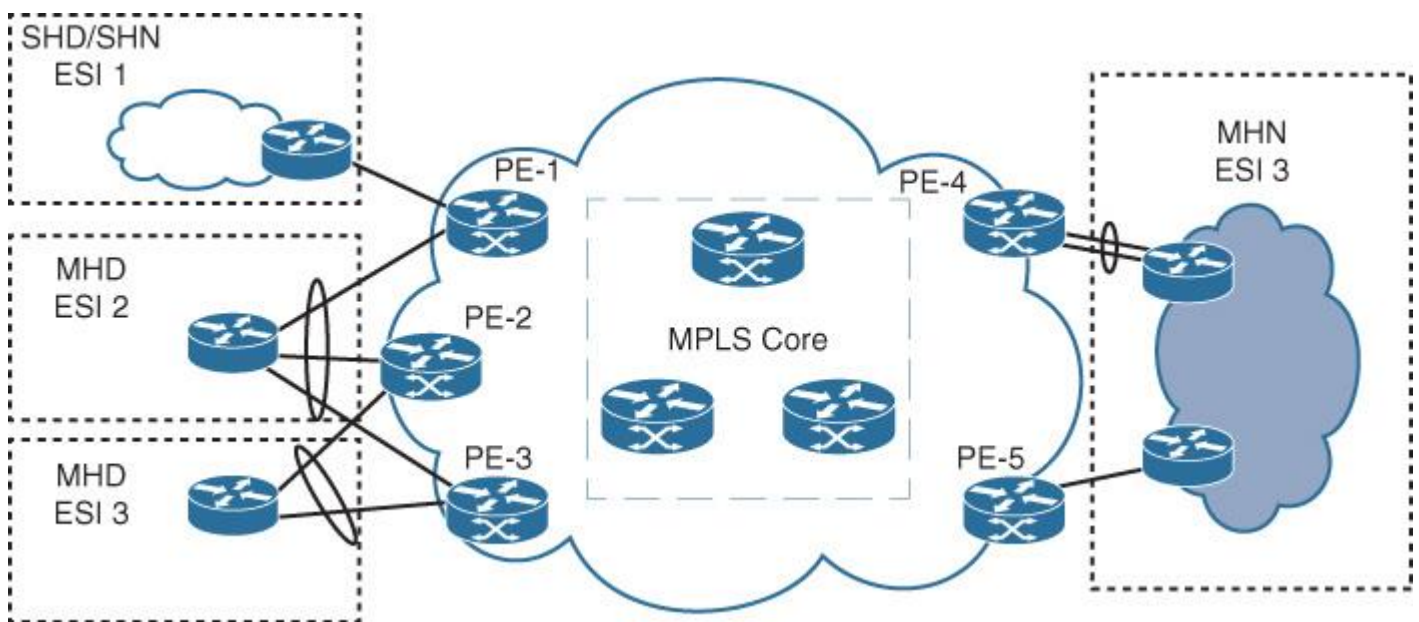


Figure 6-58 *EVPN Access Connectivity Models*

Note

In EVPN, the PE advertises in BGP a split-horizon label (ESI MPLS label) associated with each multihomed ES to prevent flooded traffic from echoing back to a multihomed Ethernet segment.

EVPN BGP Routes and Extended Communities

Because both EVPN and PBB-EVPN PEs signal and learn MAC addresses over the core via BGP, a new MP-BGP address family and BGP extended communities were created to allow PE routers to advertise and learn prefixes that identify MAC addresses and Ethernet segments over the network. Therefore, this control plane learning significantly enhanced E-LAN capability over EVPN architecture to address the VPLS (data plane learning) shortcomings discussed earlier, such as supporting multihoming with per-flow load balancing. In addition, the sequence number BGP extended community attribute offers optimized MAC mobility control plane learning and forwarding to the EVPN solution, because it facilitates and enhances the triggering of the advertising PE to withdraw its MAC advertisement in case of a MAC move (mobility) to a new network with a different ESI. This can offer a reliable solution for customers of a DCI with a stretched Ethernet segment.

Furthermore, by using BGP as a common VPN control plane, providers can now leverage their operational experience and the scalability characteristics inherent to IP VPNs for their Ethernet offerings.

EVPN Load-Balancing Models for Multihomed Devices

EVPN and PBB-EVPN support two primary load-balancing models for multihomed devices (CEs):

- **All-active load balancing (per-flow LB):** This connectivity model, shown in [Figure 6-59](#), supports multihomed devices with per-flow load balancing. Access devices are connected over a single Ethernet link aggregation bundle (mLAG) to multiple PEs, and traffic of the same VLAN can be sent and received from all the PEs by the same Ethernet segment.

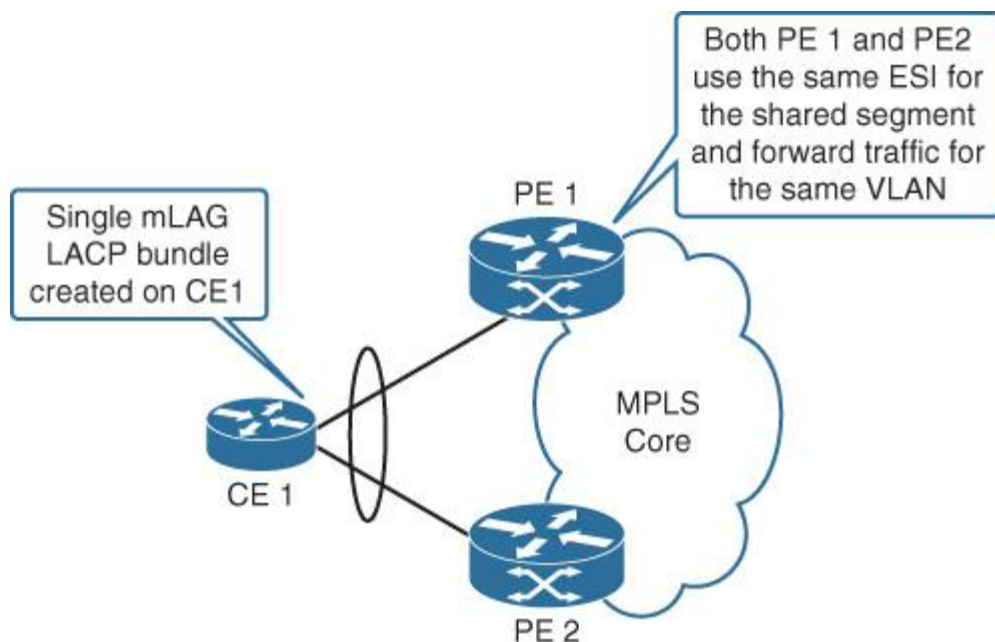


Figure 6-59 All-Active Load-Balancing EVPN

■ **Single-active load balancing (per-service LB):** This connectivity model, shown in [Figure 6-60](#), supports multihomed devices with a per-VLAN load-balancing model. The access devices in this model connect over separate Ethernet bundles to multiple PEs. PE routers, in turn, automatically perform service carving to divide VLAN forwarding responsibilities across the PEs in the Ethernet segment. The access device learns, via the data plane, which Ethernet bundle to use for a given VLAN. This is unlike the traditional VPLS model, where manual administration is required to compensate for the lack of access, autodiscovery, and automatic service carving mechanisms.

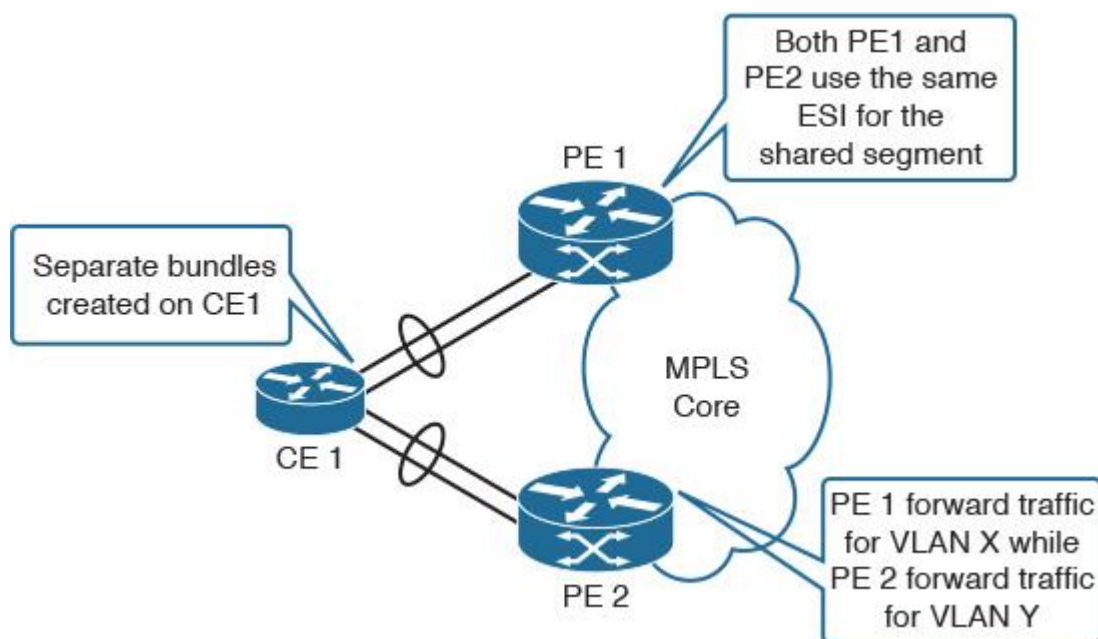


Figure 6-60 Per VLAN Load-Balancing EVPN

Note

RFC 6391 describes a mechanism that introduces a flow label that allows P routers to distribute flows within a PW.

[Table 6-8](#) provides a summarized comparison of the different VPLS design models discussed in this chapter.

	VPLS	H-VPLS	PBB-VPLS	EVPN	PBB-EVP
Scalability	Very limited	Limited	Highly scalable	Scalable	Highly scalable
Control plane protocol	BGP/LDP	BGP/LDP	BGP/LDP	BGP	BGP
Network core control plane complexity	Very High	High	Moderate	High	Low
Flow-based load balancing (CE-PE)	No	No	No	Yes	Yes
Flow-based load balancing (CE-PE)	No	No	No	Yes	Yes
Flow-based multipathing in the core	Yes	Yes	Yes	Yes	Yes
Operational complexity	High*	High*	High	Low	Moderate
Service interface VLAN-aware bundling	No	No	No	Yes	Yes

Loop prevention with multihomed CEs	STP	STP	STP	Split-horizon label (ESI MPLS Label)	Split-horizon label (ESI MPLS Label)
Fast convergence with local repairer	No	No	No	Yes	Yes
MAC mobility	Yes	Yes	Yes	Yes	Yes
Optimized control plane learning of MAC mobility	No	No	No	Yes, by the sequence number attribute	Yes, by the sequence number attribute
Targeted DCI solution**	Small (for example, enterprise controlled)	Medium	Very large	Large	Very large

*If BGP used as the control plane for VPLS, operational complexity will be reduced.

**What determines small, medium and large DCI solutions is the number of interconnected sites per customer, scale of the VMs/MACs, and the number of customers; therefore, the suggestion here can be considered as a generic and not absolute.

Table 6-8 VPLS Solution Comparison

Final Thoughts: L2VPN Business Value and Direction

This chapter covered the various L2VPN design models, protocols, and approaches, along with the characteristics of each. In addition, [Table 6-8](#) compared the different VPLS flavors available (at the time of this writing) from different design aspects. From a design perspective, all these design options and protocols are technically valid and proven solutions and still in use by many operators today. However, as a network designer, you must evaluate the scenario that you are dealing with, ideally using the top-down design approach, where business goals and requirements are at the top, followed by the application requirements that should collectively drive the functional and technical requirements.

For instance, if an enterprise needs a basic self-deployed Layer 2 DCI solution between three distributed data centers with a future plan to add a fourth data center within two years, a flat VPLS solution can be cost-effective and simple to deploy and manage, in addition to scalable enough for this particular scenario. By contrast, if a service provider is already running flat VPLS and experiencing high operational complexity and scalability challenges and they are interested in a solution that supports their future expansion plans with regard to the number of L2VPN customers while minimizing the current operational complexity then H-VPLS with BGP signaling, H-VPLS with PBB, EVPN, or EVPN PBB are possible solutions here. More detail gathering would be required to narrow down the selection. For example, is this operator offering MPLS L3VPN? Is it part of the plan to offer multihoming to the L2VPN customers with active-active forwarding? If the answer to any of these questions is yes, EVPN (with or without PBB) can be an optimal solution. Because if they are offering L3VPN, that means the same control plane (BGP) can be used for both MPLS VPN services (simplifies operational complexity and offers a more scalable solution). Adding PBB to this solution will optimize its scalability to a large extent, especially if this operator provides L2VPN connectivity to cloud-based data centers where a large number of virtual machine MAC addresses is expected to be carried over the L2VPN

cloud. If multihoming with active-active forwarding is required, EVPN here will be a business enabler, along with optimized scalability and simplified operation as compared to the existing flat VPLS. However, there might be some design constraints here (for example, if the current network nodes do not support EVPN and the business is not allocating any budget to perform any hardware or software upgrade; or if this provider has existing an interprovider L2VPN link with a global Carrier Ethernet, to extend its L2VPN connectivity for some large enterprise customers with international sites, and this global Carrier Ethernet does not support EVPN).

In both situations, the network designer is forced to look into other suitable design alternatives such as H-VPLS. Again, the design requirements and constraints must drive the decision for which solution is the suitable or optimal one by looking at the bigger picture and not focusing only on the technical characteristics of the design option or protocol.

Service Provider Control Plane Scalability

This section discusses the different design considerations that, individually or collectively, enhance the scalability of service provider type of networks.

Note

All the recommendations and design considerations covered in the enterprise architecture section about scalability of both IGP and BGP are all valid and applicable and must be taken into account. This section is only adding more design options and considerations that are more relevant to service provider grade networks.

As discussed earlier in this chapter, today's service provider networks rely mainly on different control planes and overlay protocols to offer various transport services that meet the different connectivity attributes and requirements of modern enterprises. Regardless of the services offered by the operators to their customers, the primary control protocols used for the learning and forwarding of customer prefixes while maintaining end-to-end path isolation are as follows:

- MPLS
- IGP
- BGP/MP-BGP

Therefore, for a service provider to scale its network, it must consider how to scale these control protocols in terms of number of sessions, number of nodes/routers, and utilization of router resources (memory and CPU) to a manageable and reliable scale. The following subsections cover the primary design techniques that can enhance the scalability of these protocols in large-scale networks. This, in turn, can support a more responsive and scalable service provider-grade network.

IGP Scalability Considerations

As discussed earlier, IGP in service provider types of networks is primarily used to carry reachability information for BGP next hop and is not intended to carry any customer route. Therefore, to enhance the scalability of a service provider network, the underlying control plane (IGP) must be designed to provide a reliable, scalable control plane transport that can respond to any network change within an acceptable period.

Note

The ability of any routing protocol to react to any network change (usually after a failure event) is critical and governed by multiple factors. The acceptable time for the control plane protocol to react and converge will vary from network to network and from business to business.

Ideally, a successful IGP design in service provider-grade networks should provide the following:

- Reliable and stable control plane transport
- Flexible design that takes into consideration interaction with any other control protocol across the network, such as MPLS-TE

■ A scalable design that supports the growth of the network by size without compromising its acceptable levels of stability and manageability

The concept of modular IGP design in SP-grade networks is similar to the design of modular enterprise networks, where each module can be placed in a separate routing domain, such as OSPF area or IS-IS level, to provide optimized control plane scalability and stability (fault isolation). With this approach, the SP core ideally should be the backbone of the IGP (such as OSPF area 0 or IS-IS L2) that glues all the different POP networks each into its own flooding domain, as shown in [Figure 6-61](#).

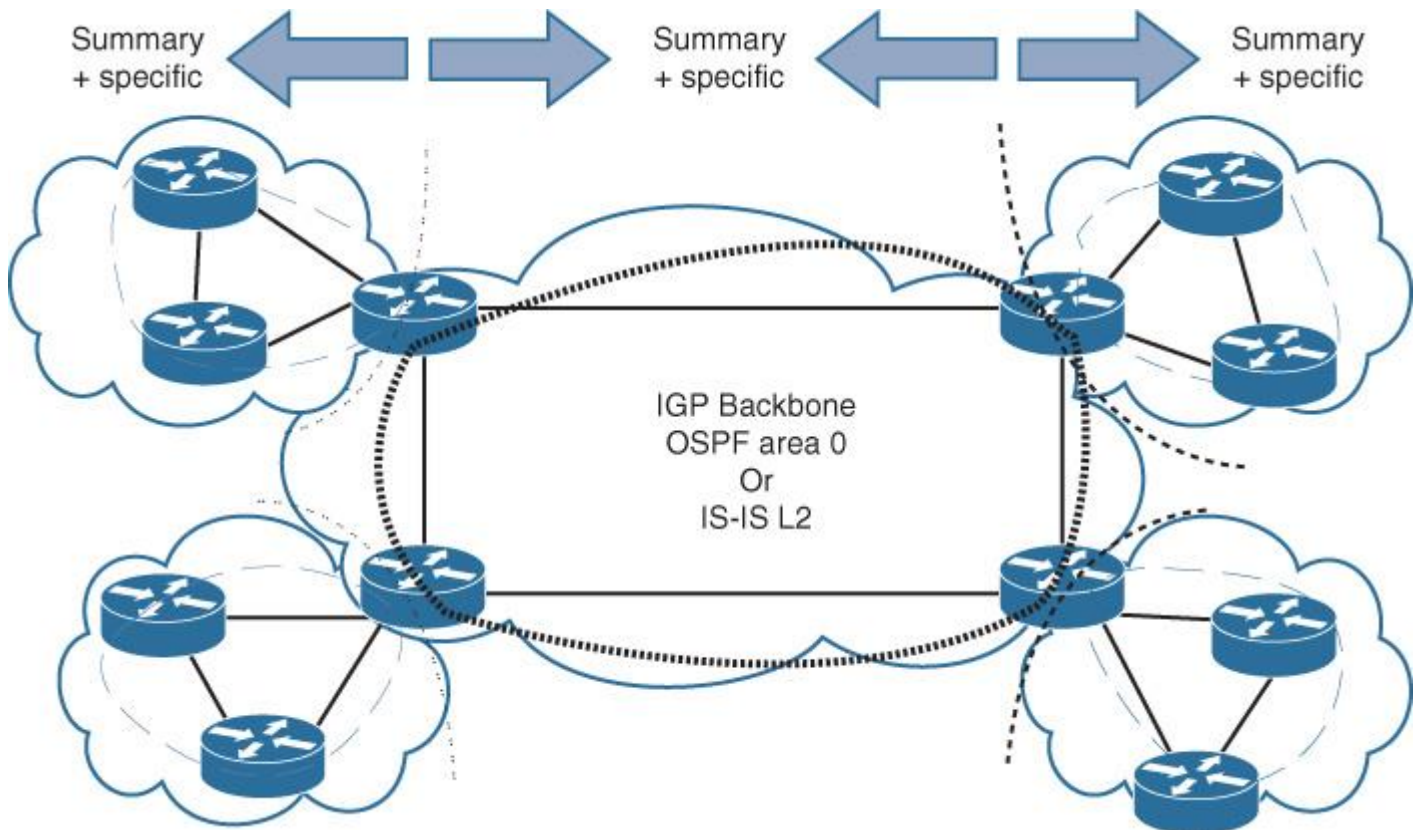


Figure 6-61 *Modular IGP Design in SP Network*

The nature of any service provider is a transport network, and it usually provides a transit path for other networks' traffic (customers). Therefore, the scalability and stability factors of its control plane are a primary factor for the P business to succeed. The following summarizes the design considerations and advantages of the modular IGP design in the SP type of networks:

- Placing the backbone in the middle and each POP in its own IGP flooding domain helps to make the overall logical architecture more modular.
- This module logical architecture is used to reduce the number of advertised routes from each POP to the core and from the core to each POP.
- Specific routes are requested in some situations. For example, if this is an MPLS SP /32 loopback address, it is required to be injected as /32. The Fast EtherChannel (FEC) element should exactly match an entry in the IP Routing Information Base (RIB) according to LDP specification RFC 3036. Therefore, in this case, a combination of summary and specific routes can be sent. Alternatively, you can use the concept known as longest-match label mapping when that choice is available (RFC 5283).
- In OSPF, you can use stub areas or not-so-stubby areas (NSSAs) in the POP; with IS-IS, the POP can be placed in L1.
- You can use IS-IS route leaking to help with injecting the /32 loopback address into the L2 POPs for MPLS VPN SPs.
- This structured design can help optimize the design and convergence time and provide fault-isolation domains. For example, if a link is flapping in one POP, it will not affect other POPs' stability in this case.

Note

Some service providers prefer simplicity over any other aspect. Therefore, a single IGP domain is sometimes used to offer a simple network, which provides better flexibility to support other services without adding any complexity to the design. Although breaking the IGP domain can add several benefits, it might introduce added complexity to other services. For instance, having multiple OSPF areas or IS-IS levels can complicate MPLS-TE design. Therefore, network designers must prioritize based on the primary goals to produce a proper design.

Route Reflection Design Options in SP Networks

Providing a scalable design of IGP is only half the job of achieving an optimal scalable, stable, and manageable control plane in service provider networks. BGP is the primary routing protocol used in service provider networks that carry customer prefixes, which is where the actual load happens. Therefore, to achieve the optimal control plane design, each IGP and BGP has to be designed in a way that can function at the desired level and support the business's future plans and current investments. For example, as discussed earlier, IGP design must provide a stable and scalable design to reliably carry next-hop reachability for BGP. At the same time, it must not add complexity to the design of other control protocols in the network. Similarly, BGP design must provide a reliable transport for the customer prefixes, considering the priority factors of the design, such as optimal path or provision of a highly scalable design, without compromising the desired level of reliability along with simplified operation.

Chapter 2, "Enterprise Layer 2 and Layer 3 Design," discussed multiple proven mechanisms that provide a more scalable and reliable BGP design. They are all applicable to BGP design in service provider types of networks. This section covers some advancements to these mechanisms, in particular to the BGP RR design, to fit very large-scale BGP networks with multiple services provided by BGP at the same time, such as Internet routing and MPLS L3VPN.

RR design in service provider networks can vary depending on the service provider architecture (for example, whether it is an MPLS-based network or native IP [non-MPLS]). Also as covered in Chapter 2, if BGP confederation is used, the design choices of BGP RR will be impacted, because BGP RR design must align with the BGP confederation sub-autonomous systems.

The design consideration of BGP route reflection in MPLS-based VPN networks varies slightly from the IP only (non-MPLS) based network, as summarized here:

- RR in an MPLS-based VPN network provides the destination PE IP of the desired L3VPN customer IP address.
- Because the forwarding is based on MPLS labels, the MPLS VPN RRs are not required to be in the forwarding path, because they will only provide the destination PE IP and the PE will use IGP to reach the other PE IP along with MPLS transport label.

This section covers the following BGP RR design options:

- Provider routers as RRs for MPLS-VPN
- Separate RR for MPLS-VPN and IPv4/v6
- Separate RR per service (MPLS-VPN and IPv4/v6)
- MPLS VPN partitioned RR
- Hierarchical RR

Provider Routers as RRs for MPLS-VPN

This design can offer cost-effective, dedicated BGP RR for MP-BGP VPN peers. However, this means that the (P) routers need to have MP-BGP VPN enabled and MPLS-VPN routes installed, along with MP-BGP VPN VPN peering reflections, as shown in Figure 6-62. This design option is feasible in the service provider grade of network with a limited number of prefixes (PEs and MP-BGP VPN instances), ideally in self-deployed MPLS-VPN by enterprise networks. For the typical service provider networks, the P routers should always be left for fast MPLS switching functions only.

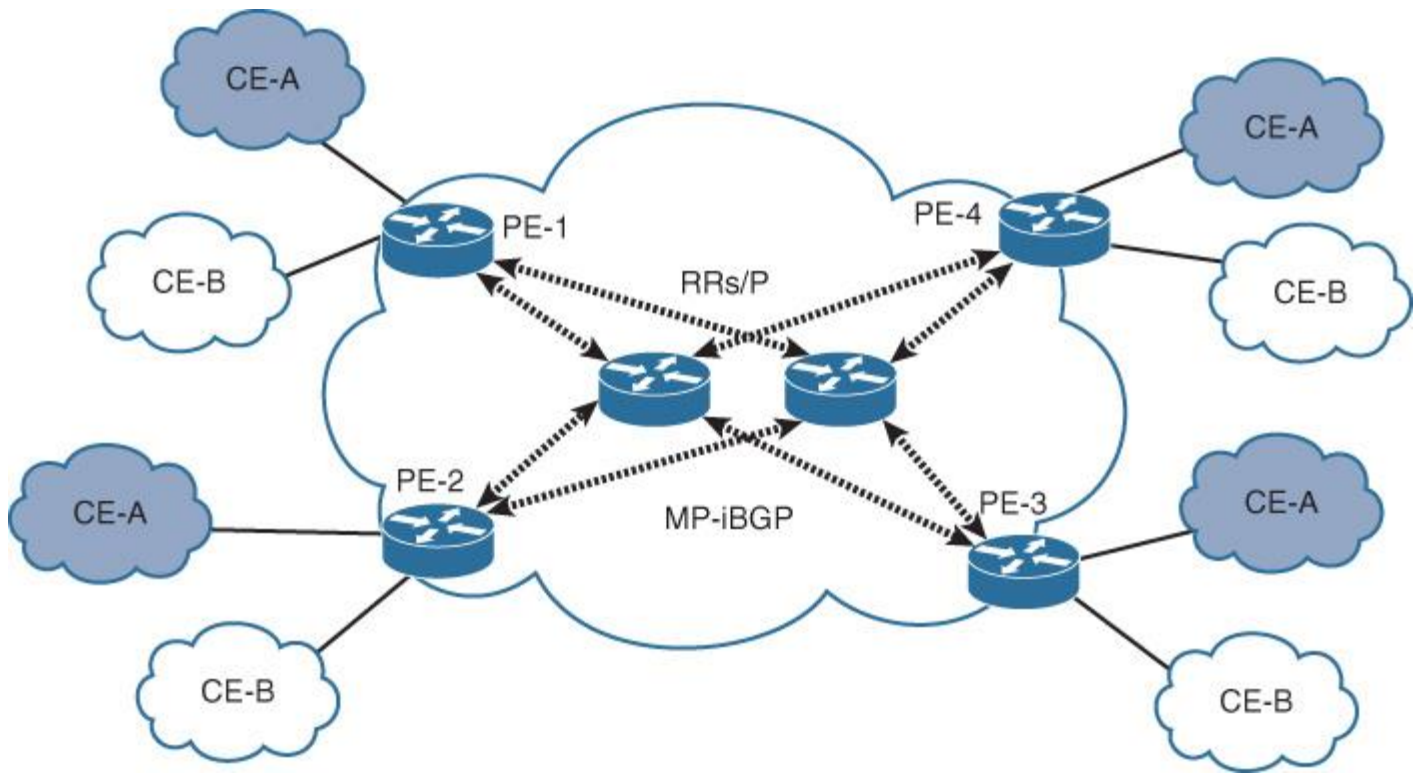


Figure 6-62 *P Nodes as BGP RR*

Separate RR for MPLS-VPN and IPv4/v6

This option offers a simple and cost-effective solution for service providers that provide both Internet (non-MPLS IP service) and MPLS VPN-based services where a dedicated RR will be used to serve both IP and MPLS VPN route reflection, as shown in the [Figure 6-63](#). However, this option still lacks the scalability for the increased load on those RRs if this service provider is expanding rapidly and if a number of PEs (BGP peers) are increasing dramatically along with a very large number of prefixes. Therefore, this design option is suitable to regional or medium-sized service provider networks and for large-scale enterprise networks with self-deployed MPLS VPN and native IP.

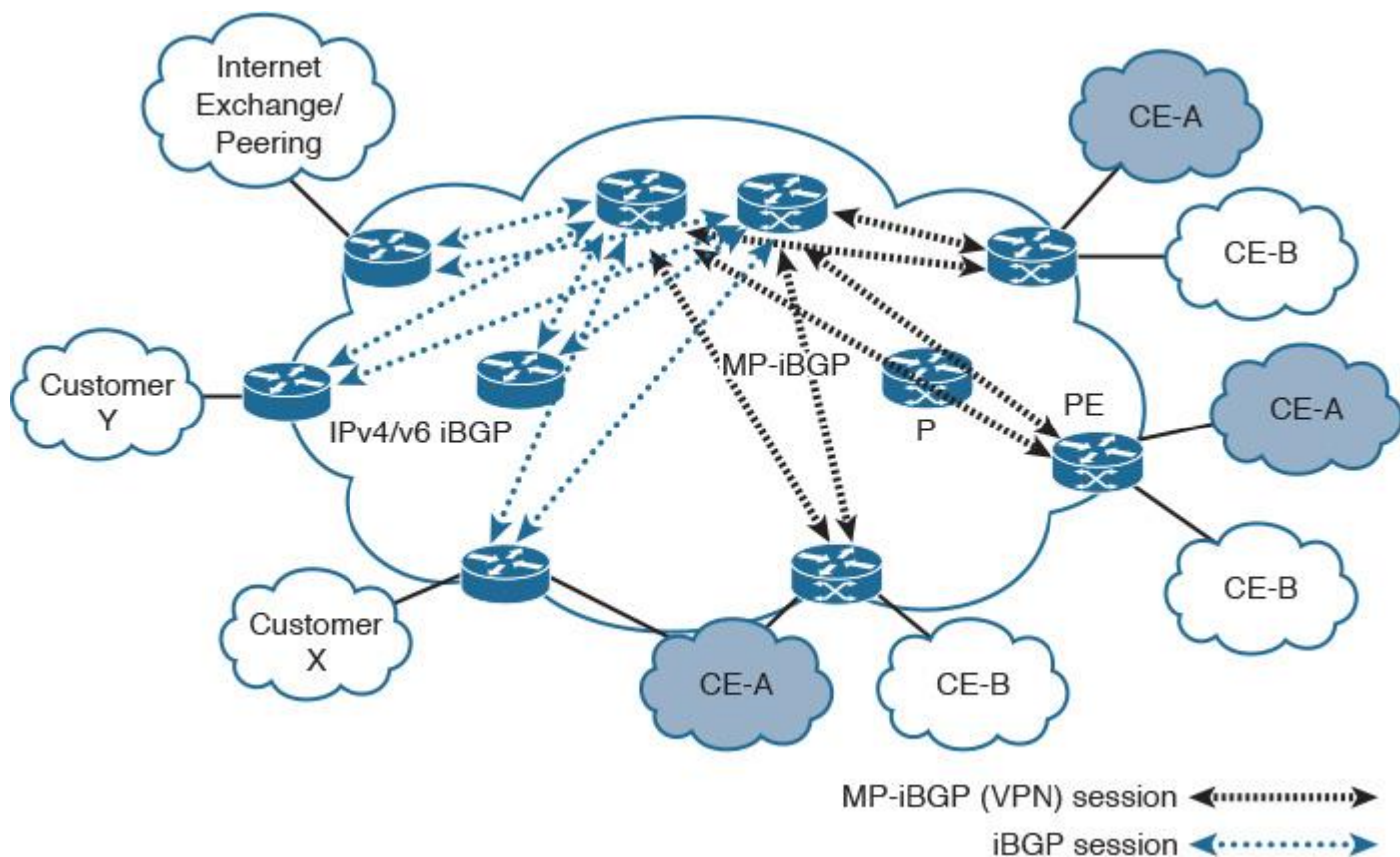


Figure 6-63 Dedicated BGP RR for VPN and IP

Separate RR per Service (MPLS-VPN and IPv4/v6)

In large-scale service provider networks, where there may be tens of hundreds of routers and different services offered (such as MPLS- and non-MPLS-based services), along with hundreds of thousands of prefixes, it is more feasible to have separate RRs per function (for example, RRs for MPLS-VPNv4, RRs for MPLS-VPNv6, and RRs for IPv4/v6 iBGP), as shown in [Figure 6-64](#).

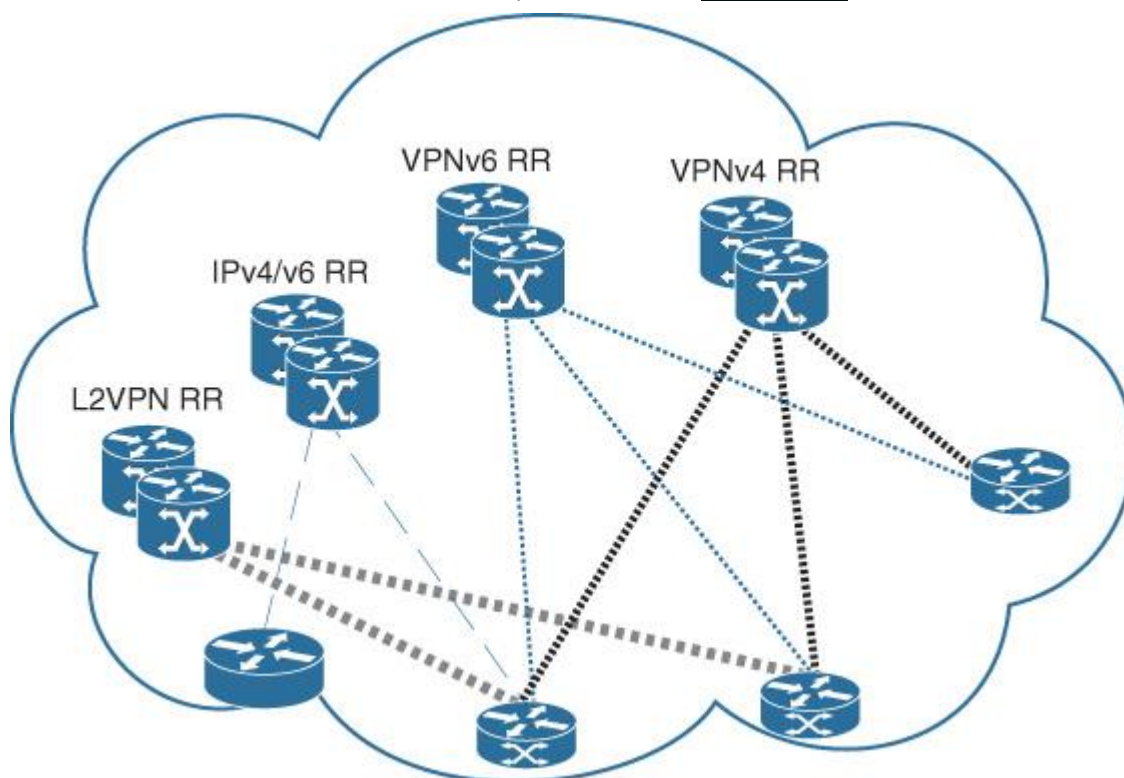


Figure 6-64 Dedicated BGP RR per Service

In contrast, in smaller-scale networks, the same RRs can service multiple functions such as MPLS-VPNv4/VPNv6. In fact, it is a design decision where network designers must have a good understanding of the current state of the organization and its future plans to select the most suitable design option. For instance, if there is a limited number of MPLS-VPNv6 customers yet the plan is to increase IPv6 offerings, it makes more sense to separate the RR functions of MPLS-VPNv4 and MPLS-VPNv6, for the following reasons:

- Optimizes performance and scalability
- Eliminates any single point of failure or dependencies between the two worlds (V4 and V6)
- Offers more structured, flexible, and simplified management and policy control

Hierarchical RR

As discussed earlier in [Chapter 1](#), “[Network Design Requirements: Analysis and Design Principles](#),” the optimal scalable design should be reliable and manageable at the same time. One of the key factors to achieve this is by reducing the number of prefixes to be processed and stored by the nodes. The concept of BGP RR can significantly optimize BGP design (MPLS based and native IP) because by default it usually performs implicit route aggregation by advertising the single best route only. However, in large BGP environments (usually service provider-grade network), multiple layers of BGP speakers are distributed across multiple regions (such as POPs). For example, in [Figure 6-65](#), the iBGP sessions of the edge nodes are terminated within the same region (POP) at the POP/core border nodes (RR aggregation nodes per POP) to provide a more modular BGP design that scales and adapts easily without adding significant operational complexity to the overall design. However, in this particular scenario, the mesh among the RRs across the core can introduce increased sessions and prefix states to the border nodes of each POP. For instance, the network shown in [Figure 6-65](#) has two RRs per POP. Based on this design, if a new prefix is advertised (like Prefix X in POP 3), each RR will usually reflect this update to all its clients and normal e/iBGP peers. As a result, each RR (such as the PE_A in POP 1) will receive several routes for a single prefix (the more POPs in the network, the more prefixes received). This design can become even worse when there is a large number of BGP prefixes and a large number of POPs, which can impact the overall network performance and limit the design scalability and manageability in very large-scale networks with multiple tiers and regions.

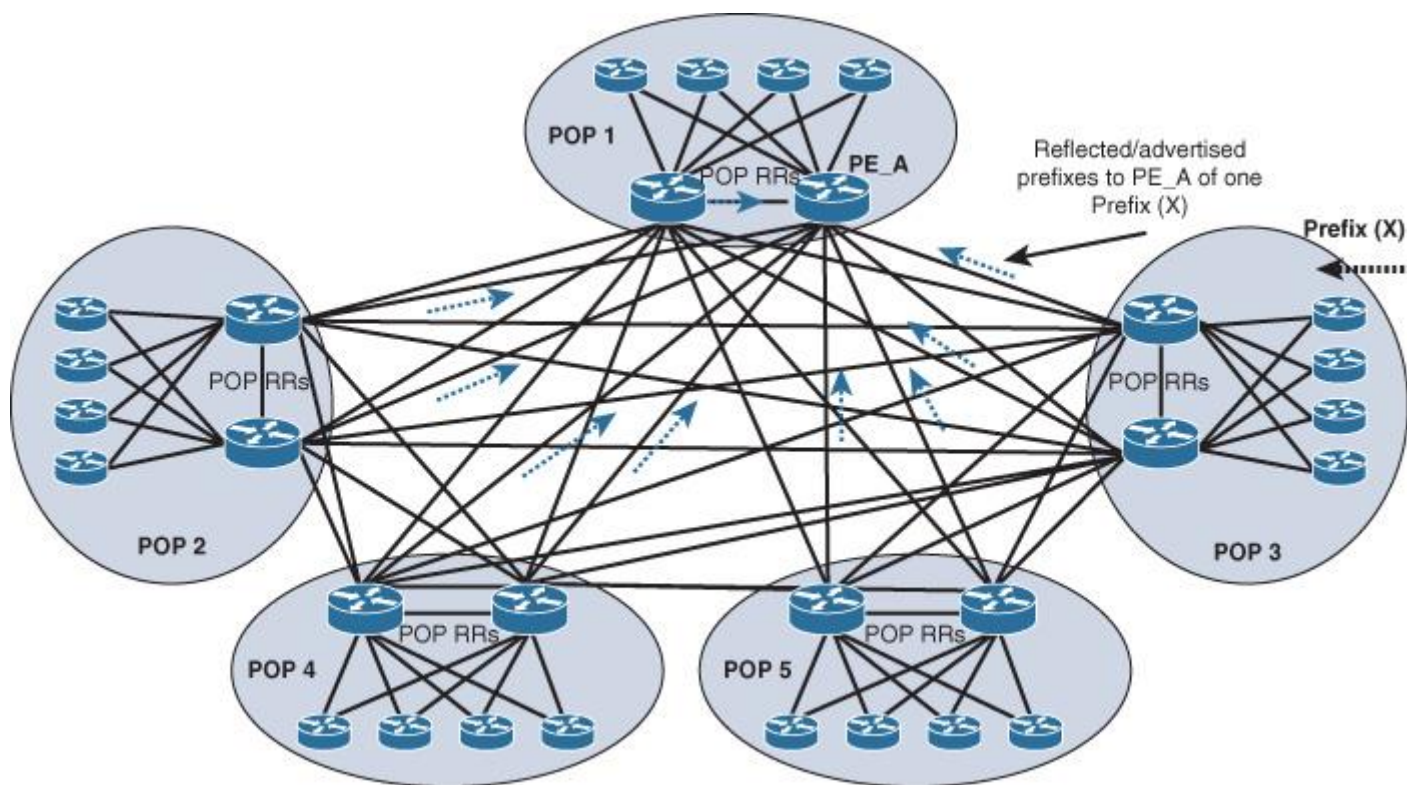


Figure 6-65 Single-Tier RR in a Large BGP Environment

Therefore, by introducing another tier of RRs, the design can be optimized to scale better by aggregating the full-mesh sessions between the RRs of the POPs, relaxing the full mesh of the level 2 RRs, as shown in [Figure 6-66](#).

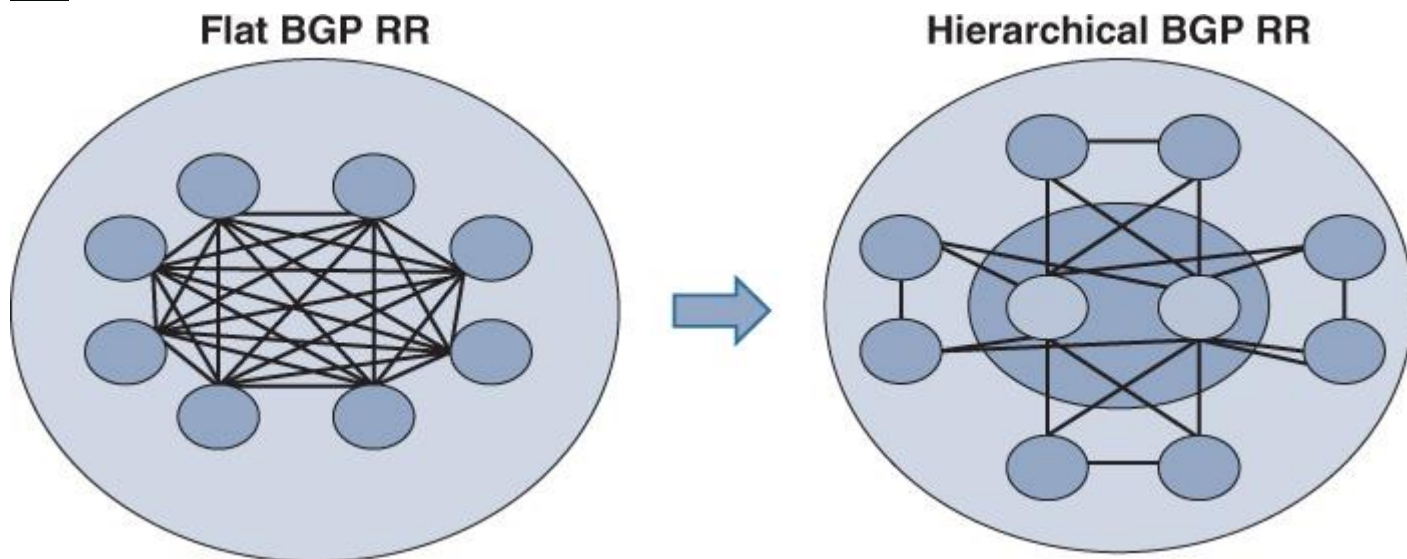


Figure 6-66 Multitiered BGP RR

This will usually result in a more refined number of iBGP sessions at multiple layers and a significant reduction in the number of routes each level 2 RR at the (POP border) will receive. As shown in [Figure 6-67](#), regardless of the number of POPs, each POP border (RR) will receive two routes per remote prefix (one from each level 1 RR, assuming the level 1 RRs deployed with different CLUSTER_IDs). In other words with this design, level 1 RRs will reduce the number of prefixes being carried to the BGP speaker at the lower tiers.

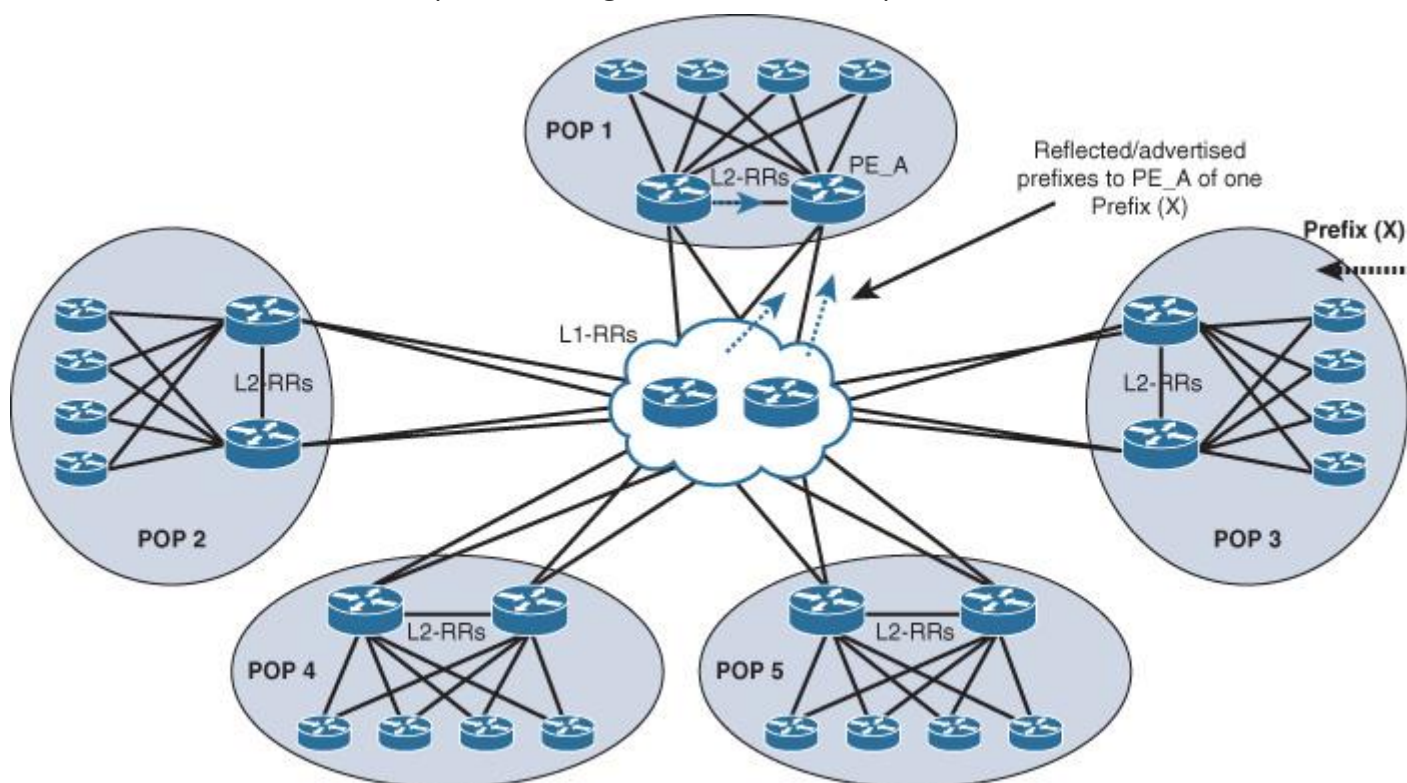


Figure 6-67 Multitiered BGP RR Optimization

Although this hierarchical BGP RR design approach offers a more structured and optimized design to handle very large-scale BGP environments, having multiple layers of RRs can break hot-potato routing requirements (if it is a requirement). Also multiple tiers of RRs may introduce route oscillation and a noticeable increased network convergence time following any failure event. Therefore, network designers must evaluate the